FAKULTA APLIKOVANÝCH VĚD
ZÁPADOČESKÉ UNIVERZITY
V PLZNI

Edgebreaker

Komprese trojúhelníkových sítí

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2026-11-04

1. Motivace

1.1 Dobový kontext

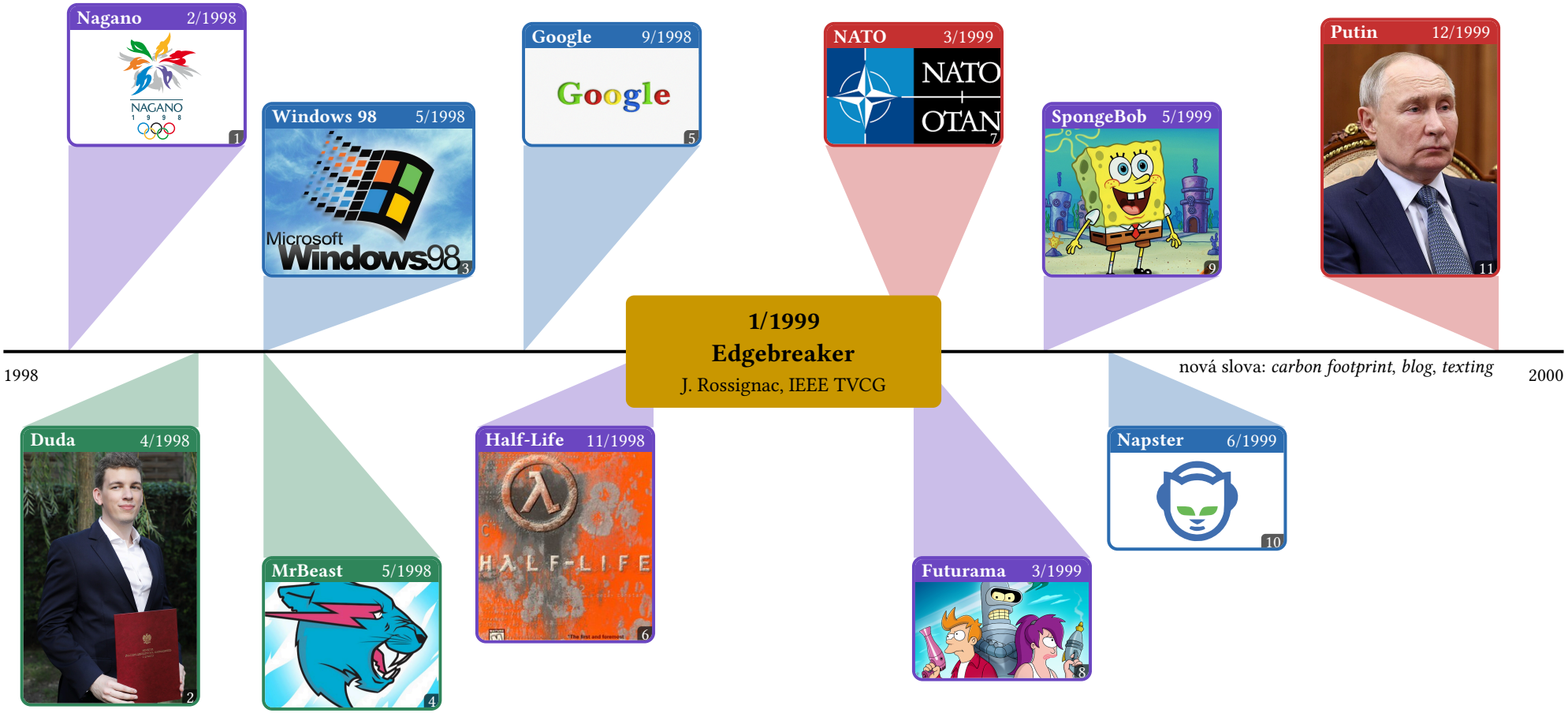


Figure 1: Dobový kontext vzniku algoritmu Edgebreaker.

1.1 Dobový kontext



Figure 2: Jarek Rossignac.

zdroj: http://kelvinkuo.com/wp-content/uploads/2010/03/KKK_6258.jpg

disk vs RAM

2. Definice problému

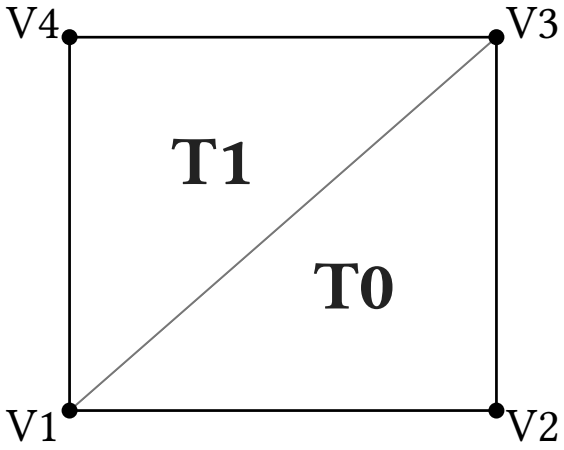


Figure 3: Trojúhelníková síť (mesh).

2.1 Mesh: geometrie + konektivita

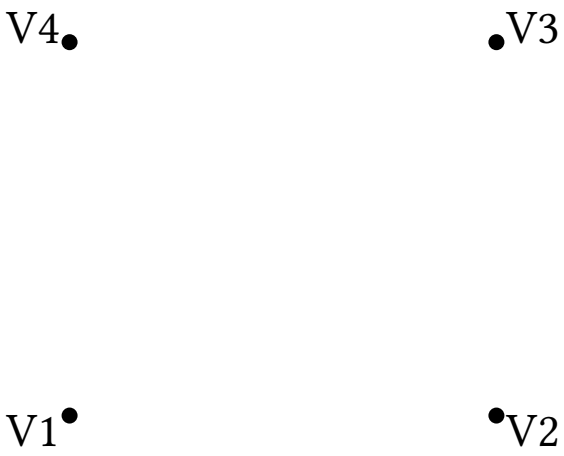


Figure 3: Trojúhelníková síť (mesh).

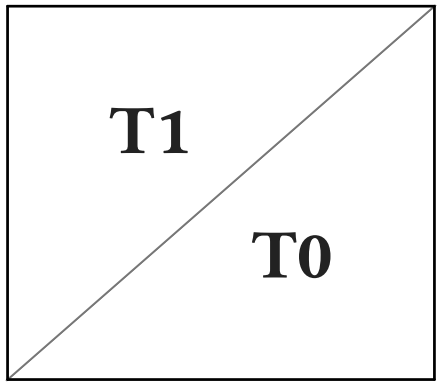


Figure 3: Trojúhelníková síť (mesh).

2.2 Jak schovat informaci?

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pořadí vertexů

[V1 V2 V3]

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[V1 V2 V3]
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[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

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[V1 V2 V3]
[V1 V3 V2]
[V2 V1 V3]
[V3 V1 V2]
[V2 V3 V1]
[V3 V2 V1]

CLERS řetězec

2.2 Jak schovat informaci?

pořadí vertexů	CLERS řetězec
[V1 V2 V3]	CLRSLRCLEE
[V1 V3 V2]	
[V2 V1 V3]	
[V3 V1 V2]	
[V2 V3 V1]	
[V3 V2 V1]	

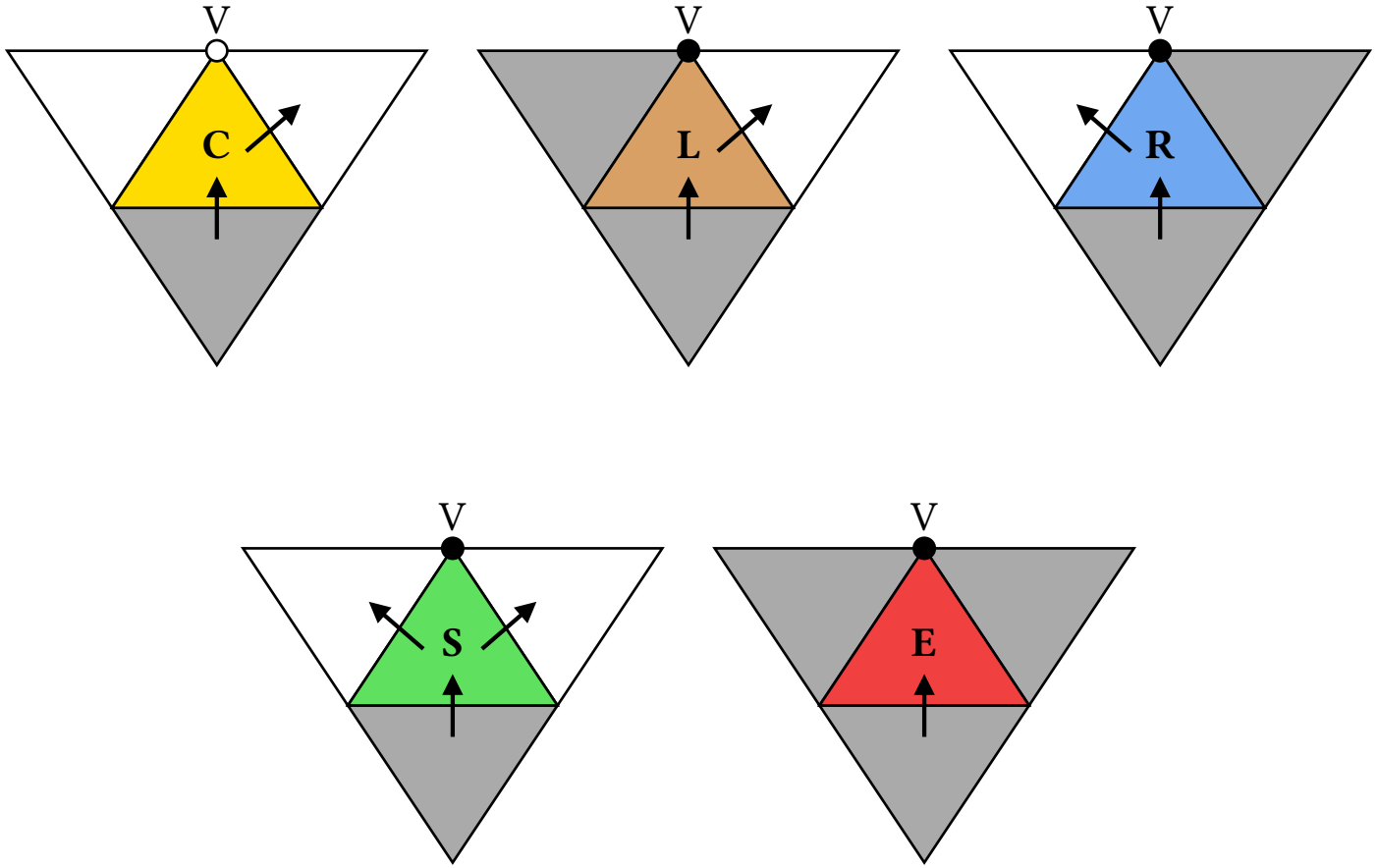


Figure 4: Základní CLERS operace.

3. Komprese

3.1 Omezení

Mesh musí být **orientovatelný manifold** homeomorfní s koulí/kruhem.

Lehce upravená verze algoritmu zvládne i *díry* a *handly*, obecně meshe s vyšším genusem.

Historie =

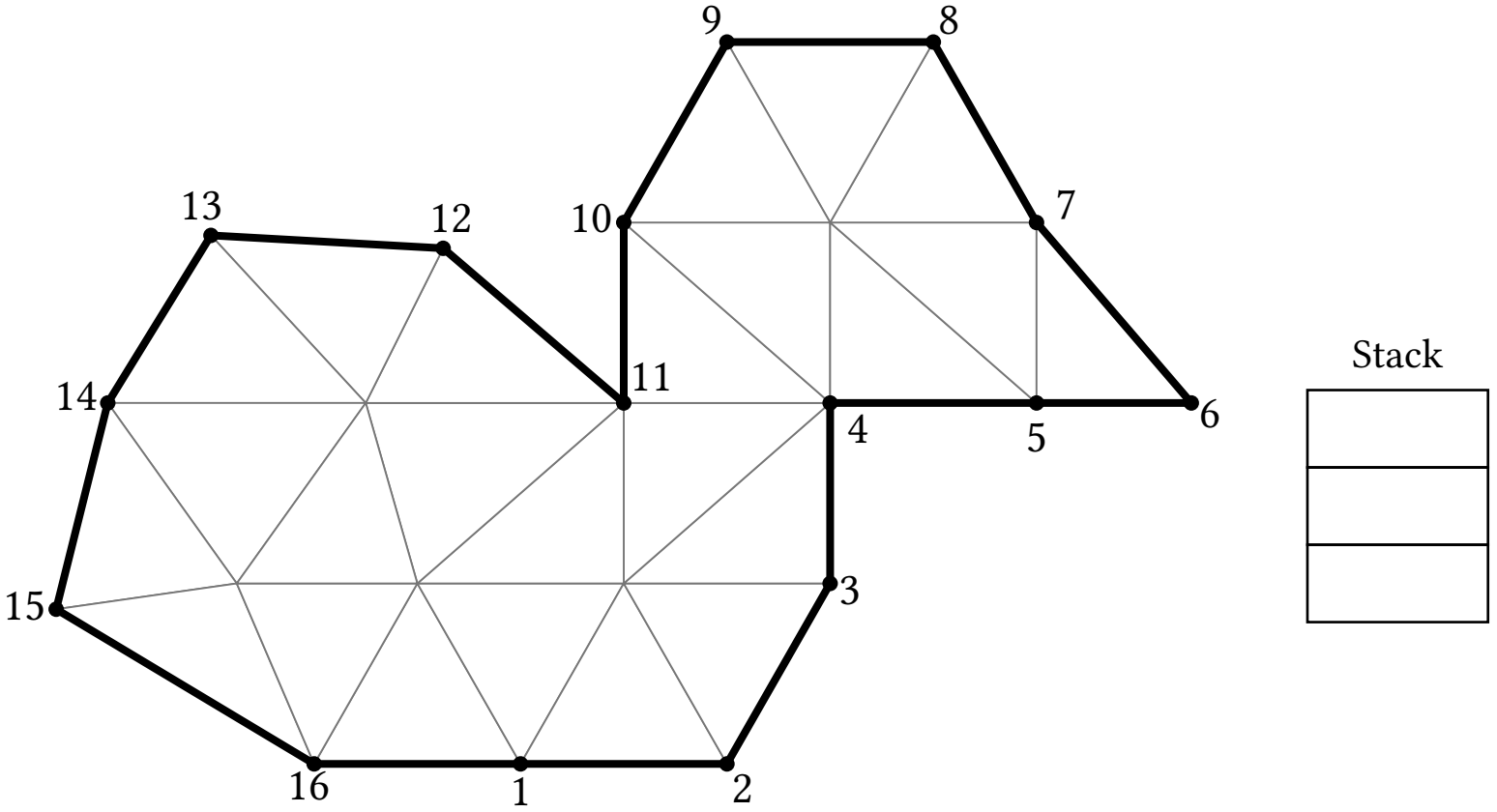


Figure 5: Jednoduchá komprese.

Historie = C

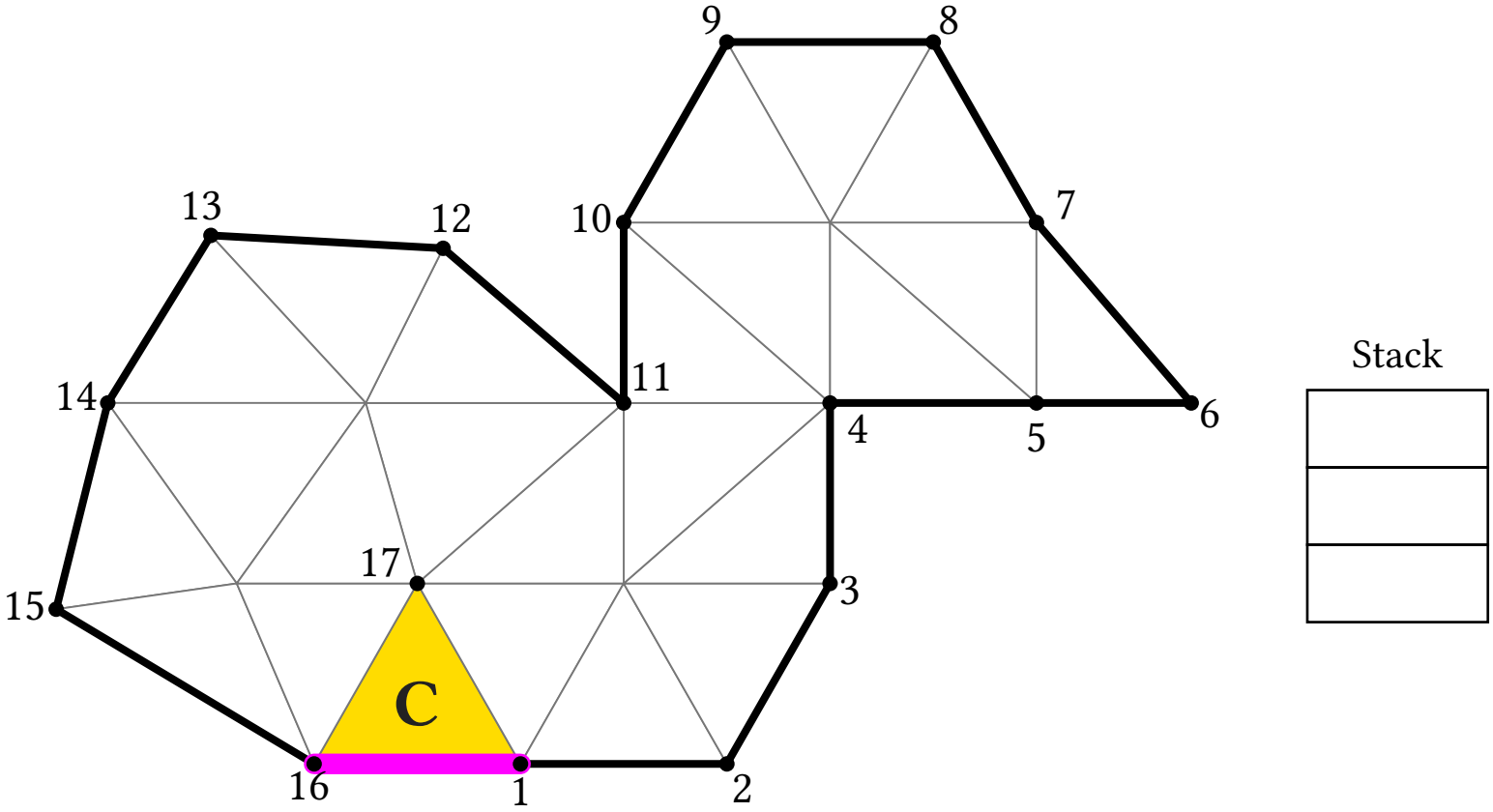


Figure 5: Jednoduchá komprese.

Historie = CC

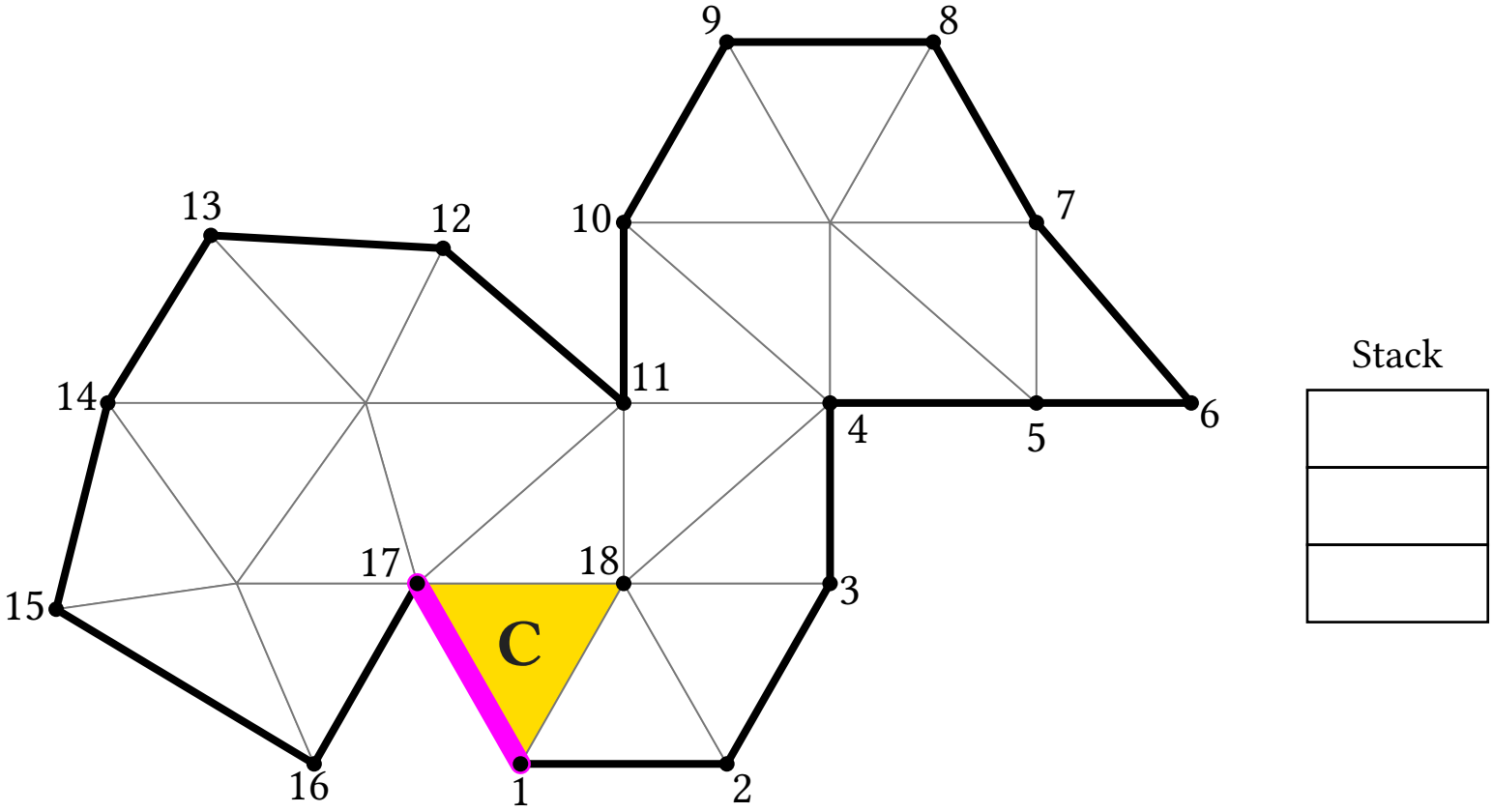


Figure 5: Jednoduchá komprese.

Historie = CCR

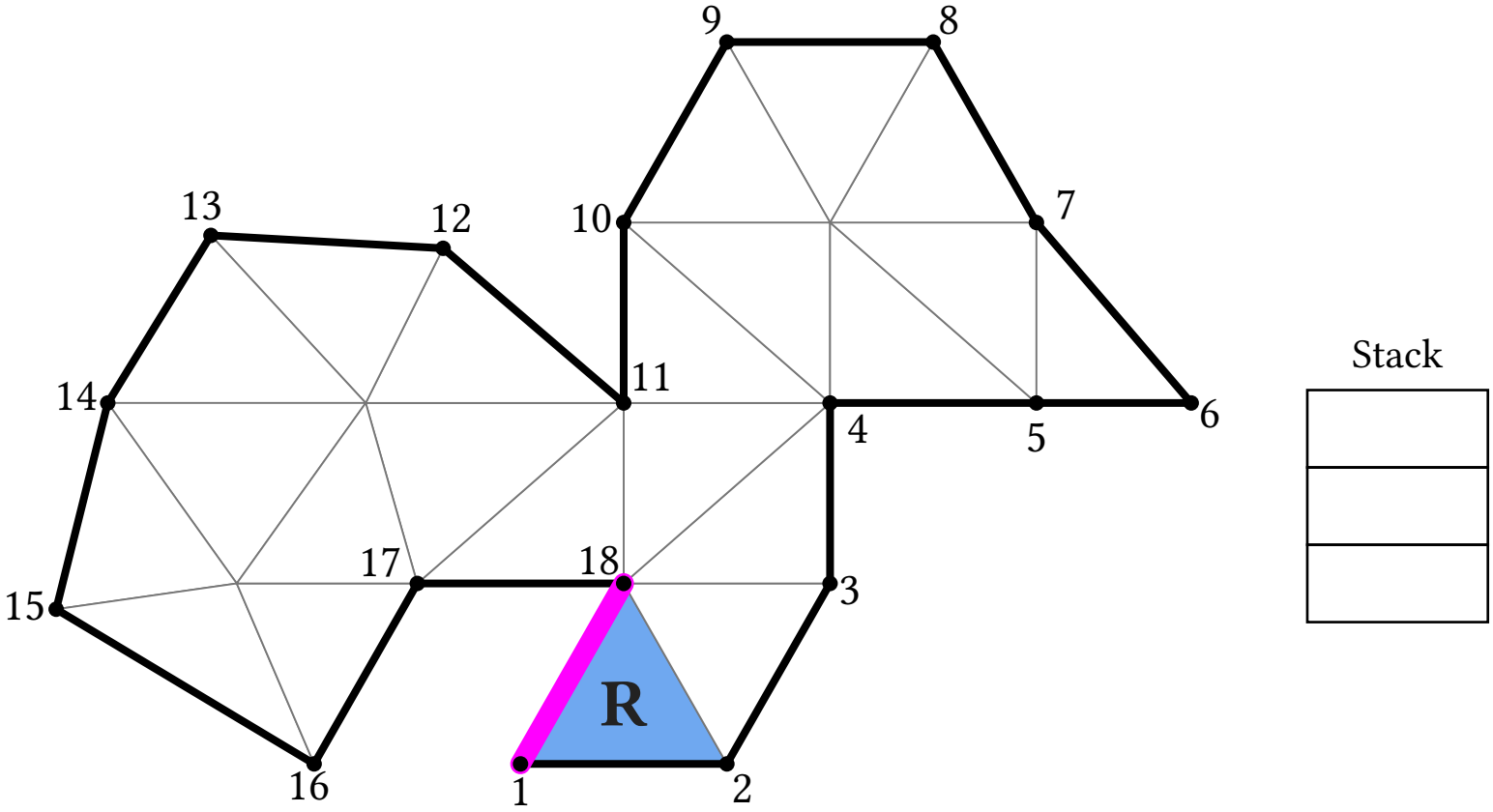


Figure 5: Jednoduchá komprese.

Historie = CCRR

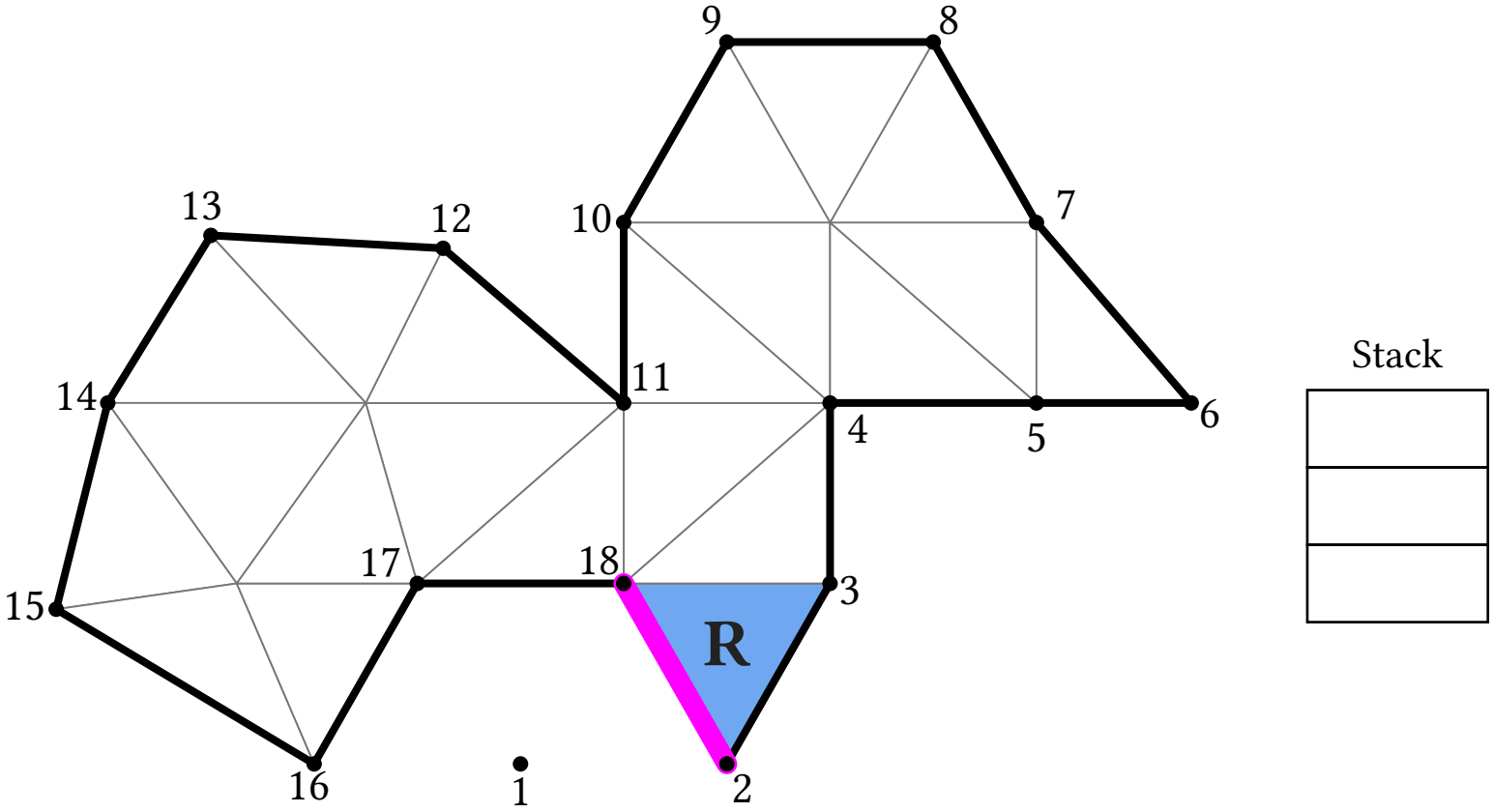


Figure 5: Jednoduchá komprese.

Historie = CCRRR

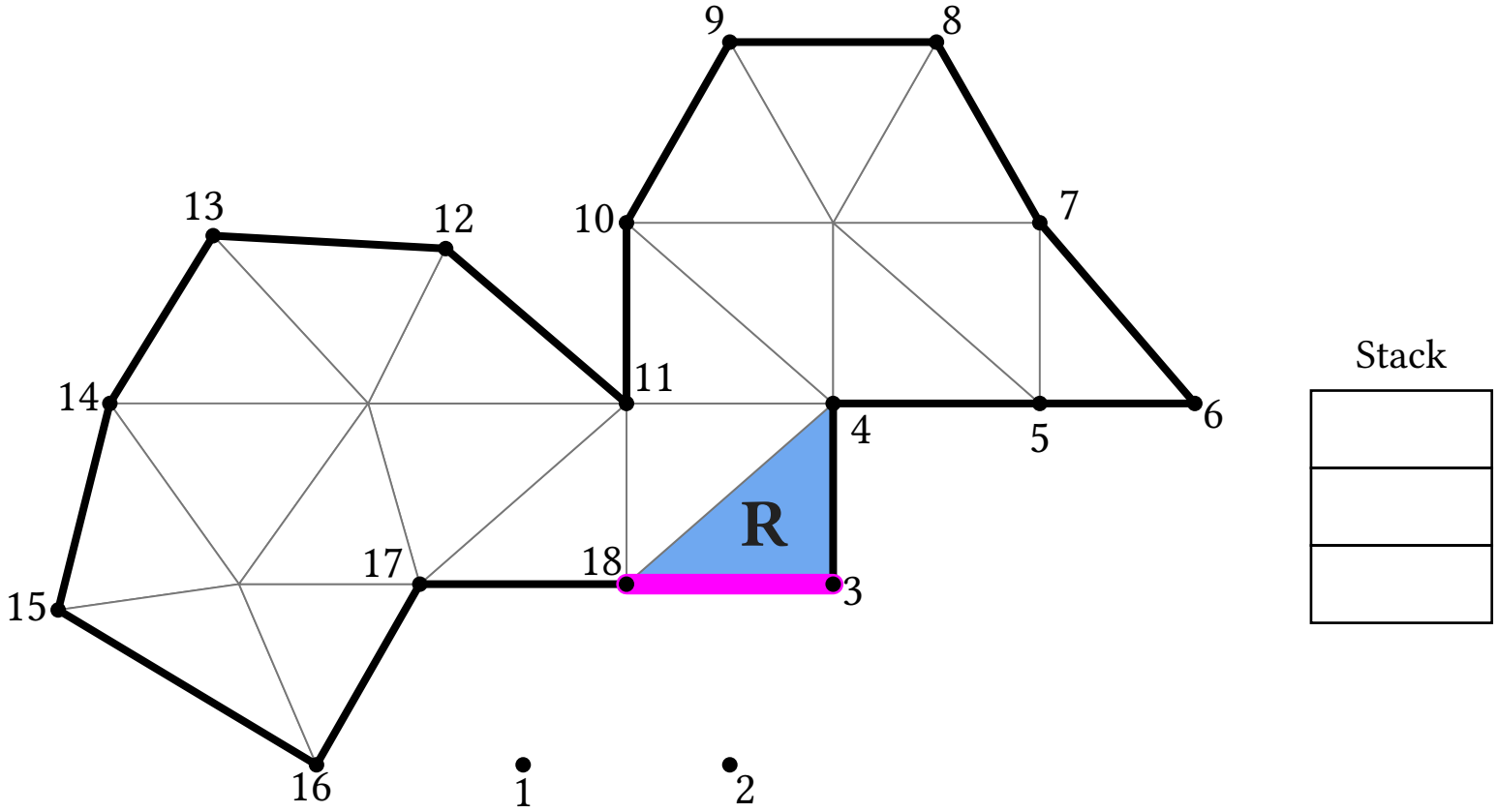


Figure 5: Jednoduchá komprese.

Historie = CCRRRS

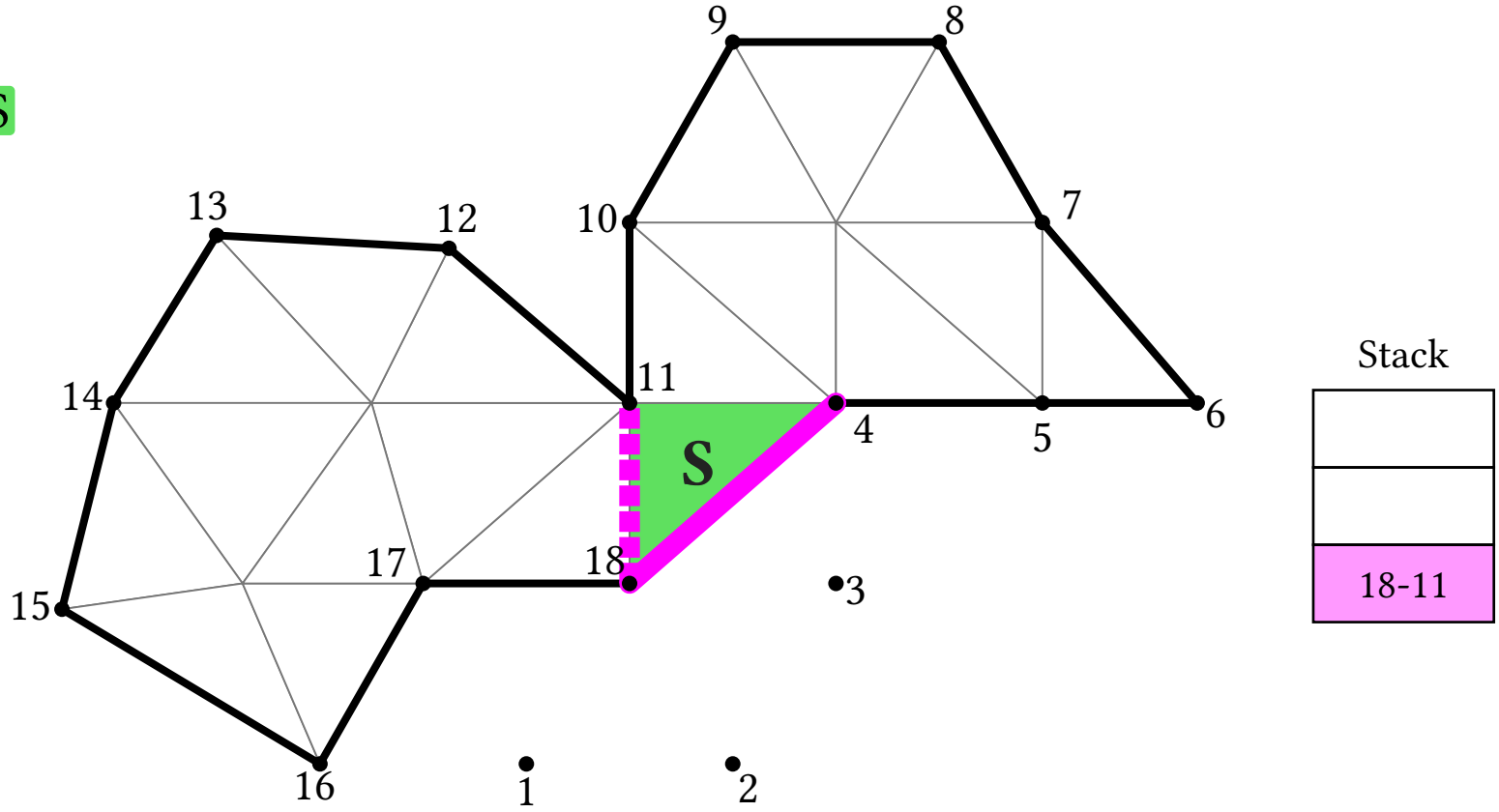


Figure 5: Jednoduchá komprese.

Historie = CCRRRSL

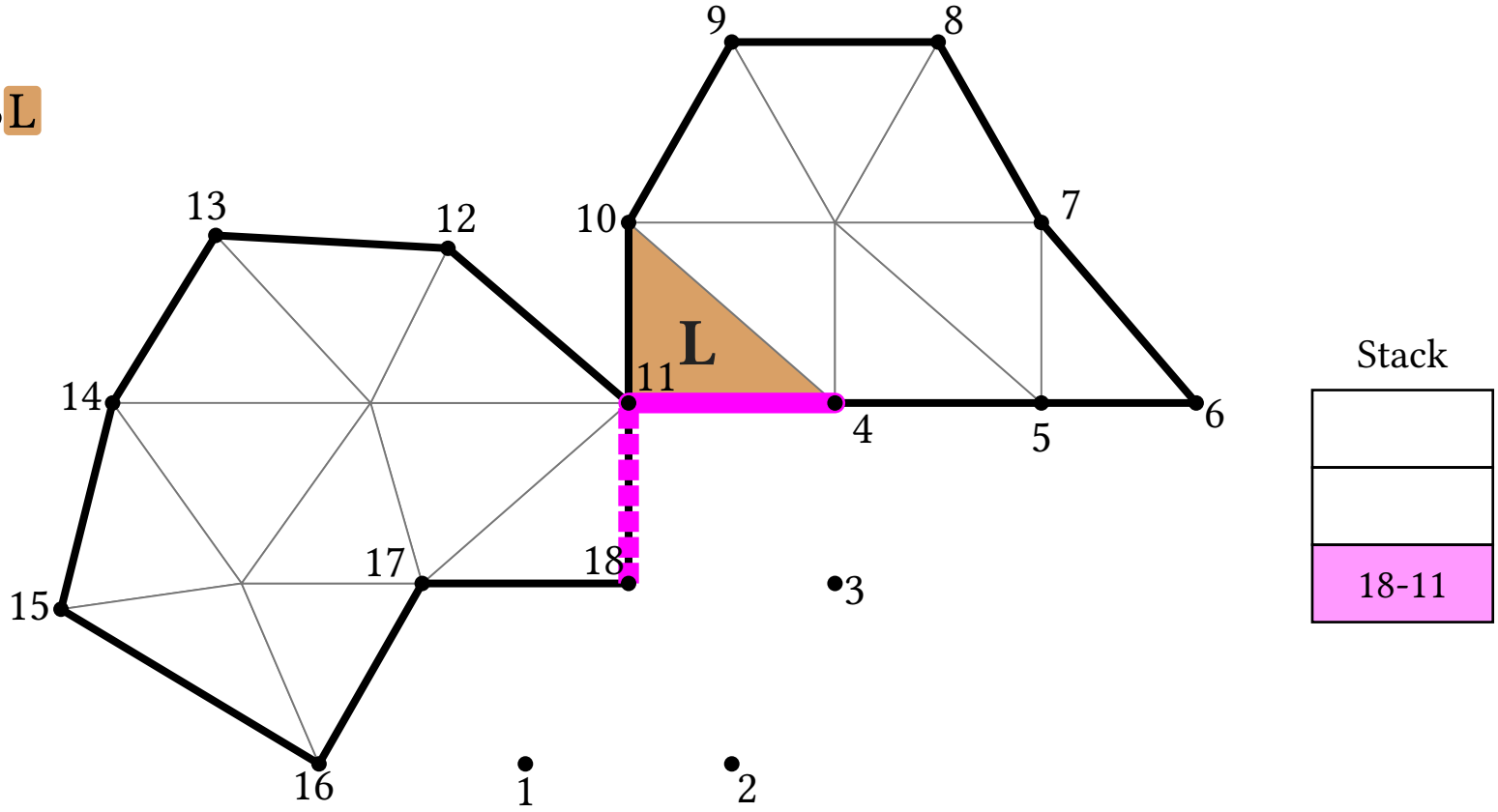


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLC

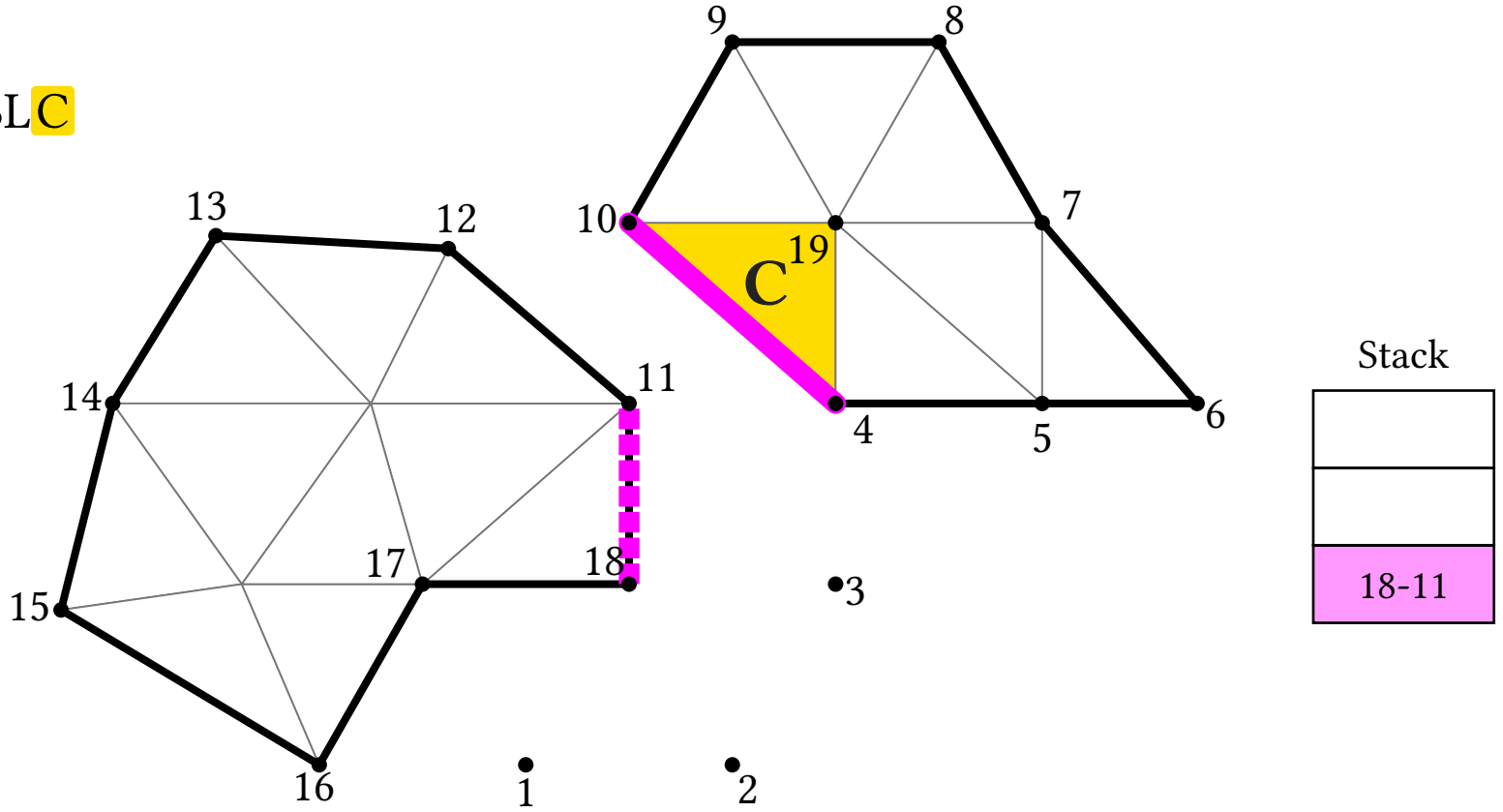


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCR

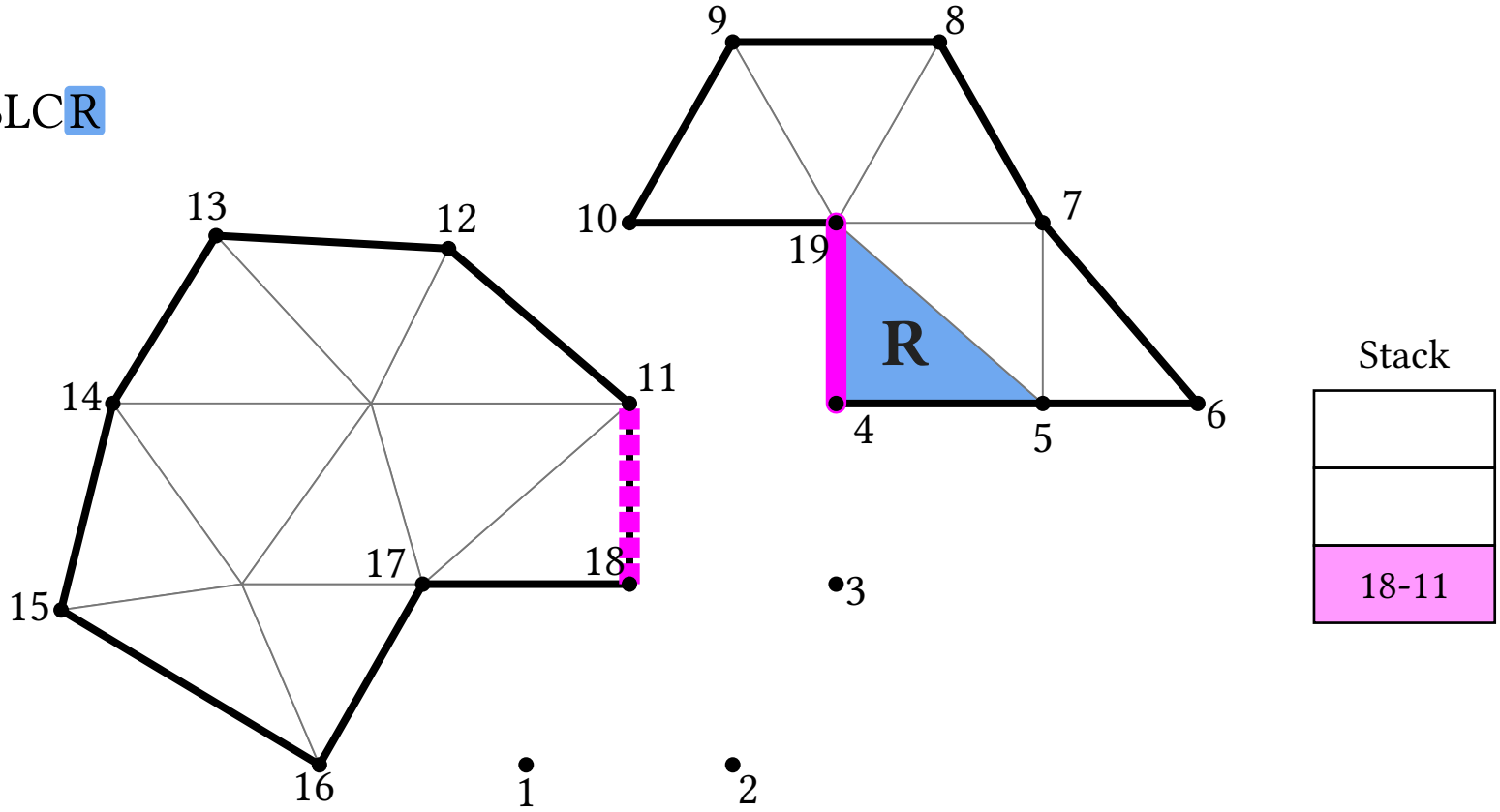


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRS

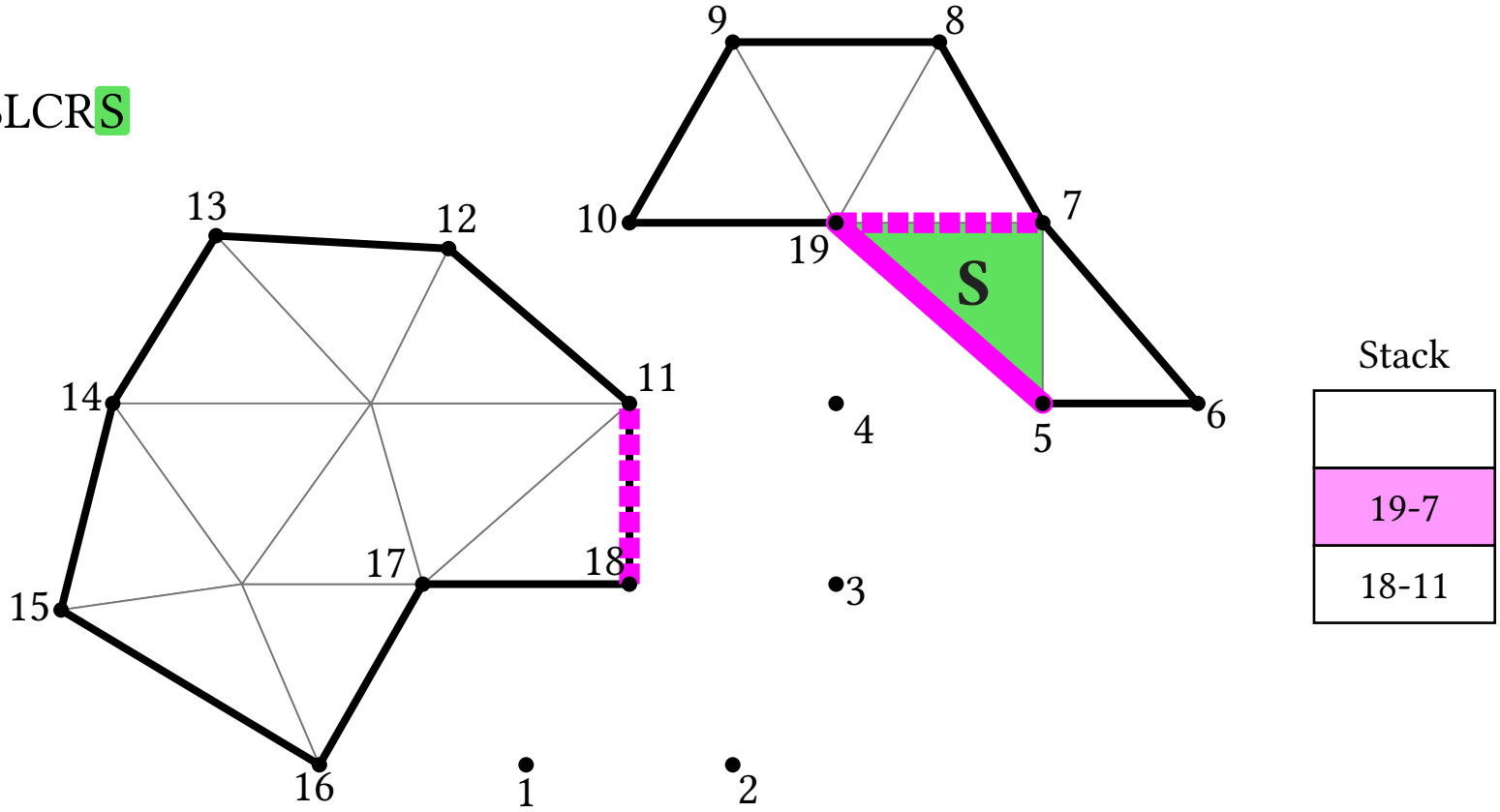


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSE

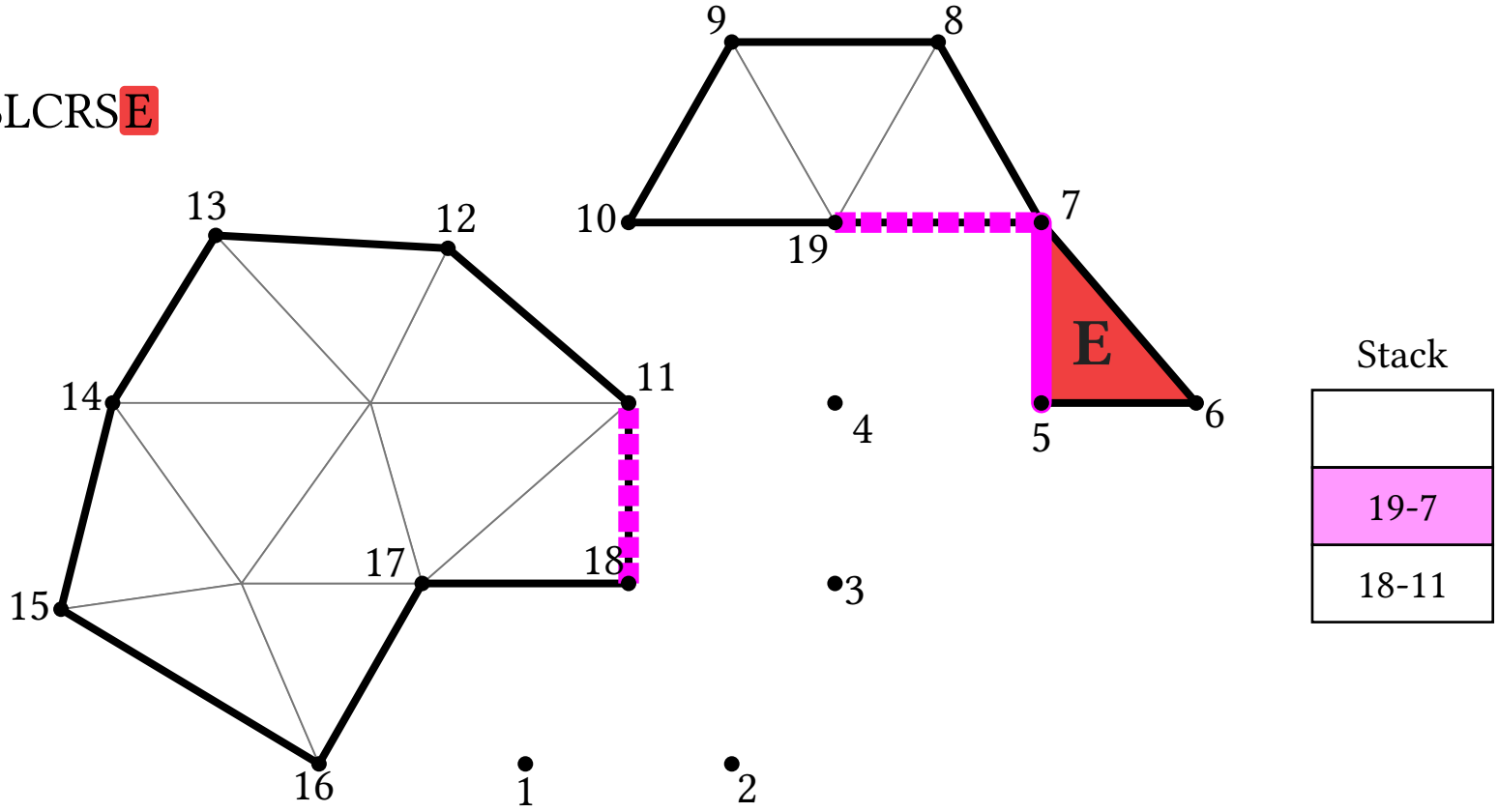


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSER

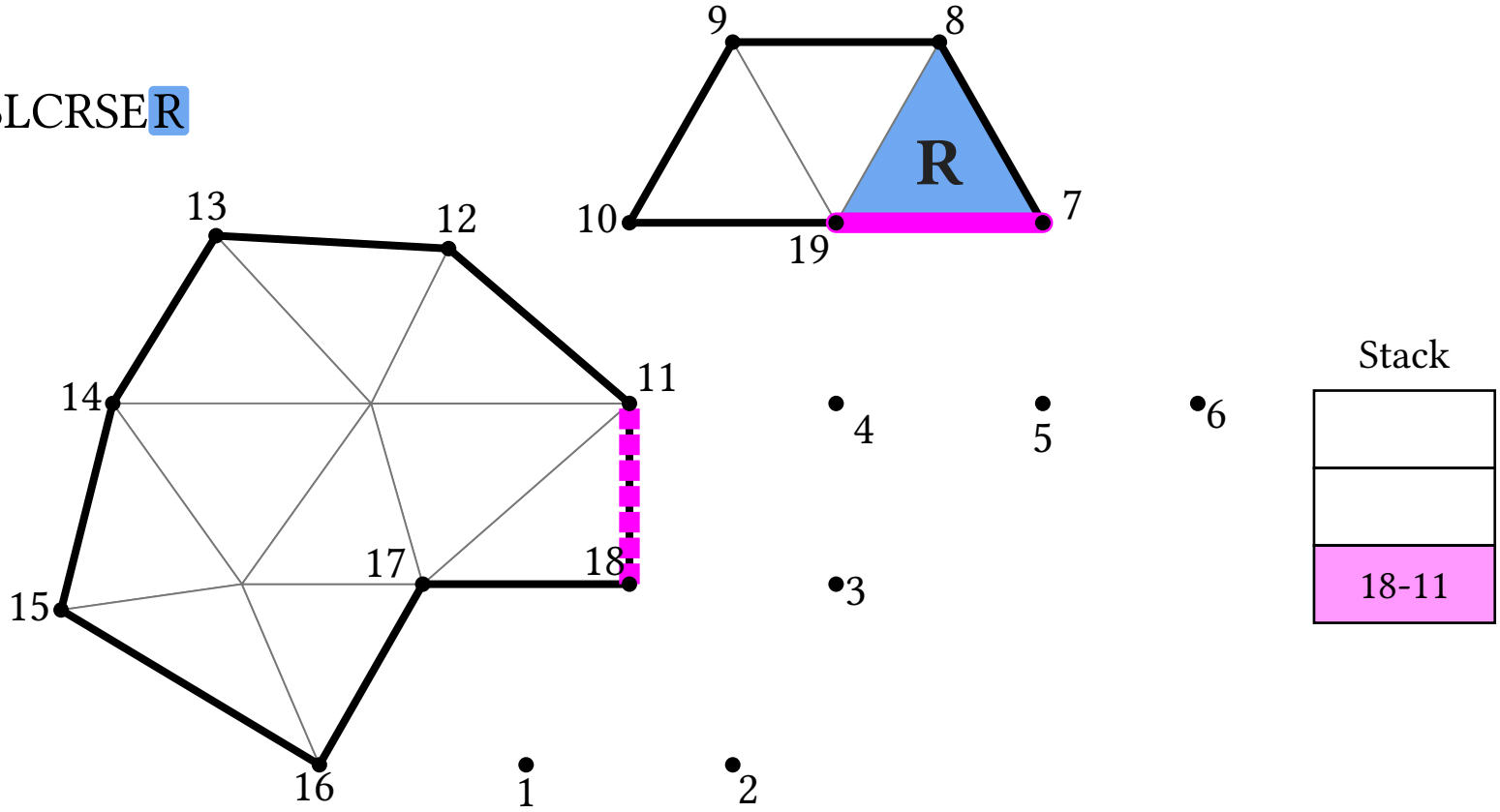


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERR

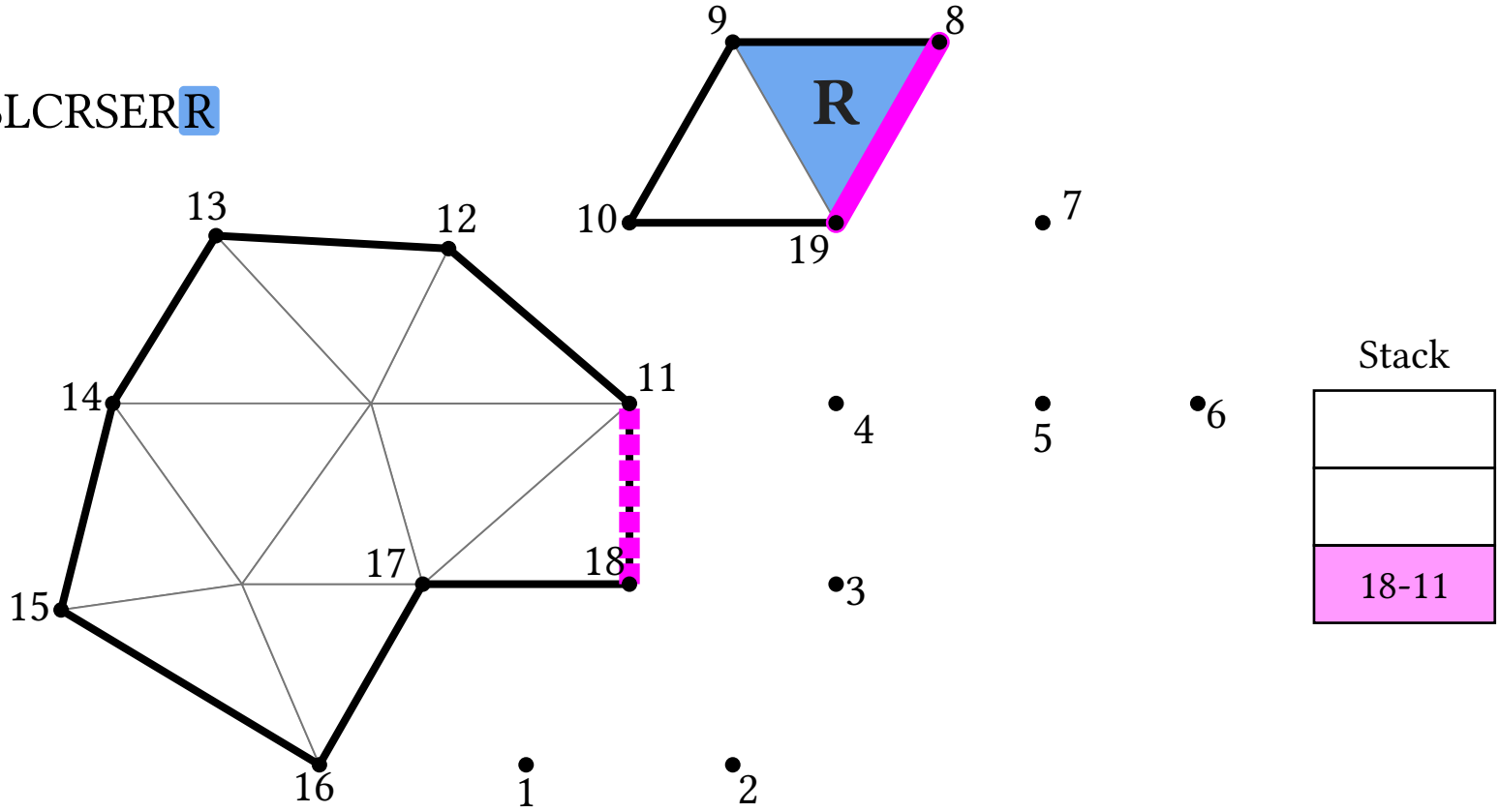


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRE

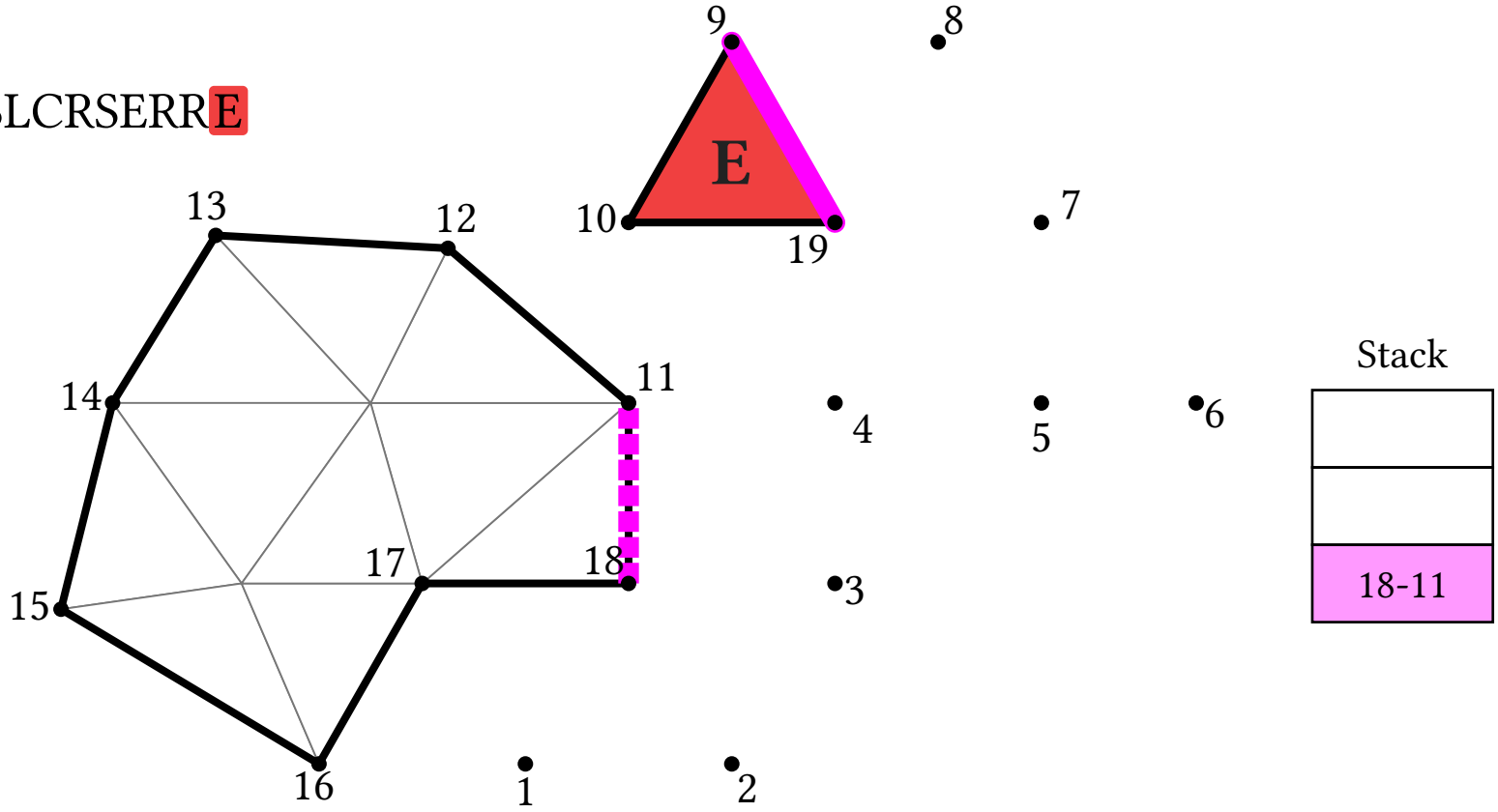


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERREL

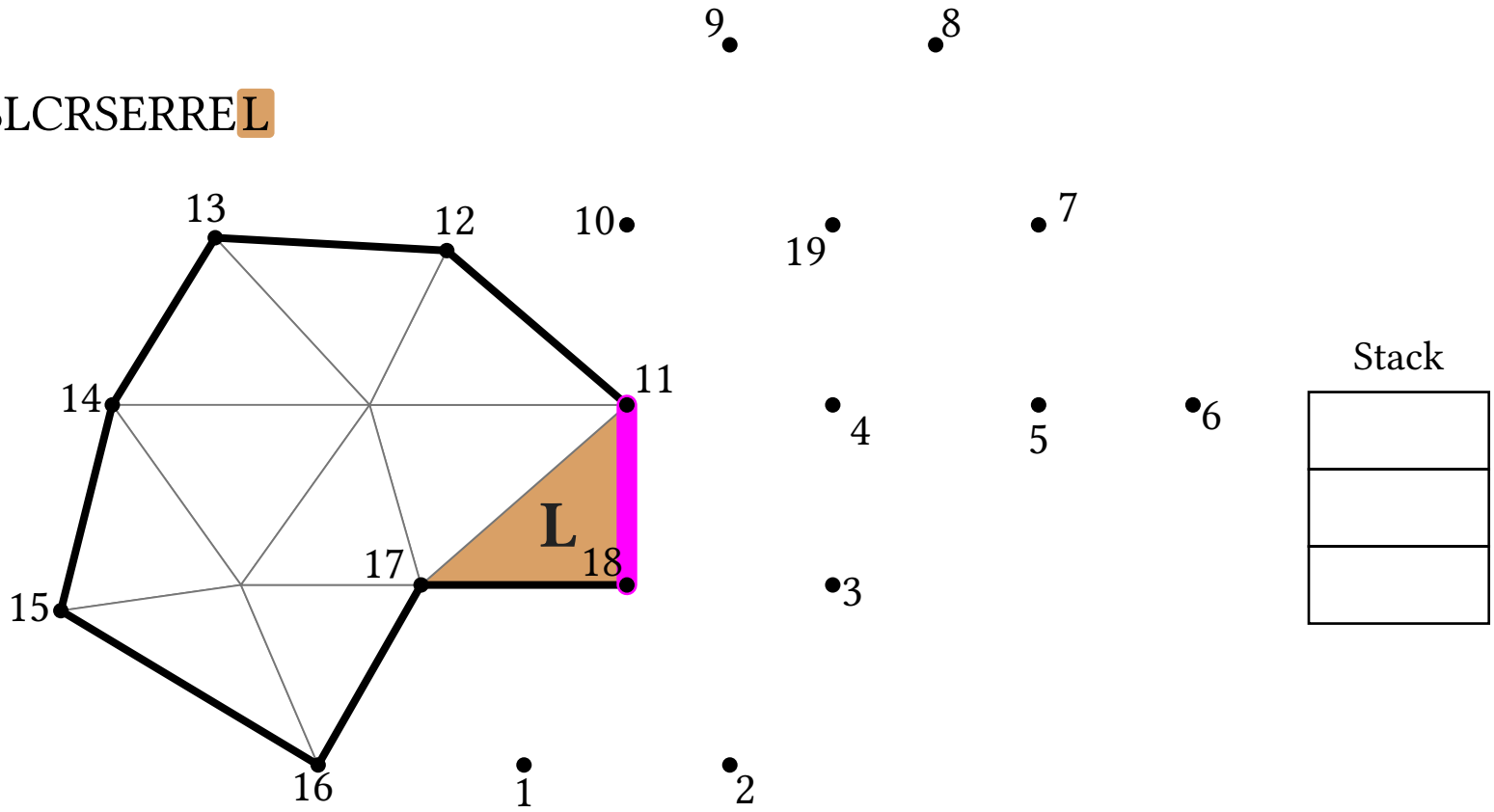


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERREL**C**

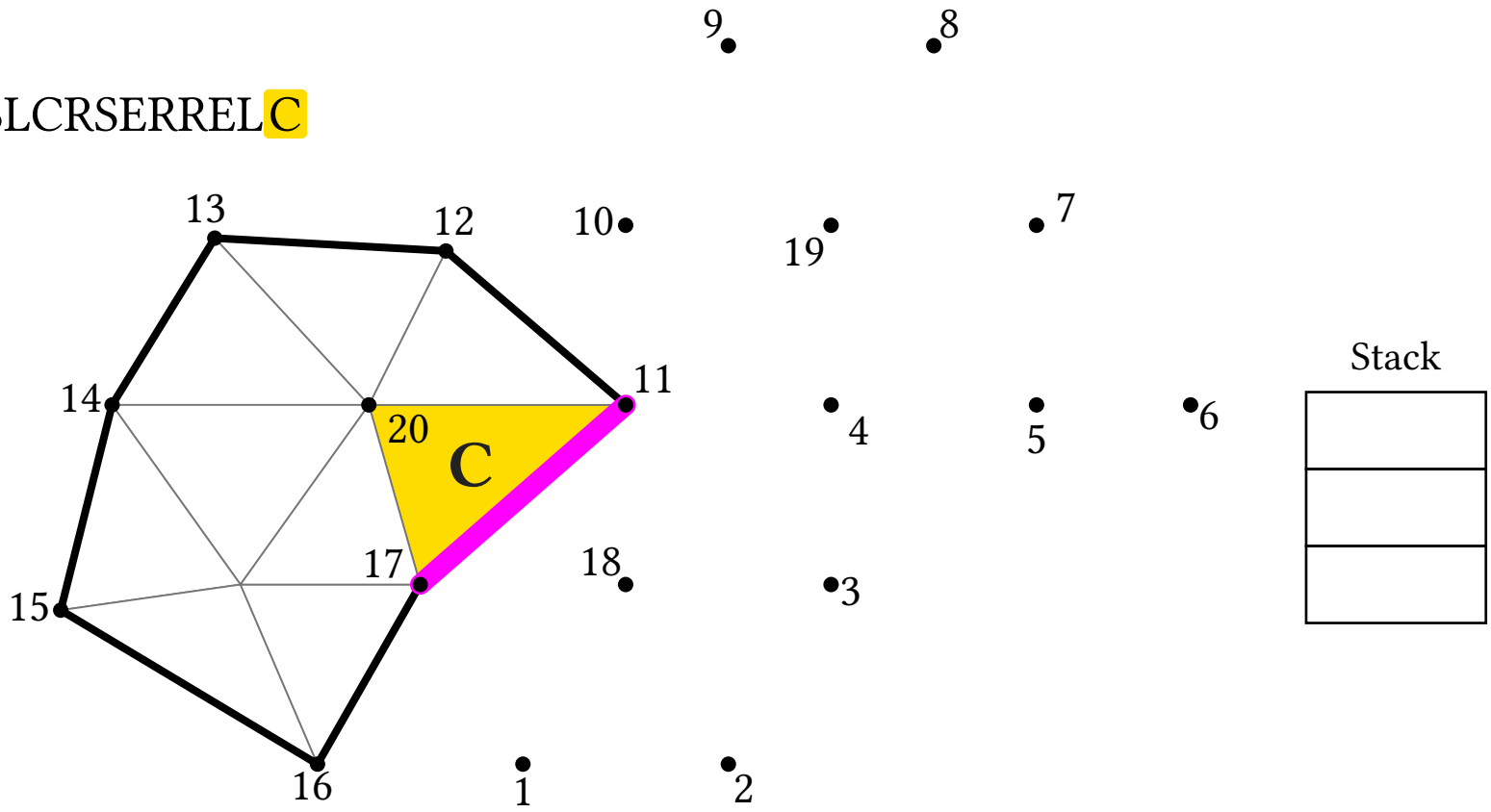


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCR

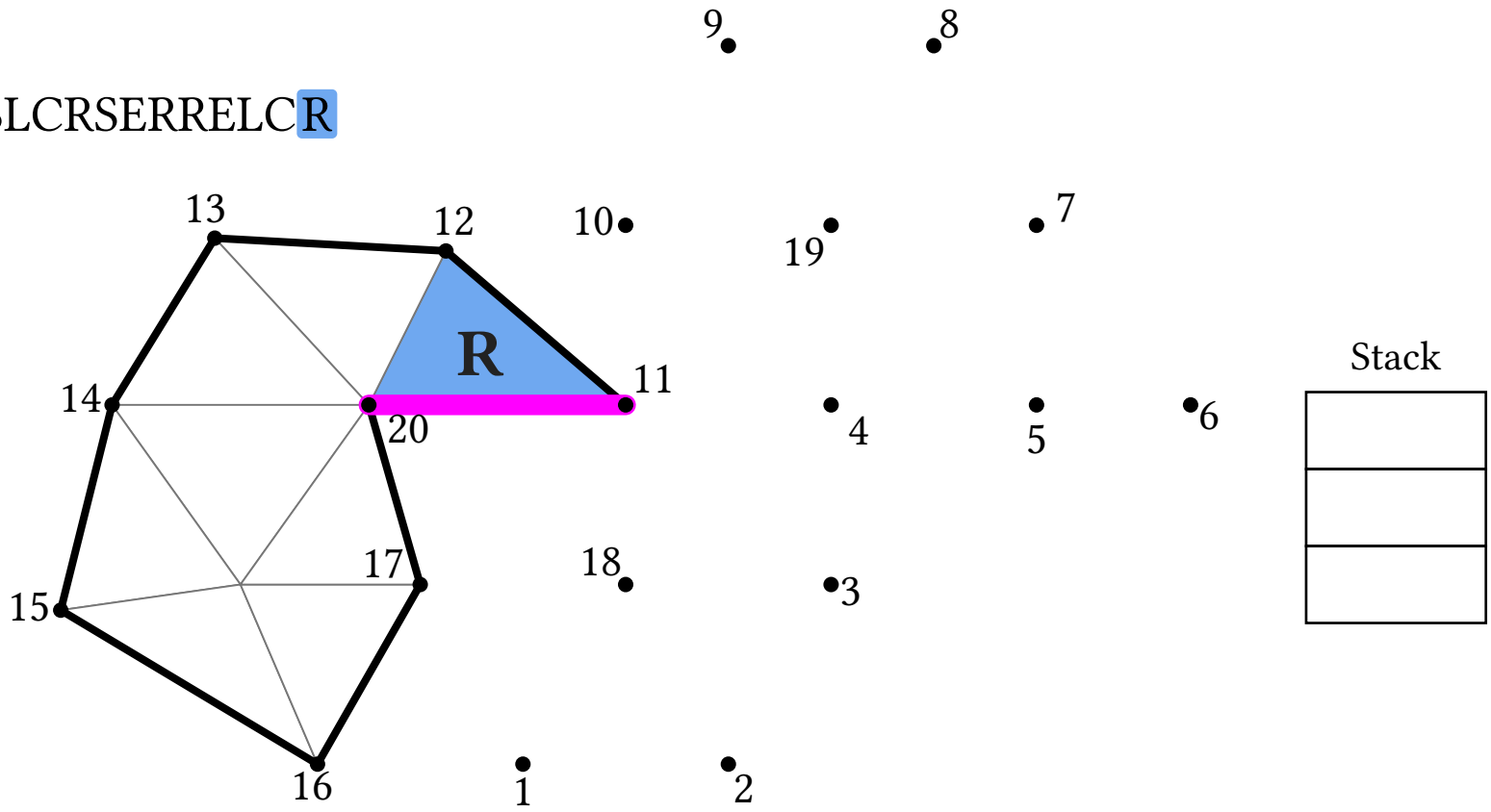


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRR

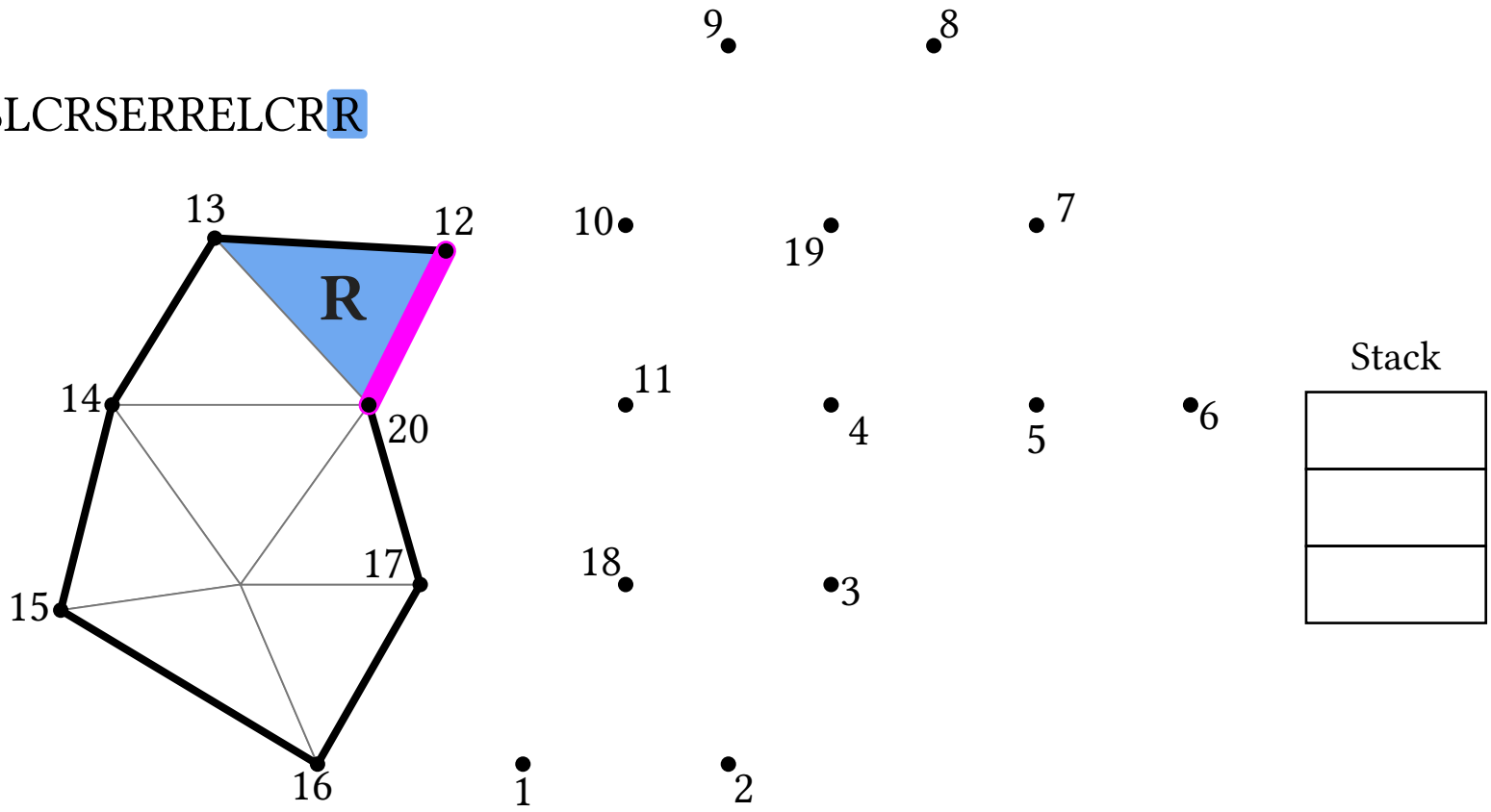


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRR

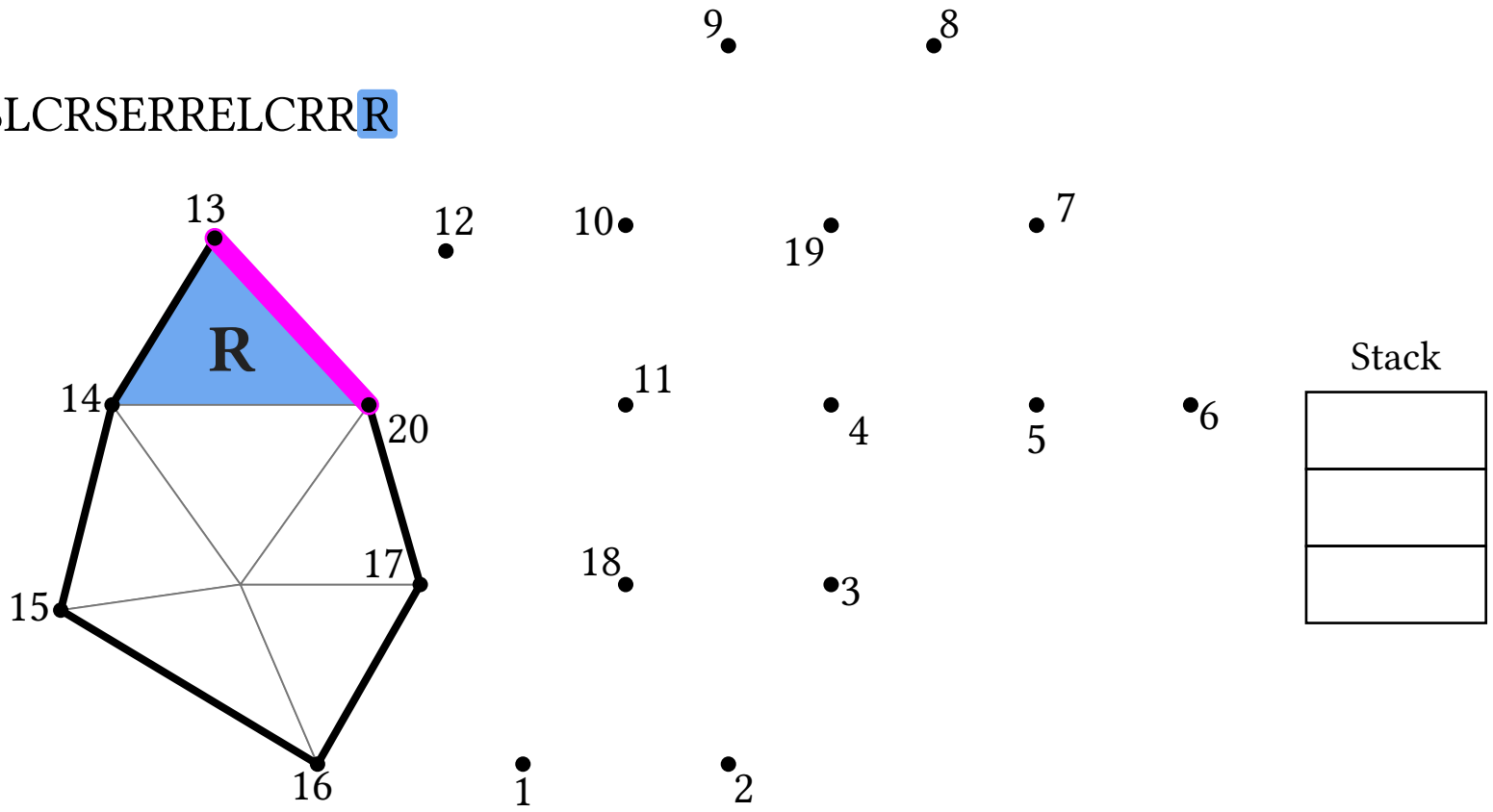


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRC

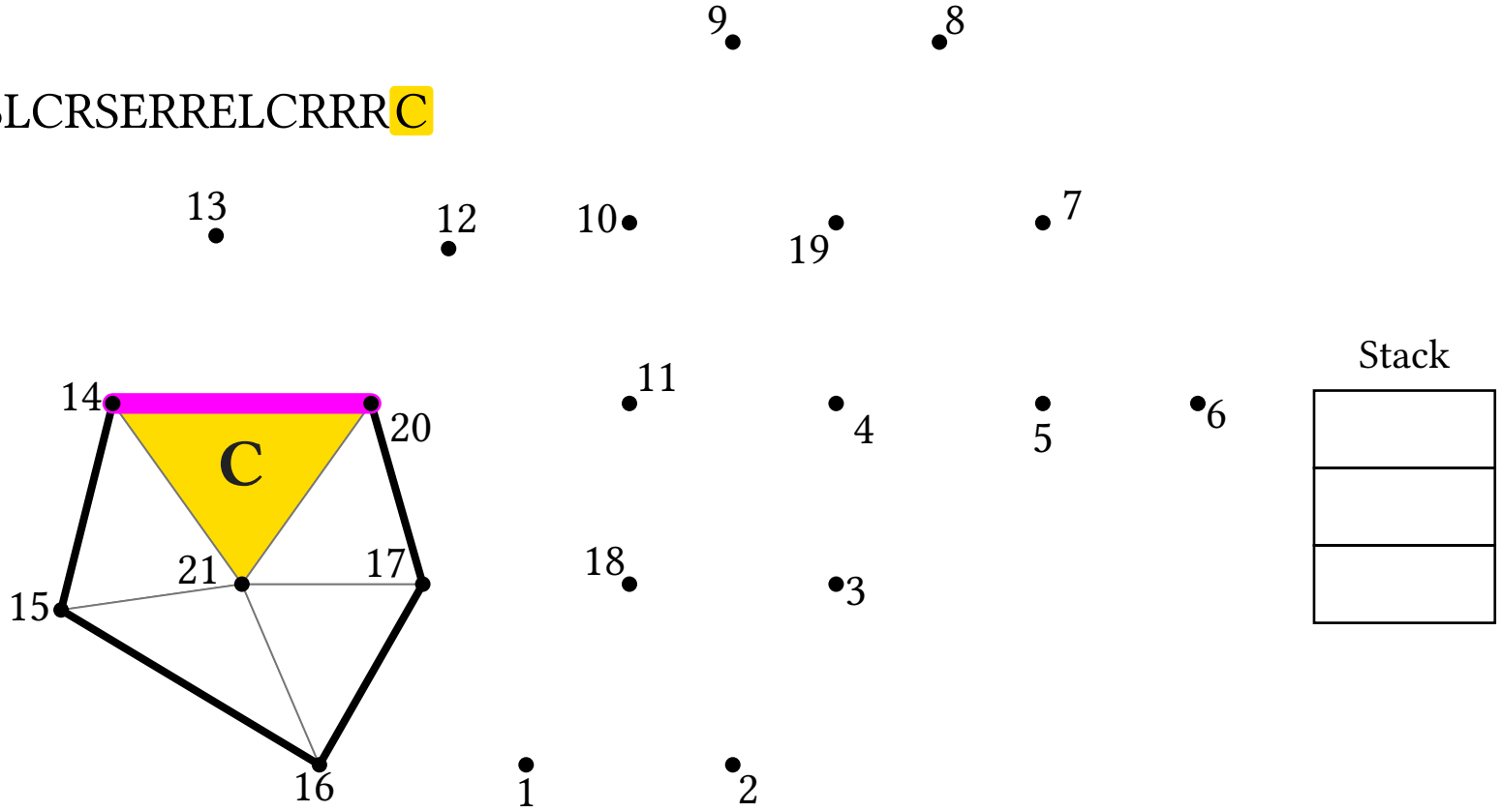


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCR

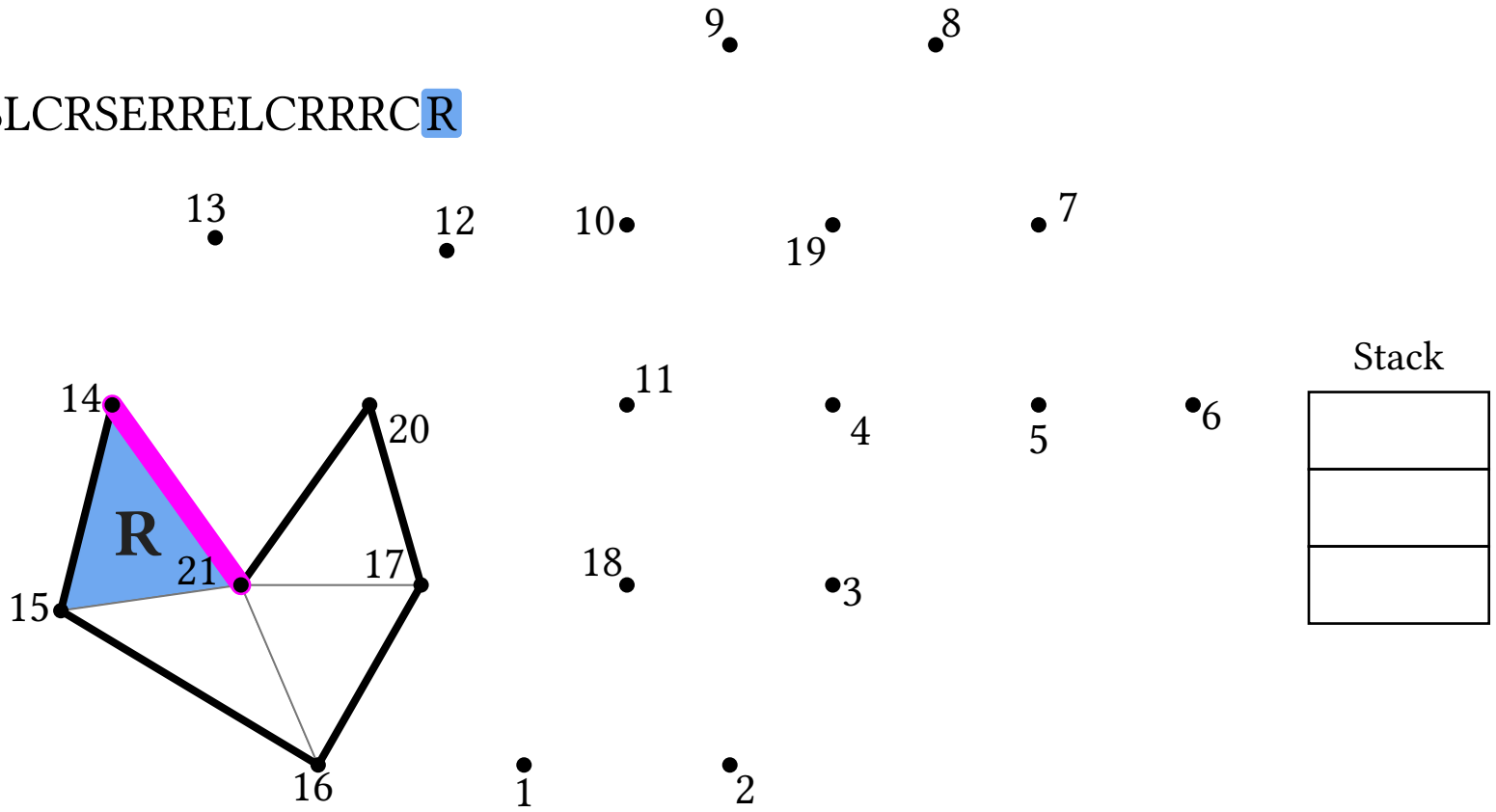


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRR

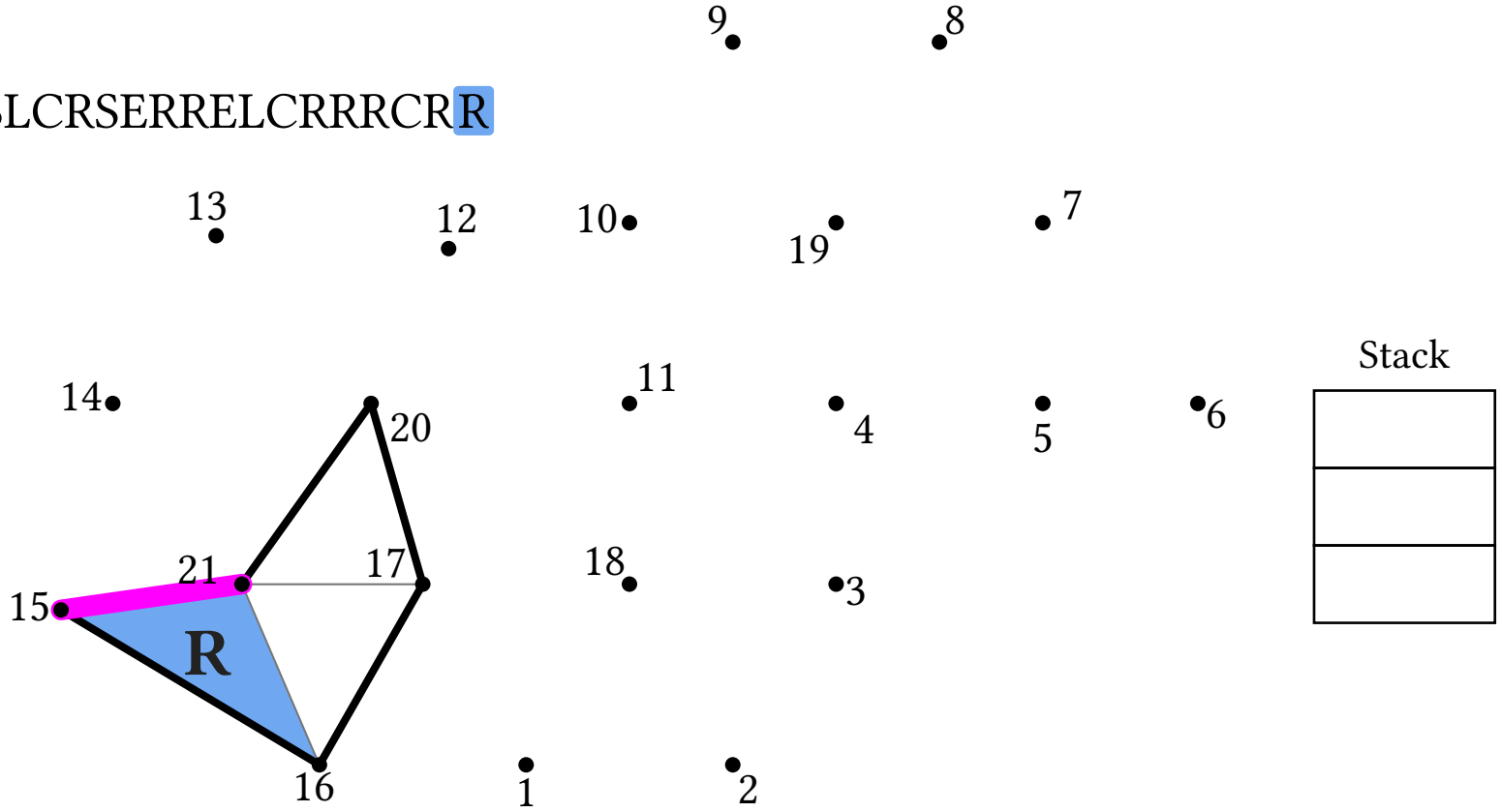


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRRR

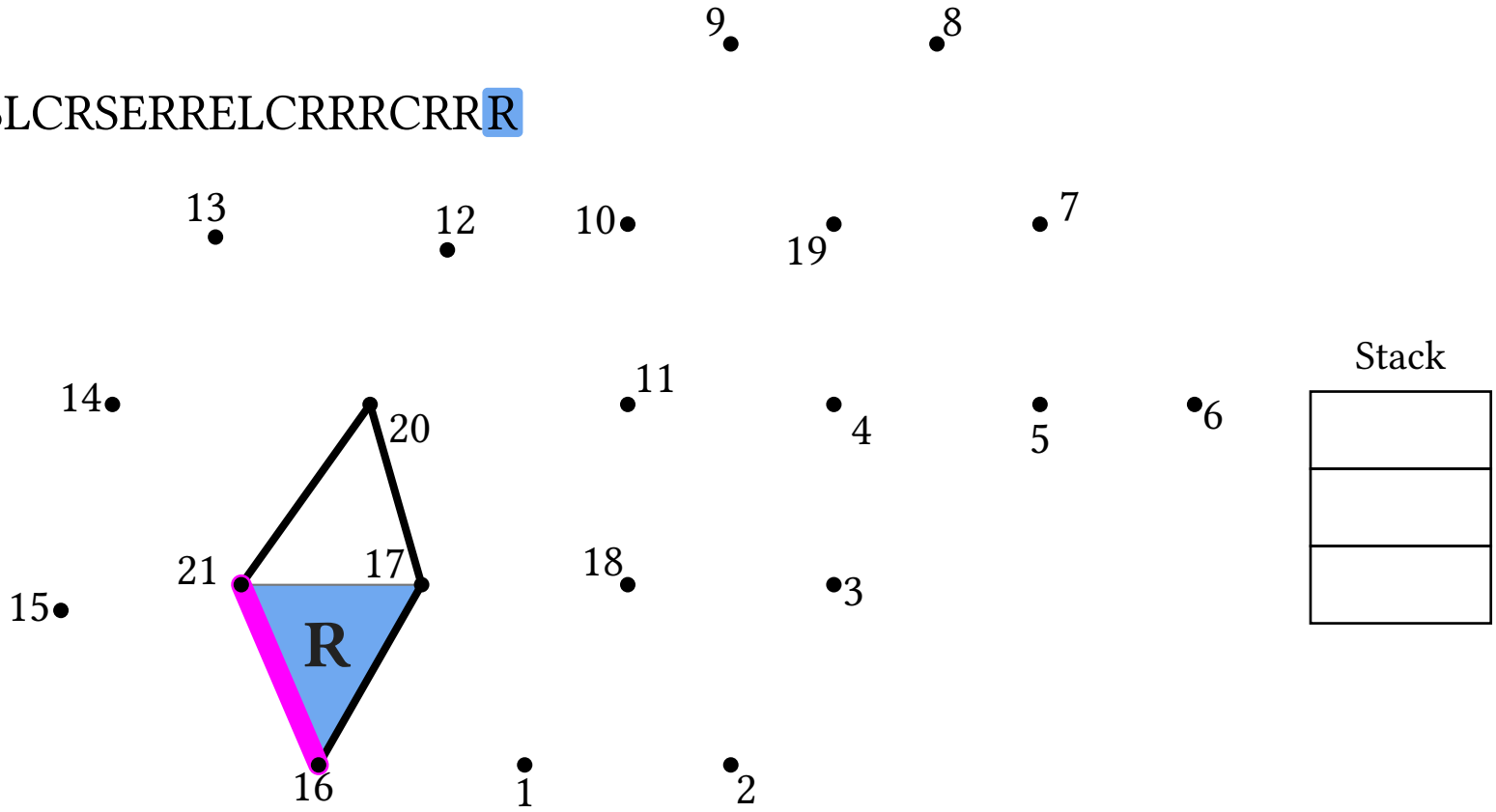


Figure 5: Jednoduchá komprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

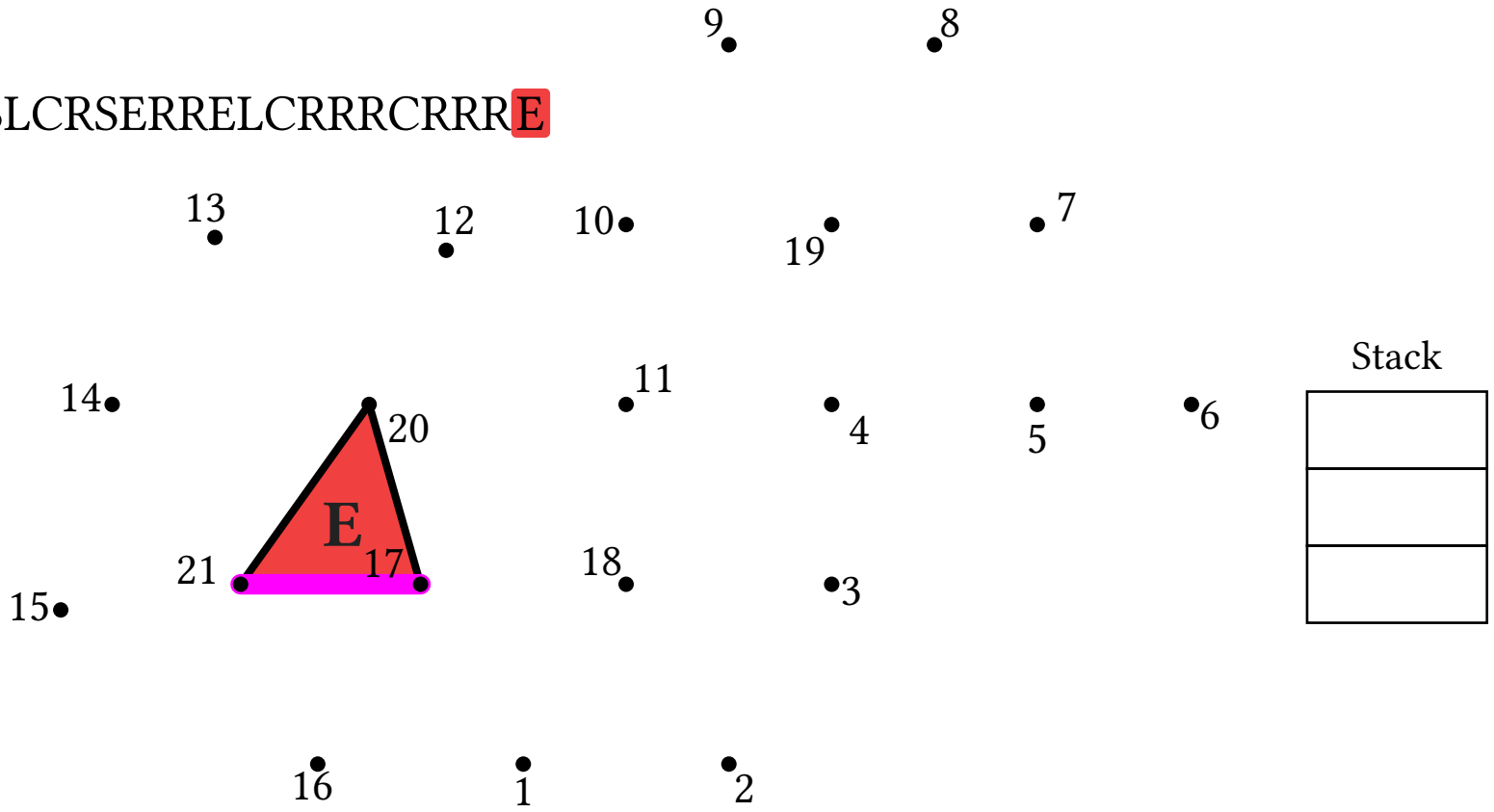


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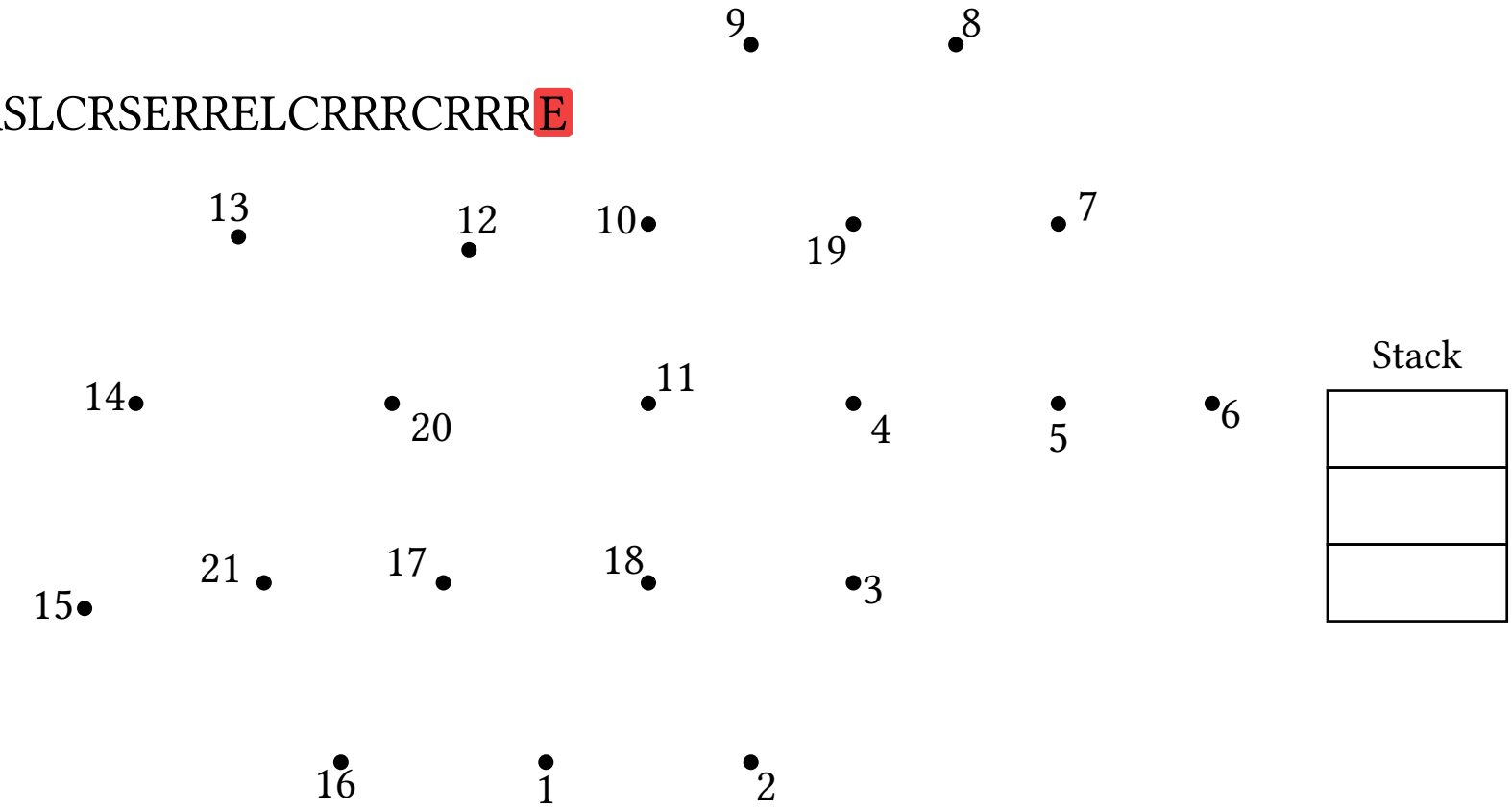


Figure 5: Jednoduchá komprese.

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

Kódování: C = 0, S = 100, R = 101, L = 110, E = 111

3.3 Výsledný soubor

Geometrie vrcholů:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21

CLERS řetězec:

CCRRRSLCRSERRELCRRRCRRRE

Kódování: C = 0, S = 100, R = 101, L = 110, E = 111

(CLERS) 0110111101100 bitové pole:

0010110110110011001011001111011011111100101101101010110110111

4. Dekomprese

4.1 Možnosti

Existuje víc algoritmů, implementuji spirale-reversi a wrap&zip.

Demonstrováno na wrap&zip, je intuitivnější.

Historie = CCRRRSLCRSERRELCRRRCRRRE

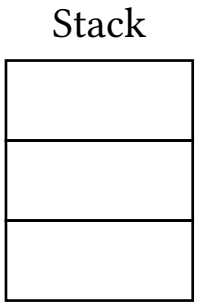


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

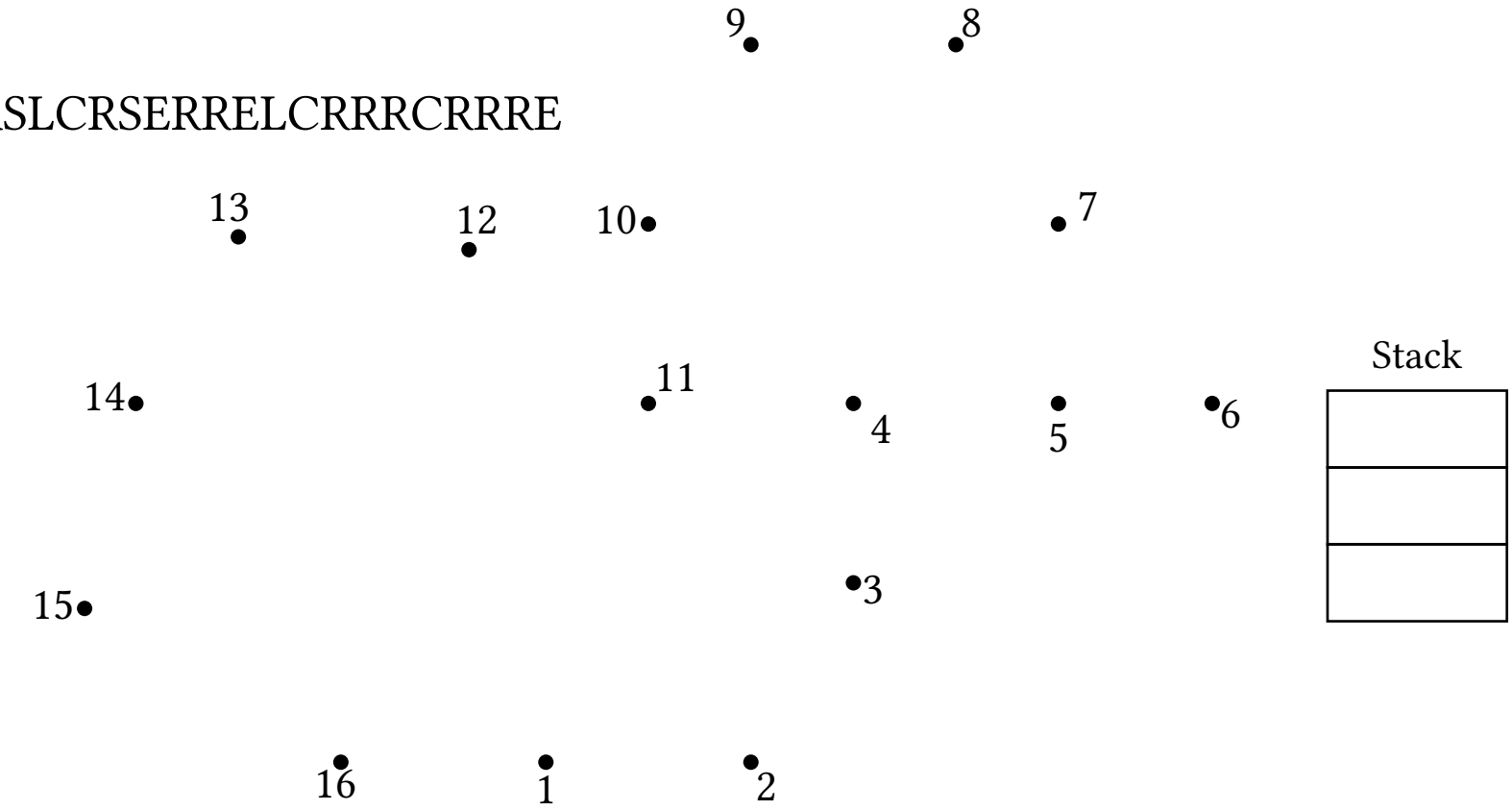


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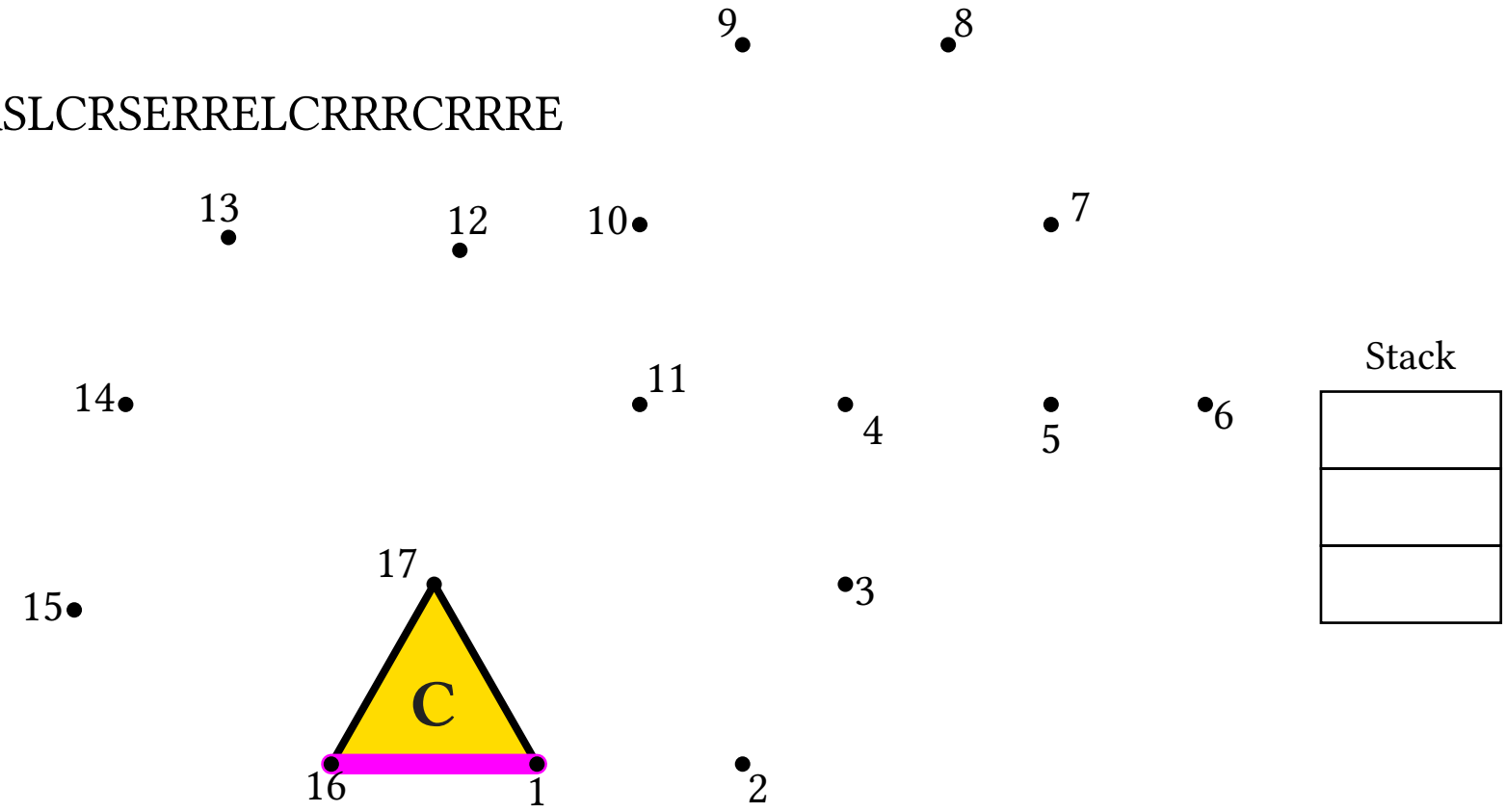


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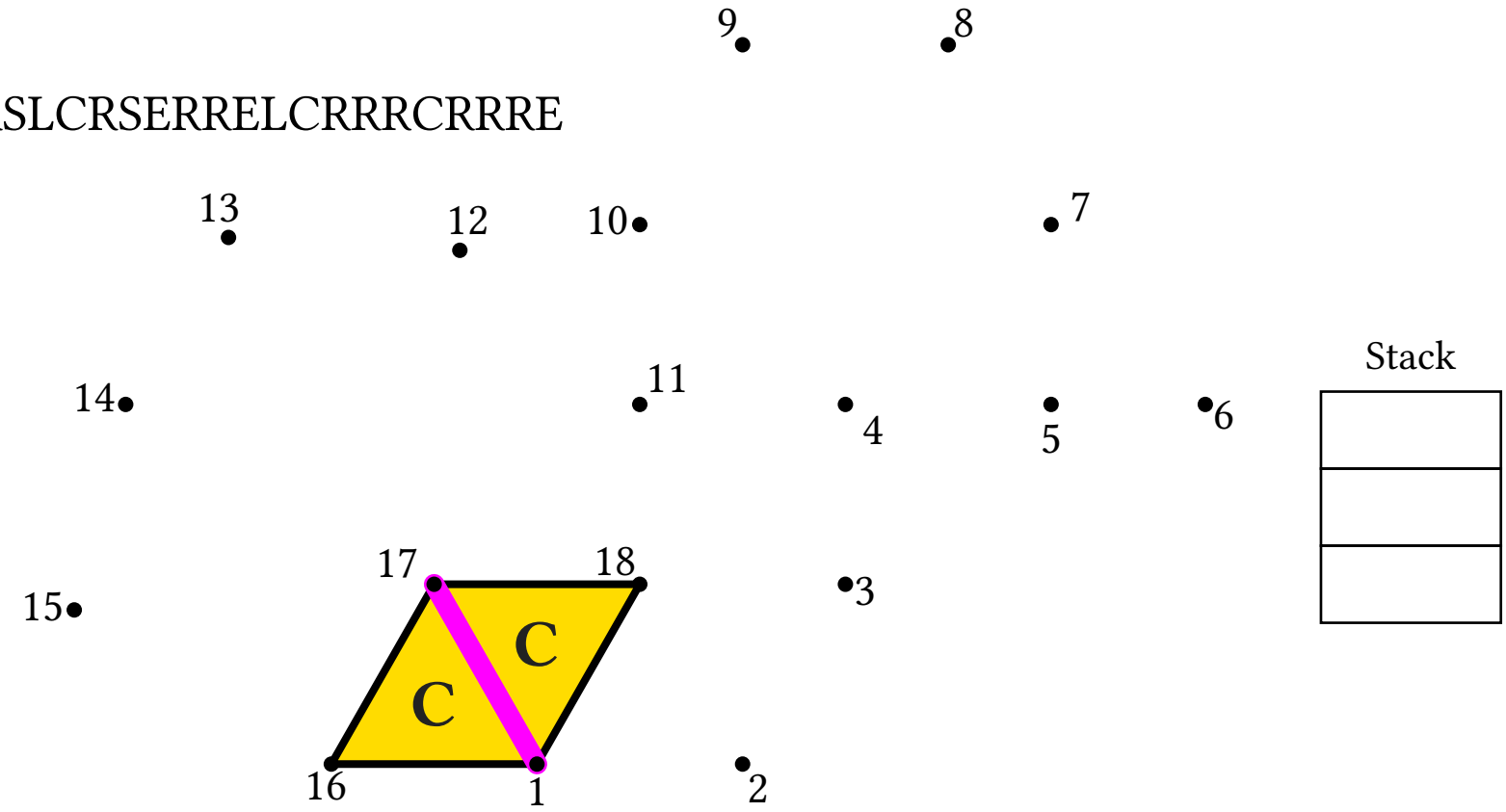


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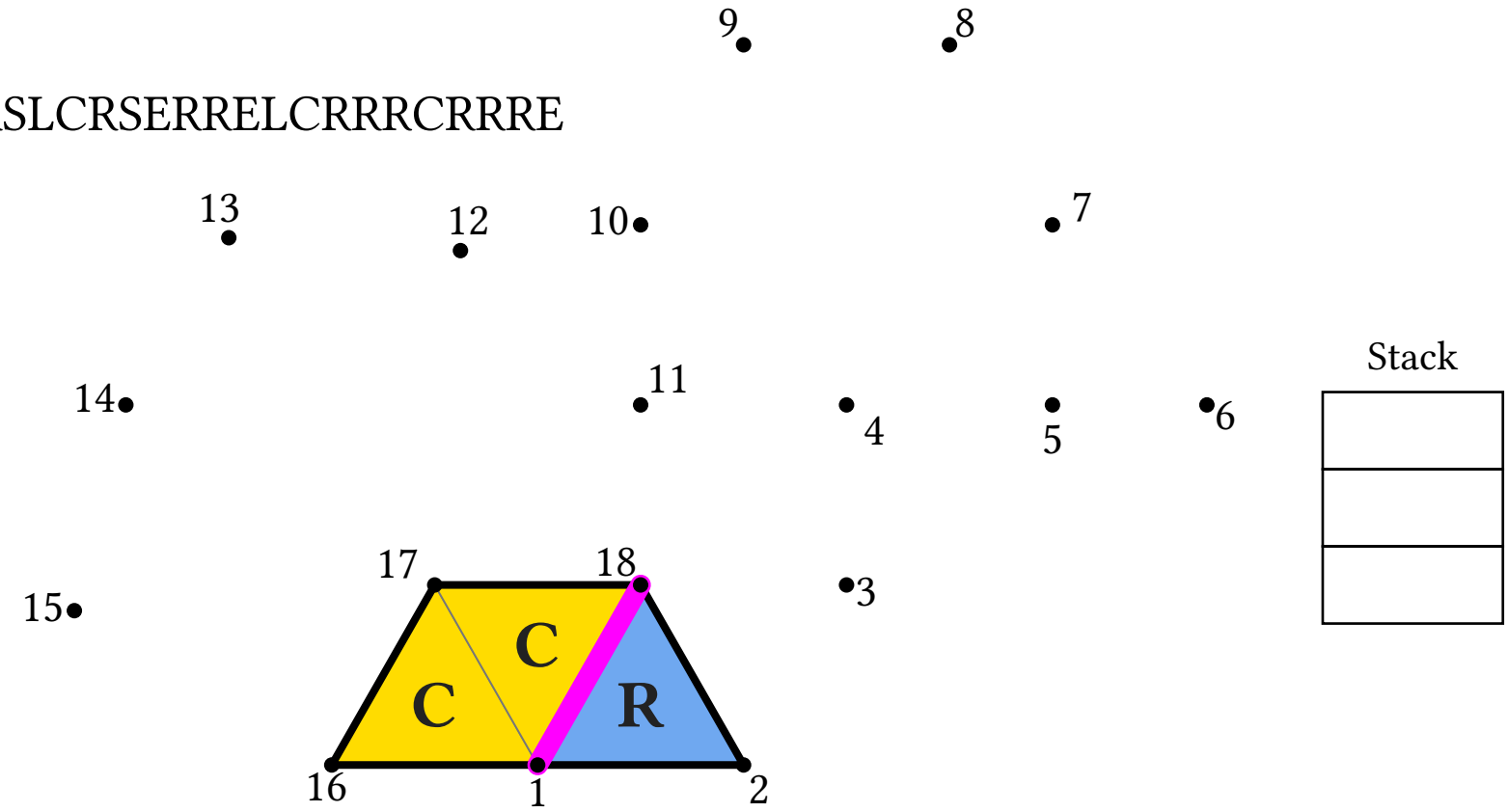


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Historie = CCRRRSLCRSERRELCRRRCRRRE

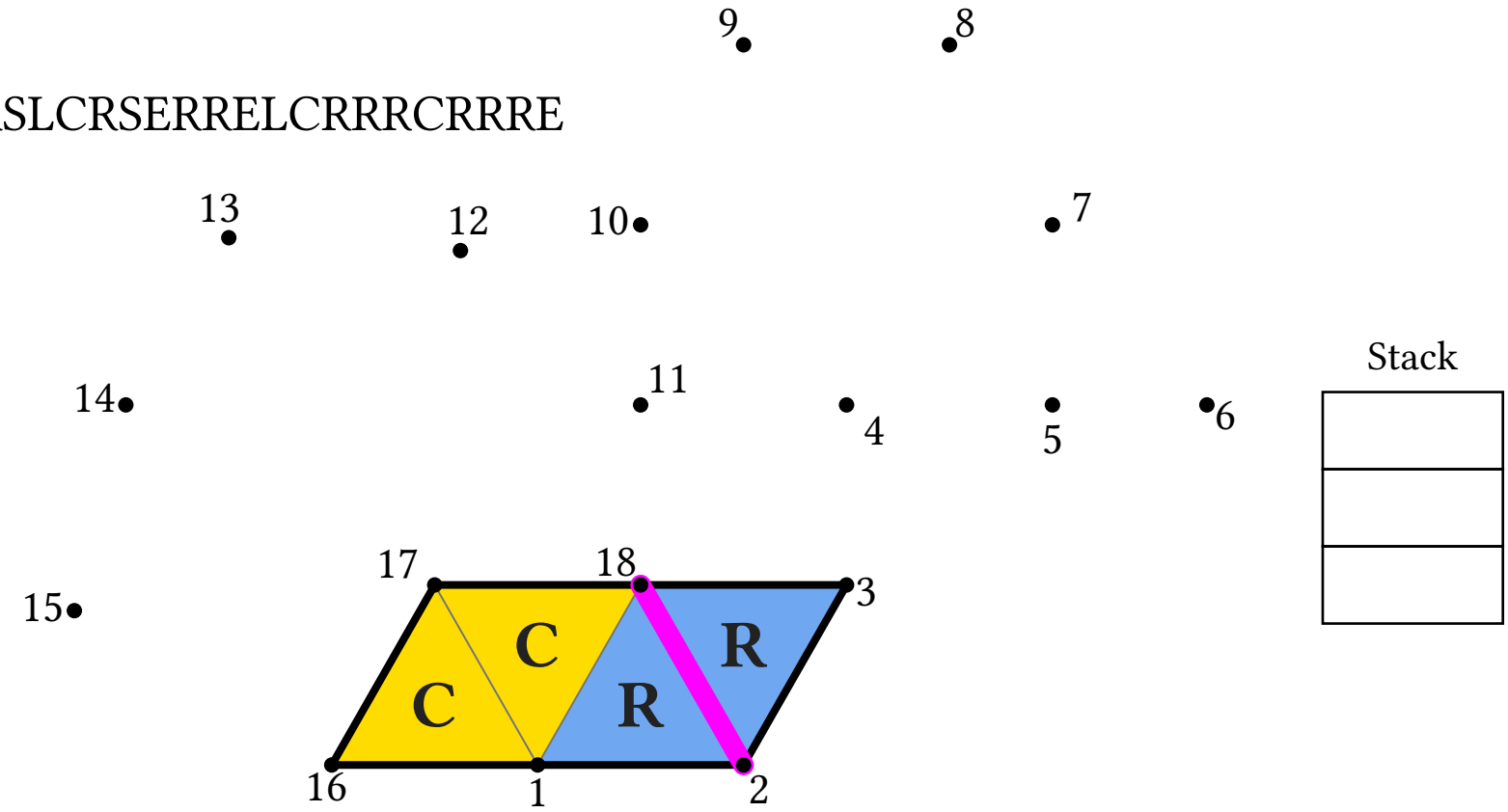


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

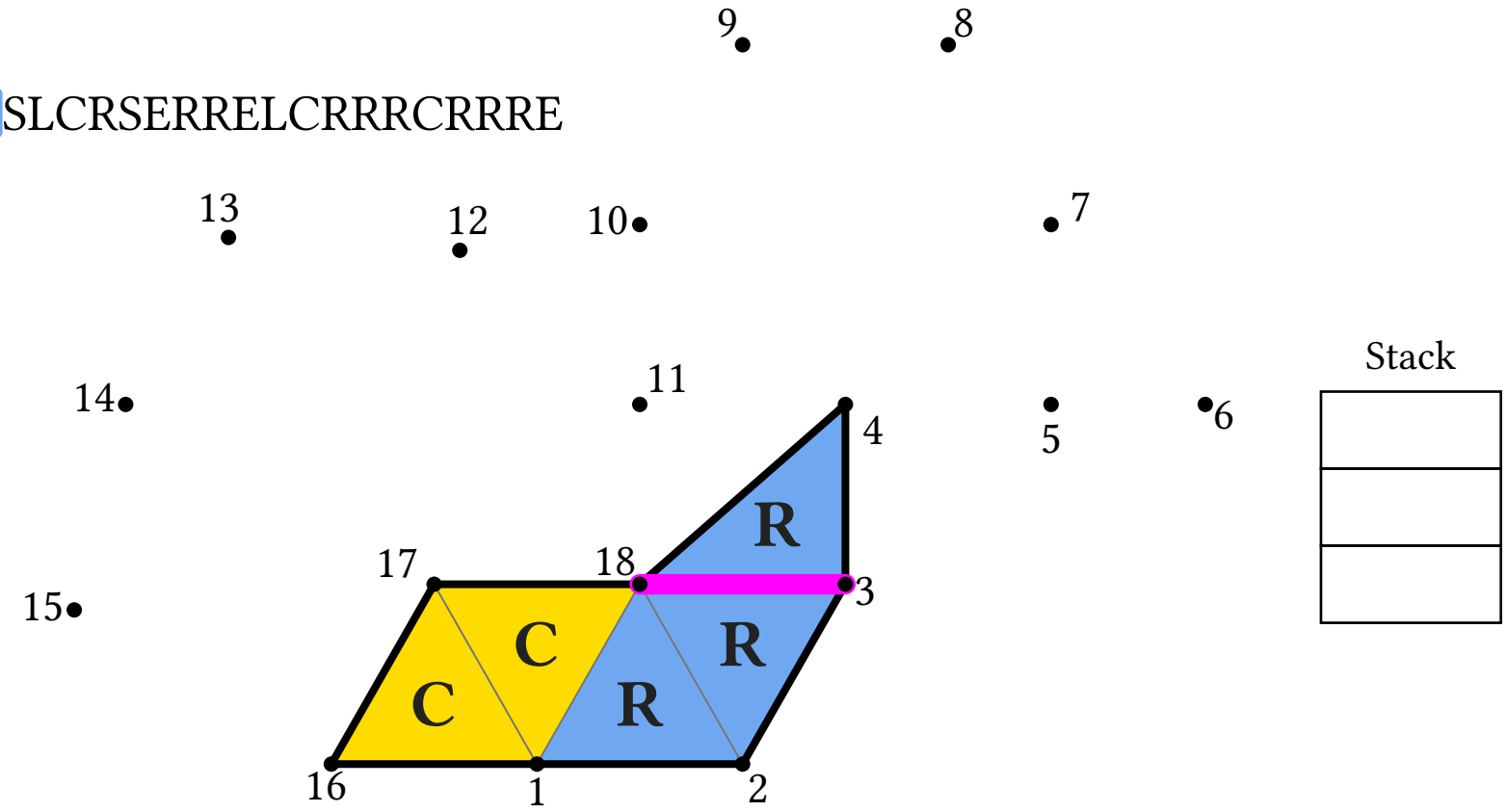


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

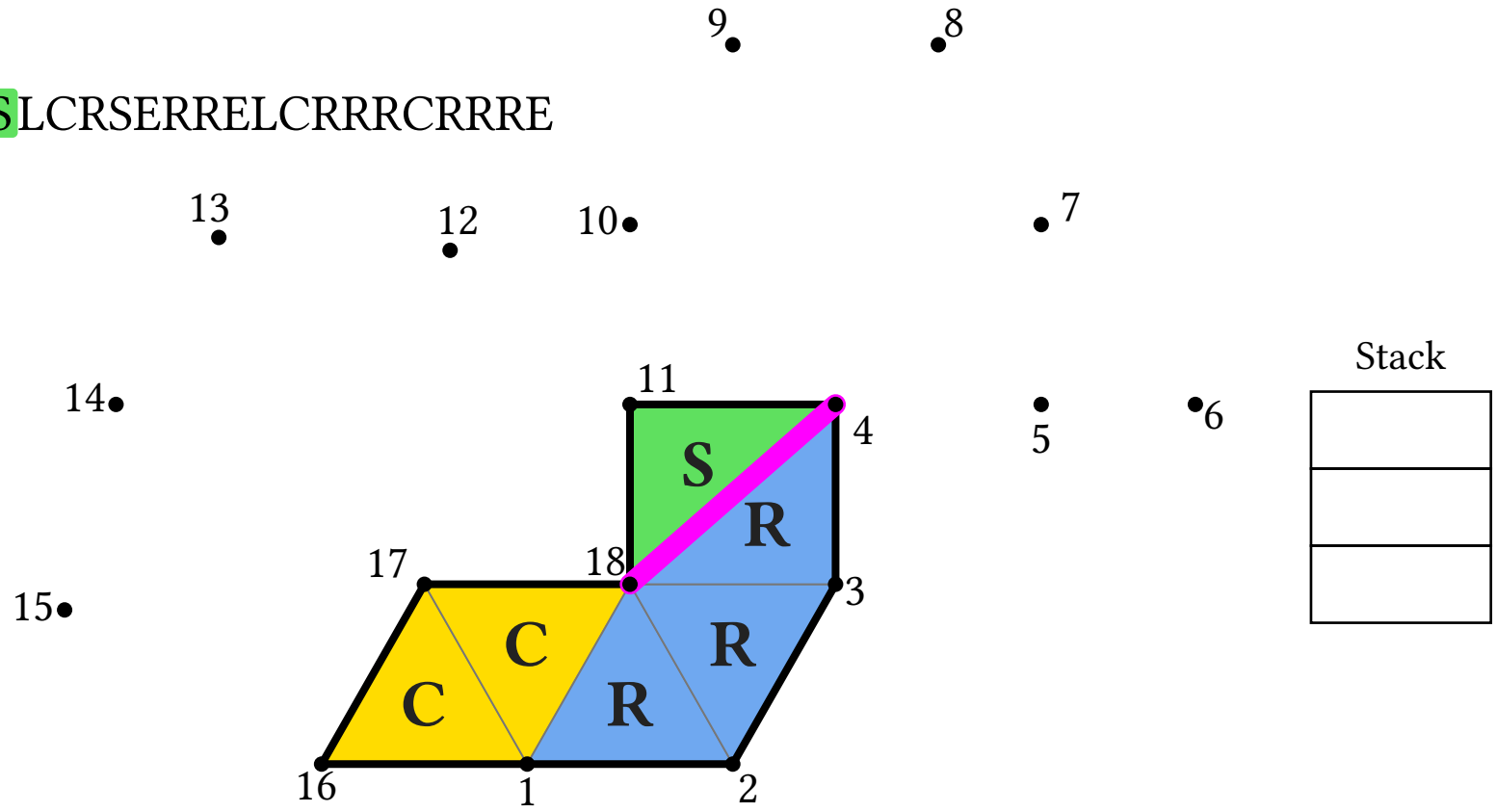


Figure 8: Dekomprese.

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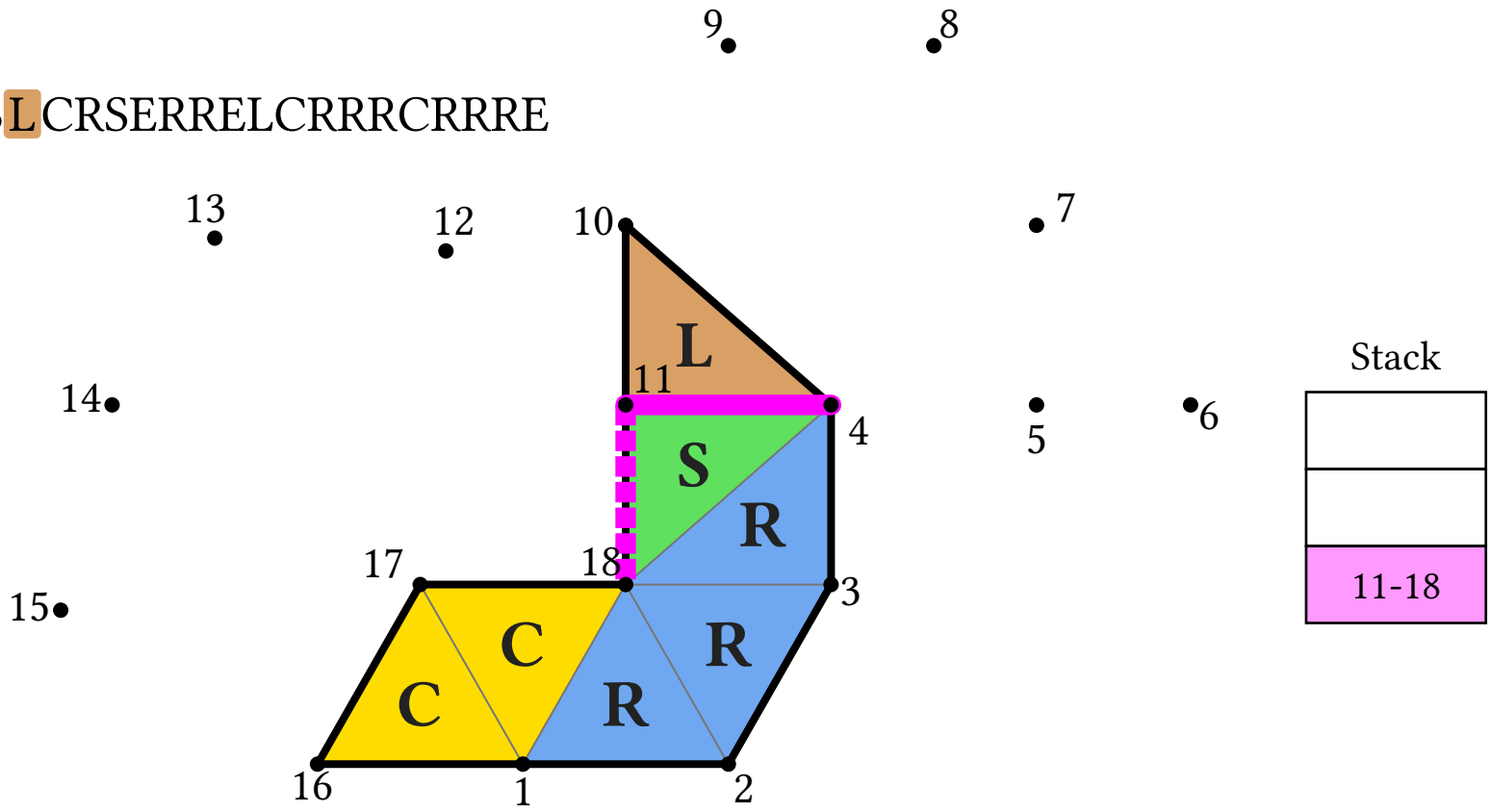


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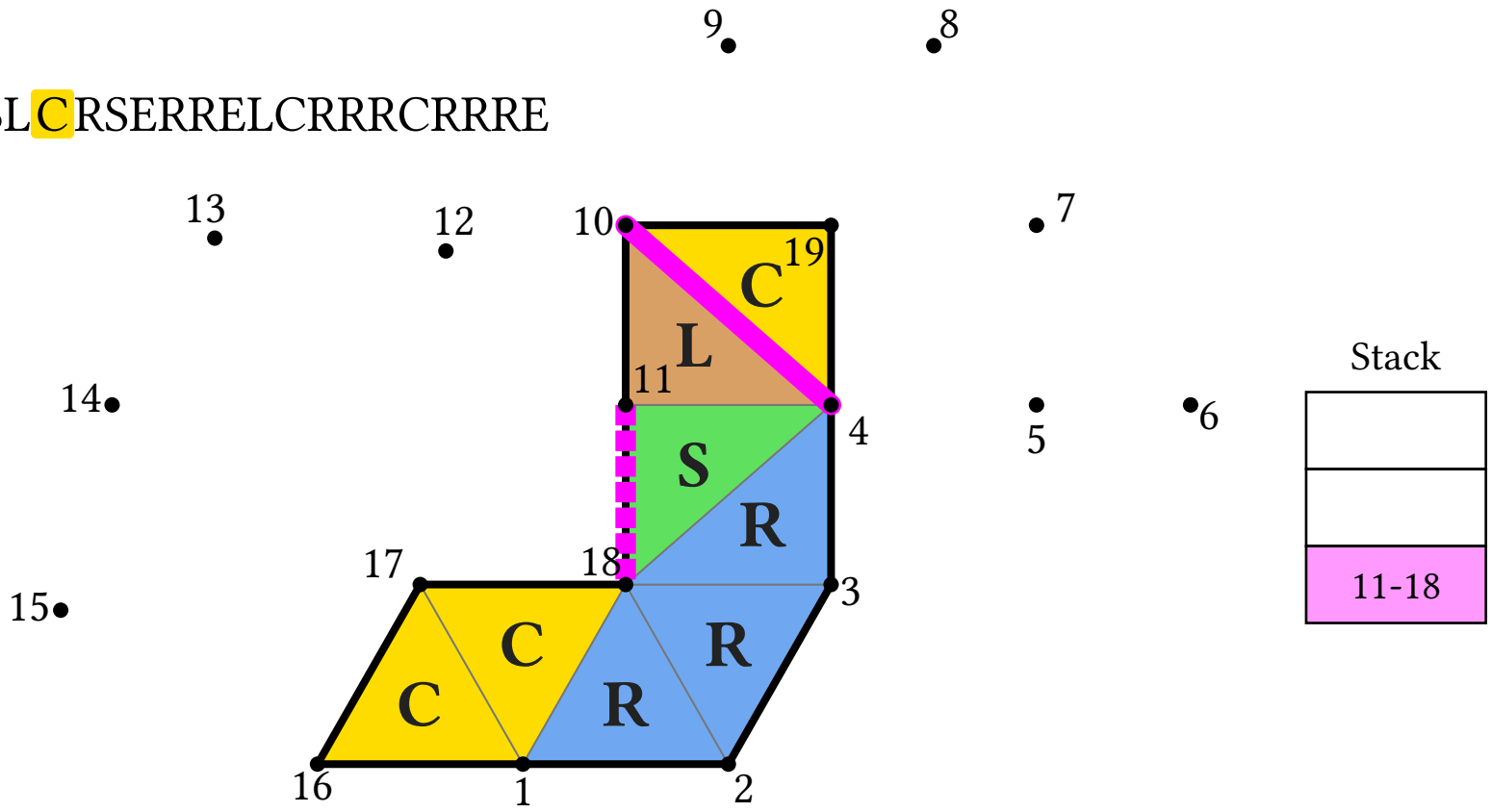


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

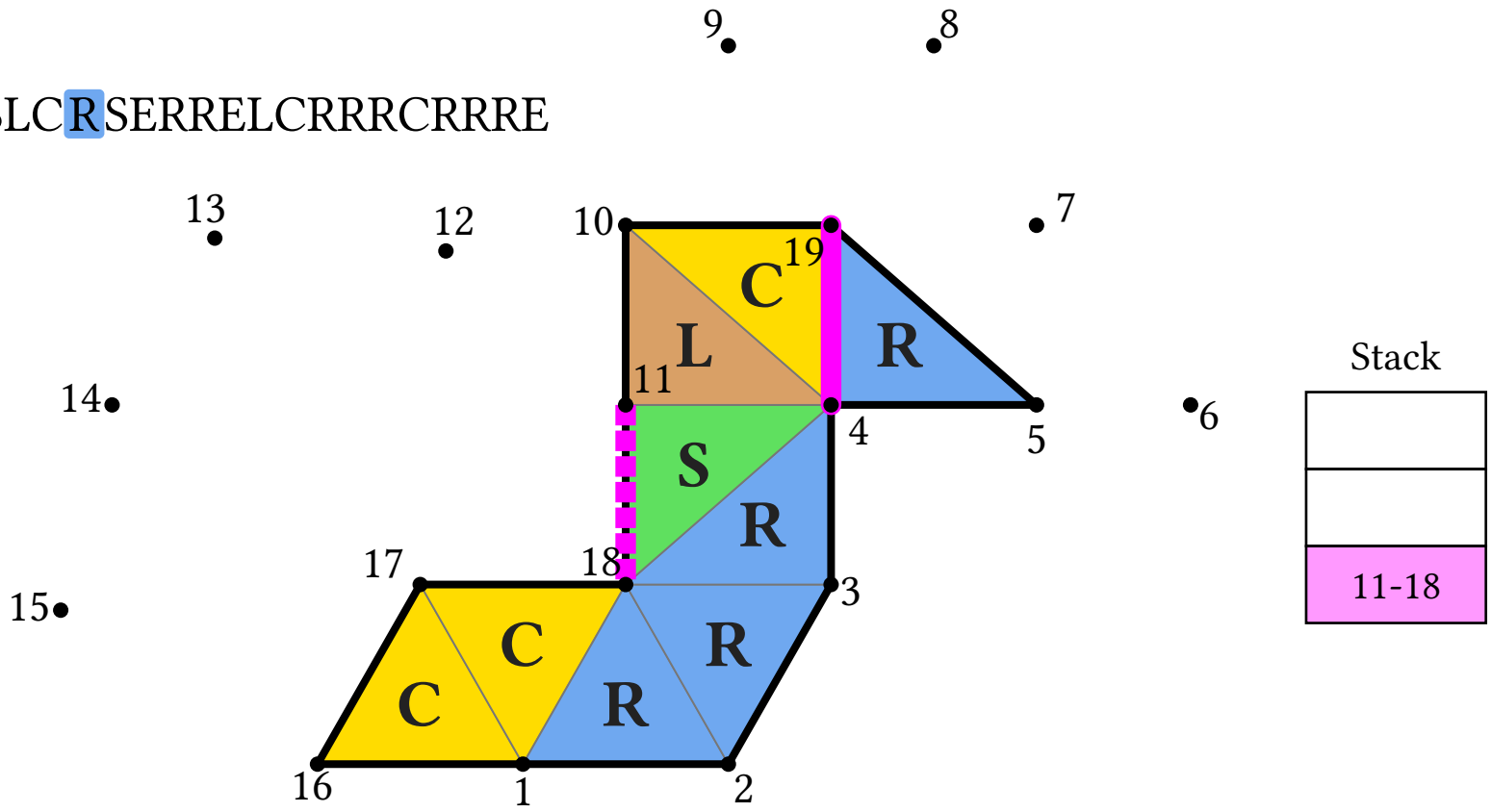


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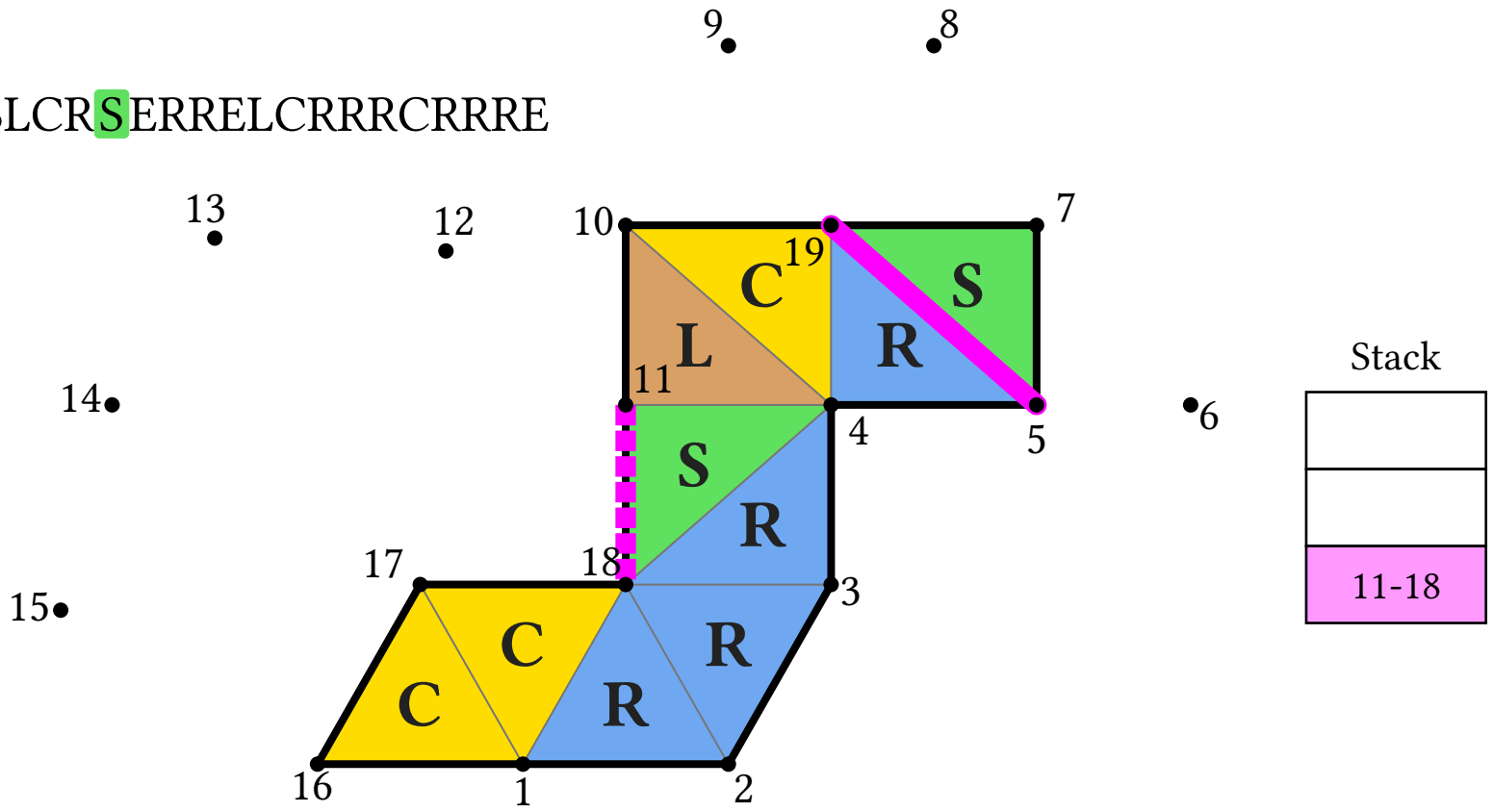


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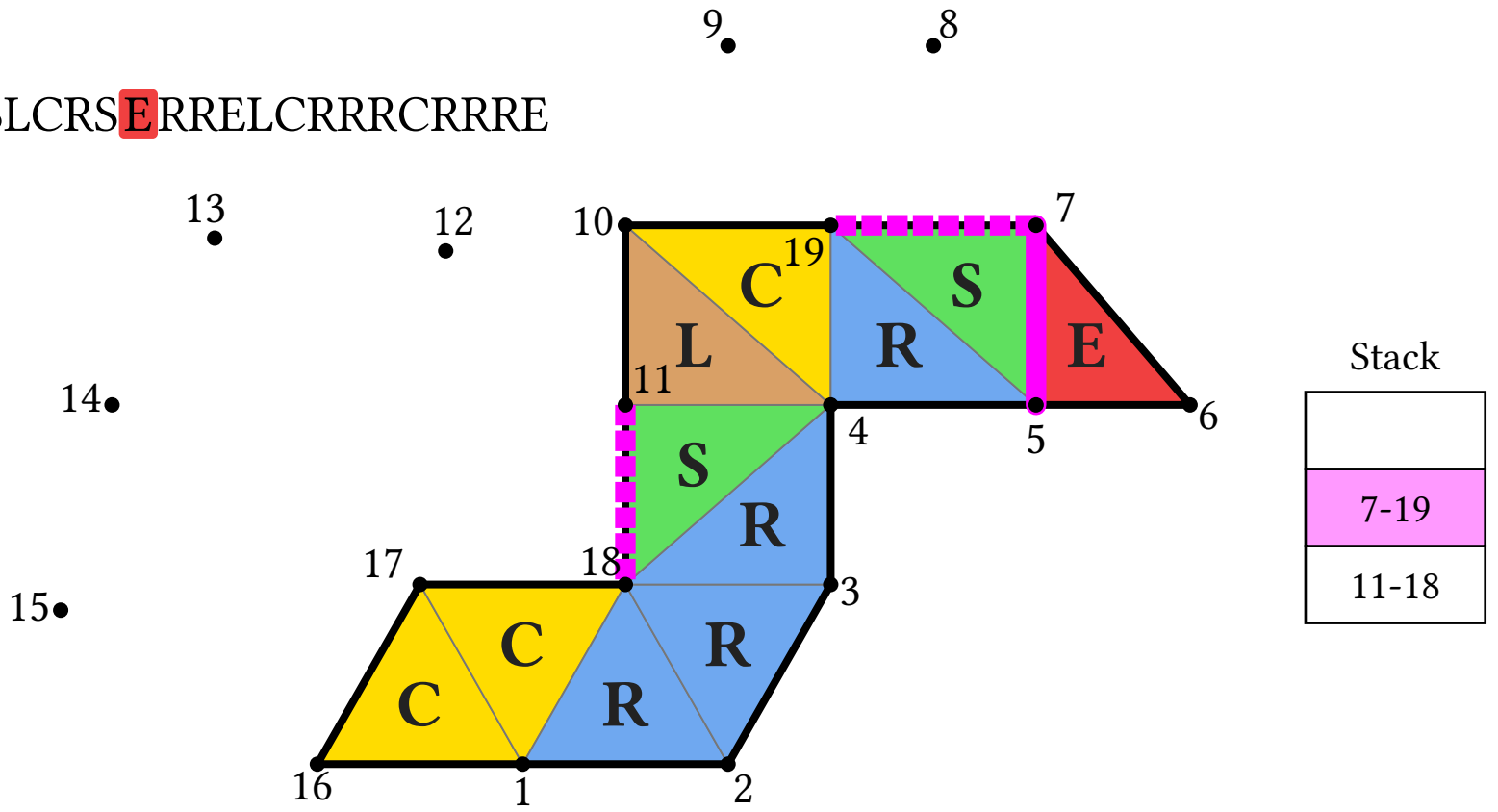


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

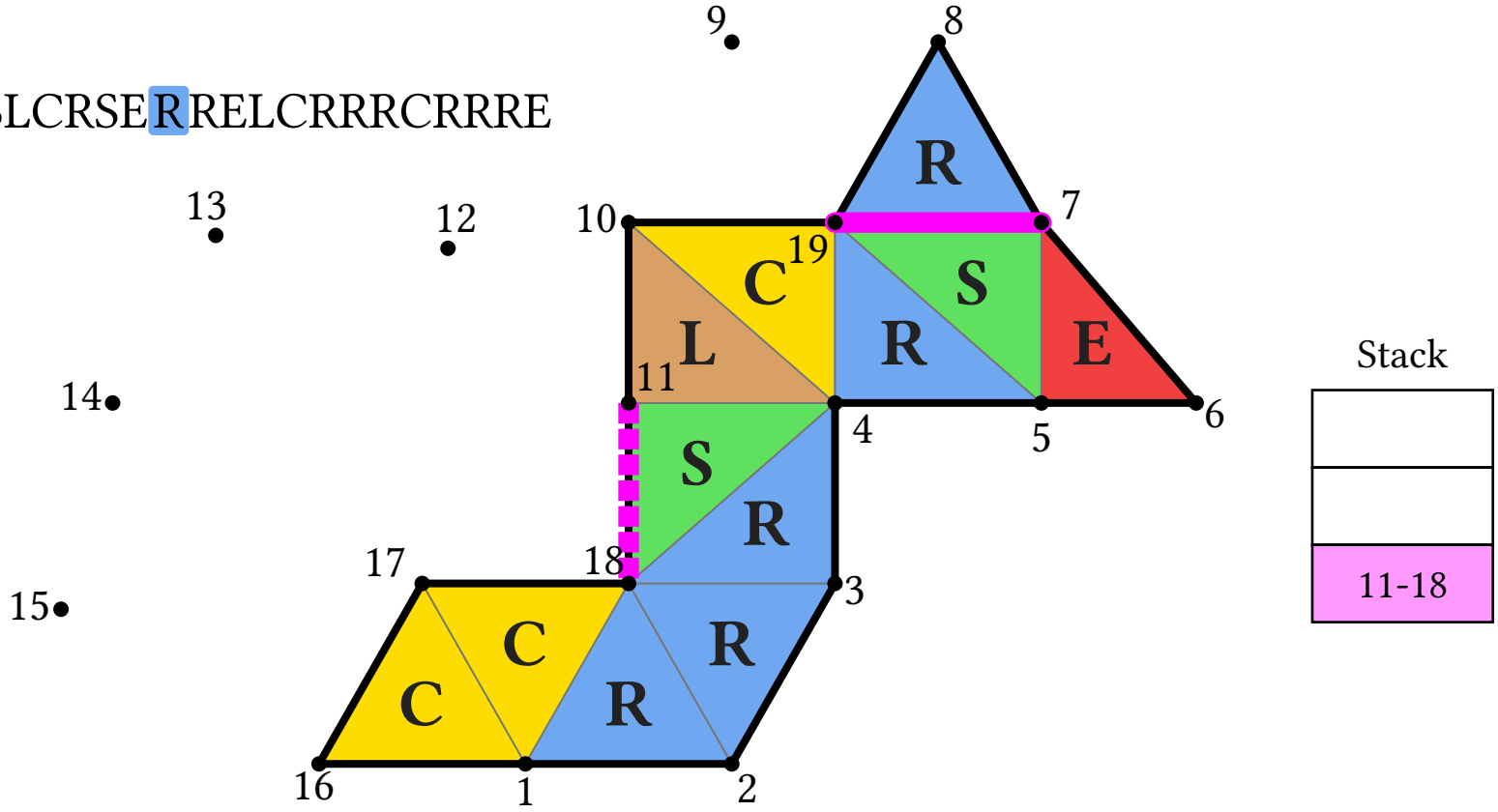


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

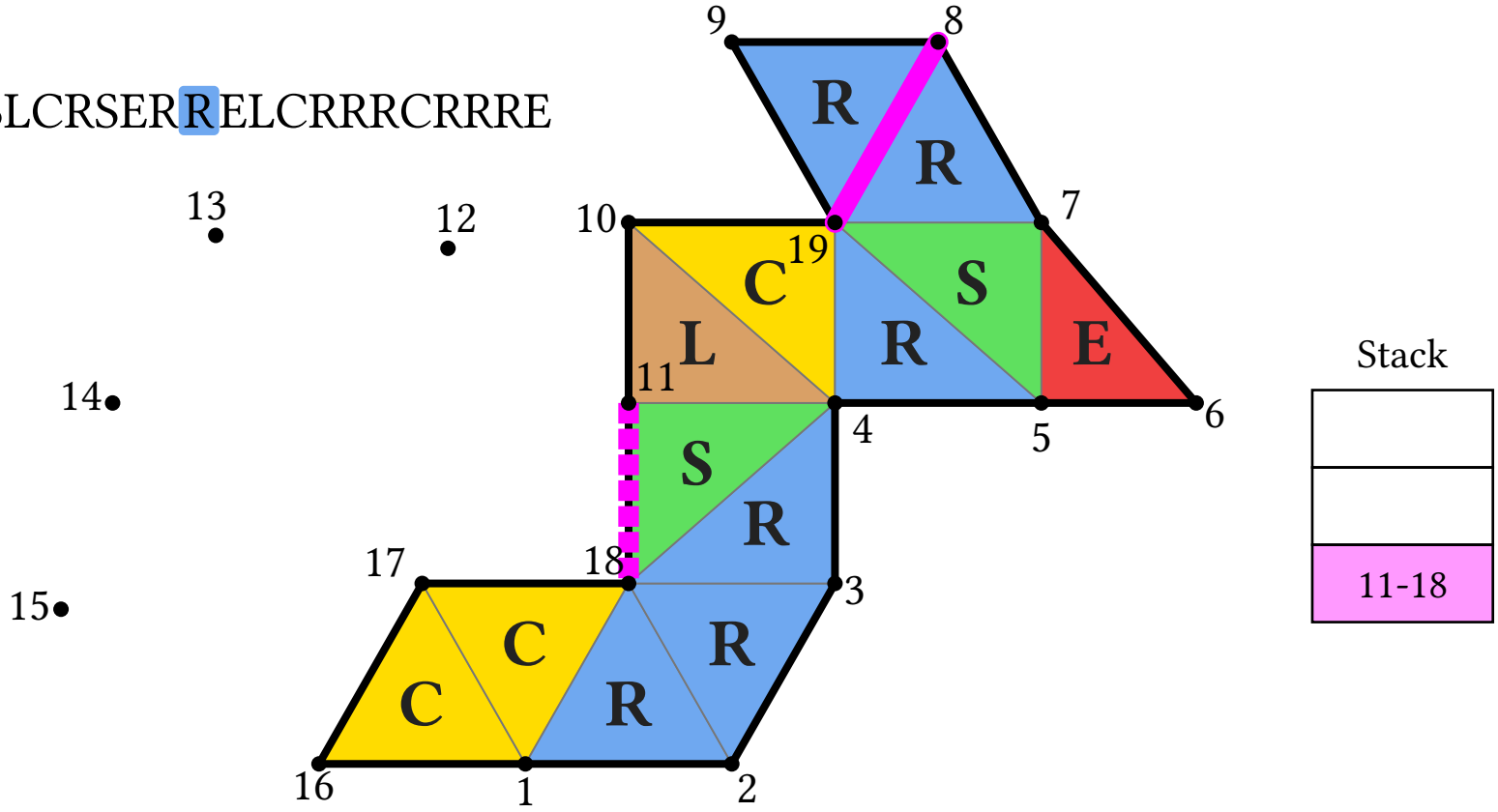


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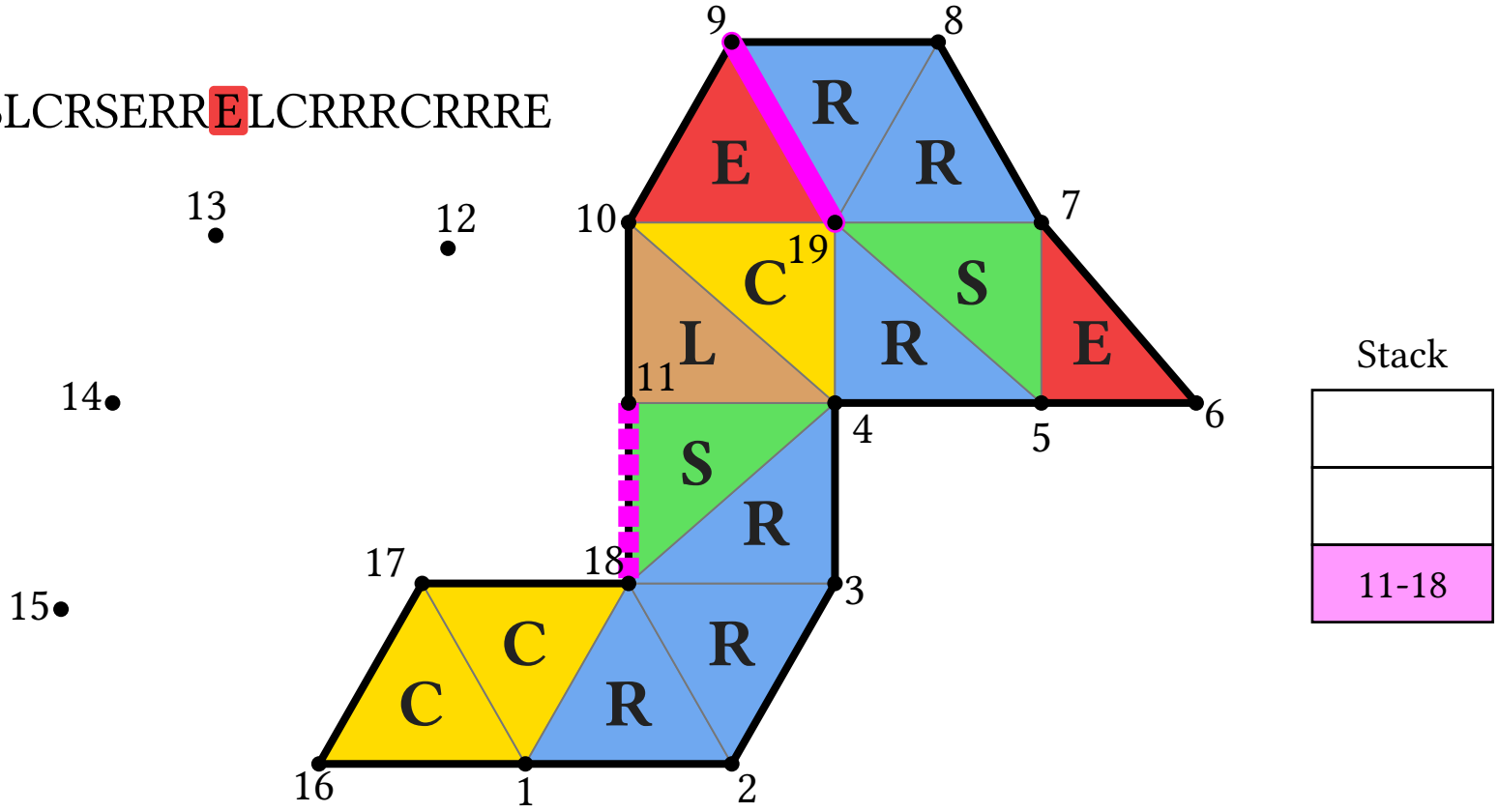


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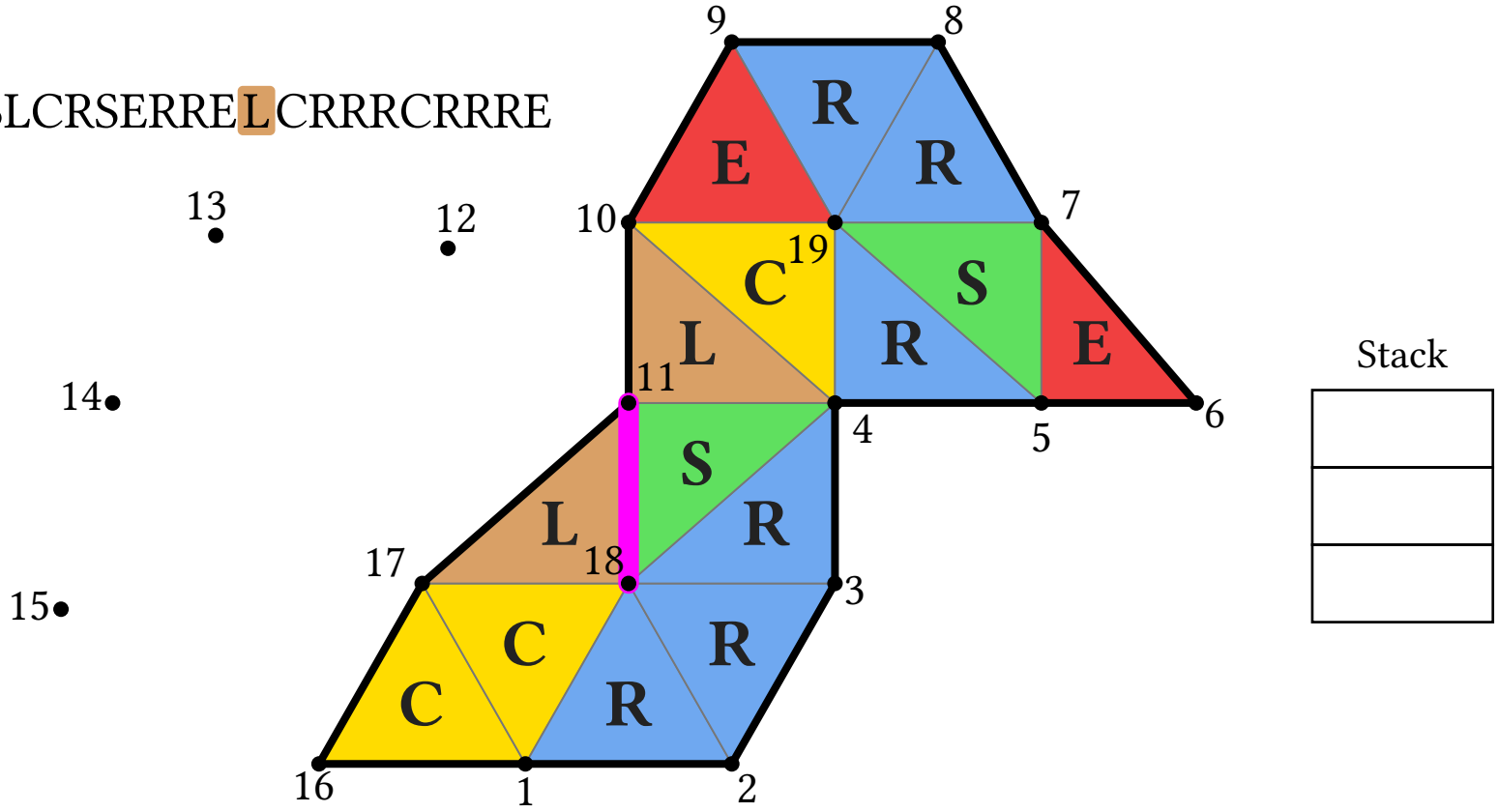


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERREL**C**RRRCRRRE

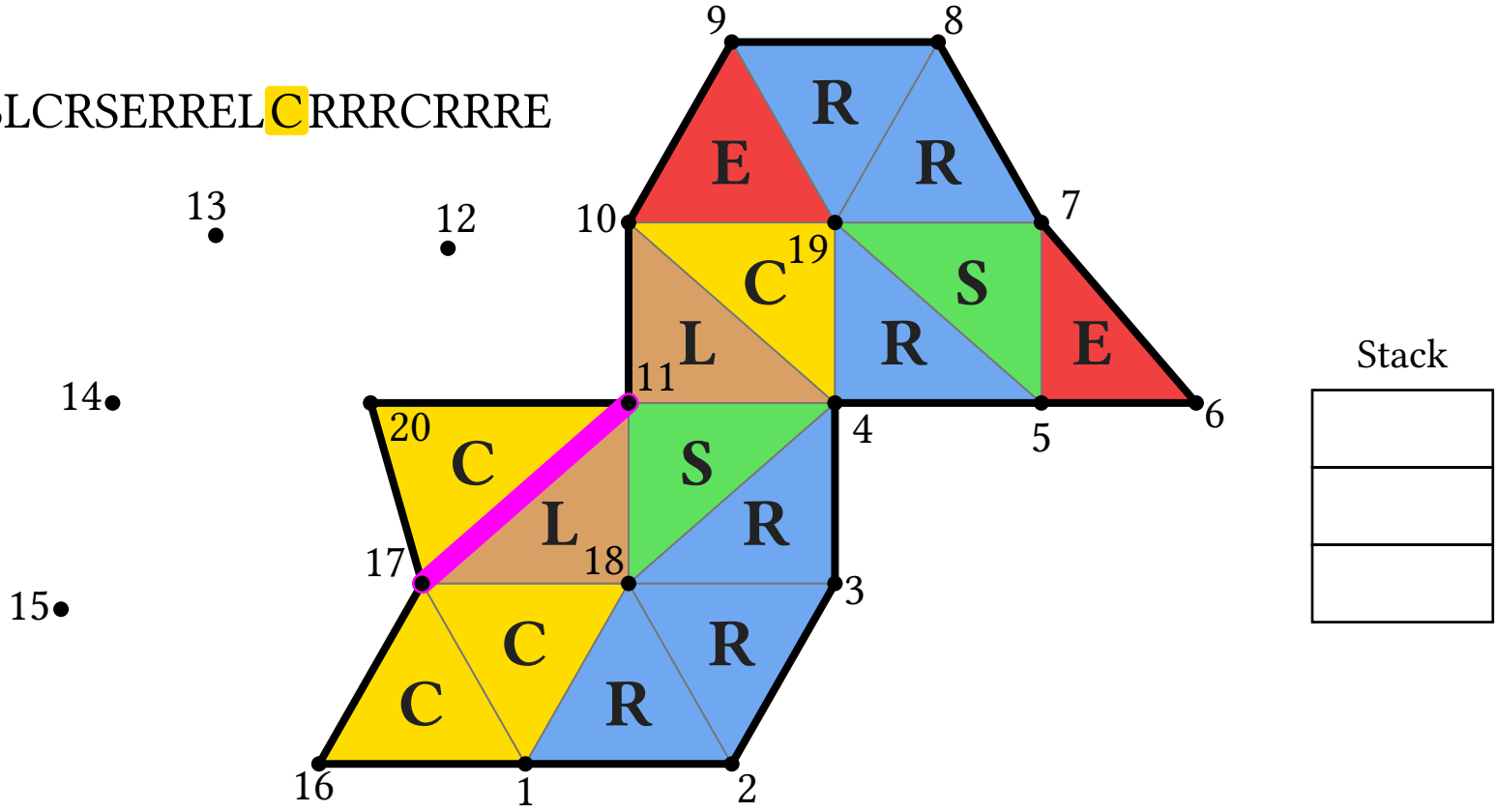


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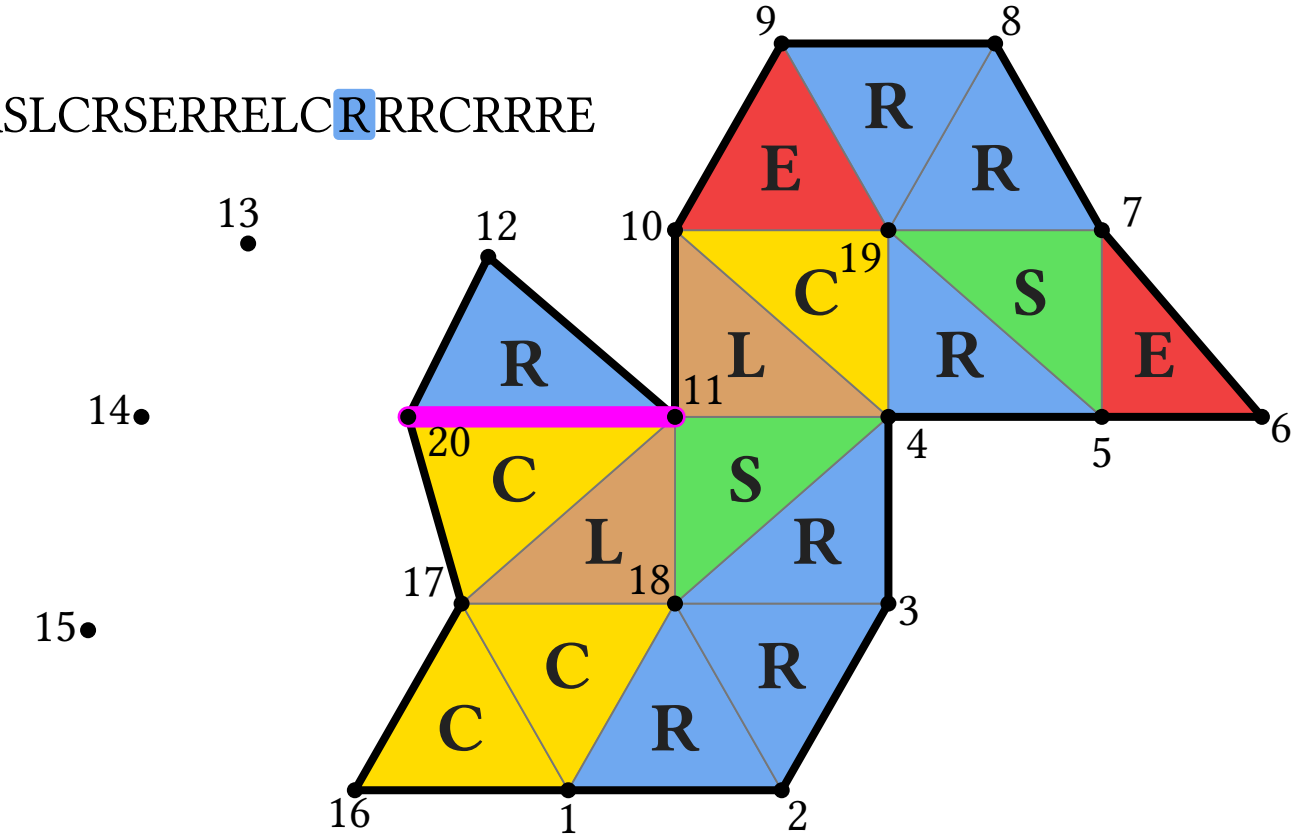


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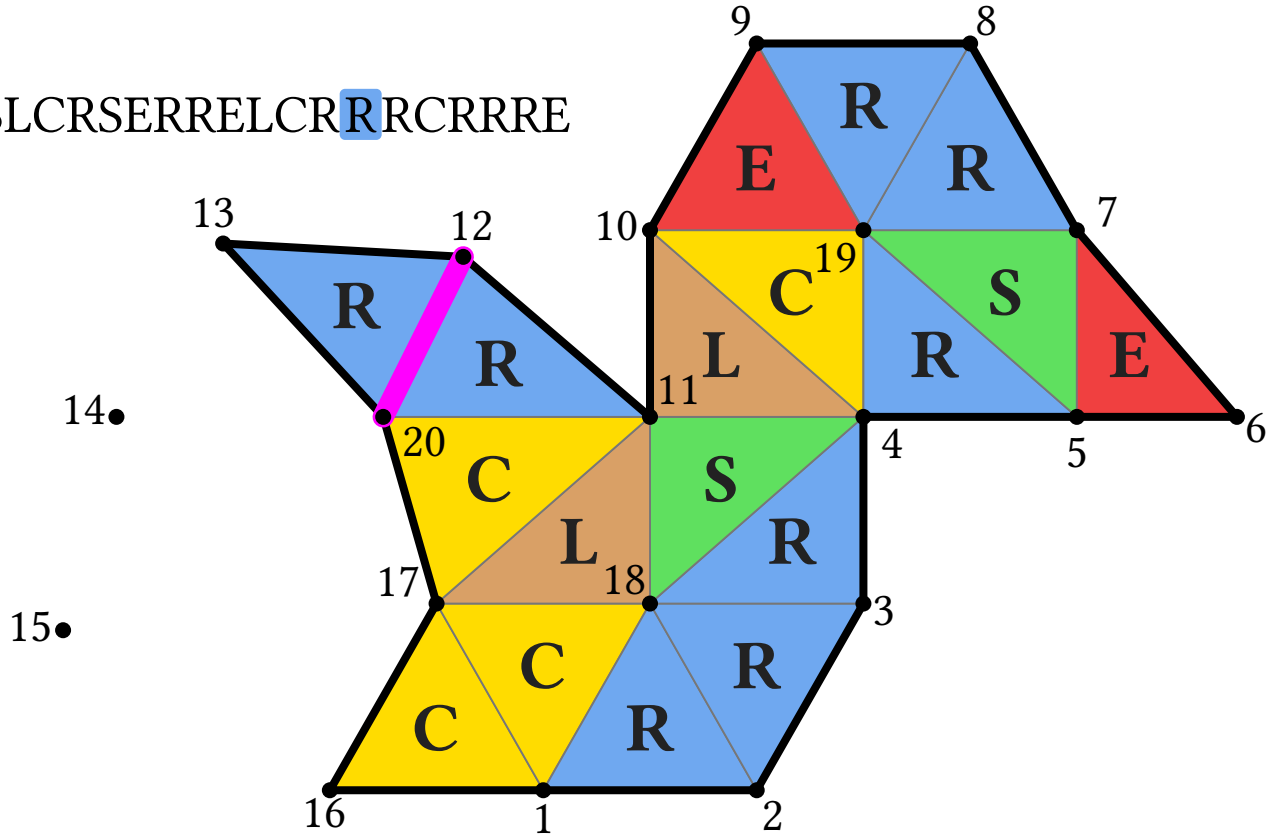
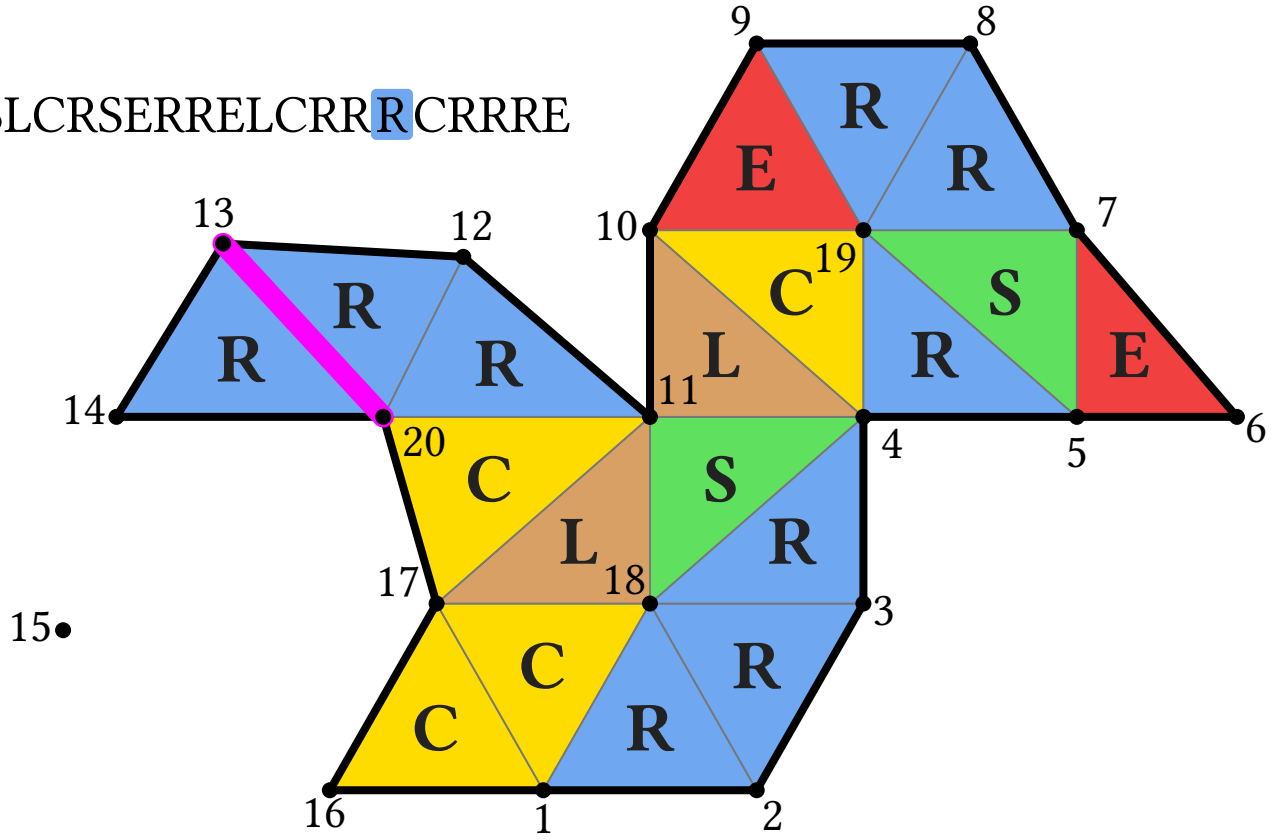


Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE



Stack

Figure 8: Dekomprese.

Historie = CCRRRSLCRSERRELCRRRCRRRE

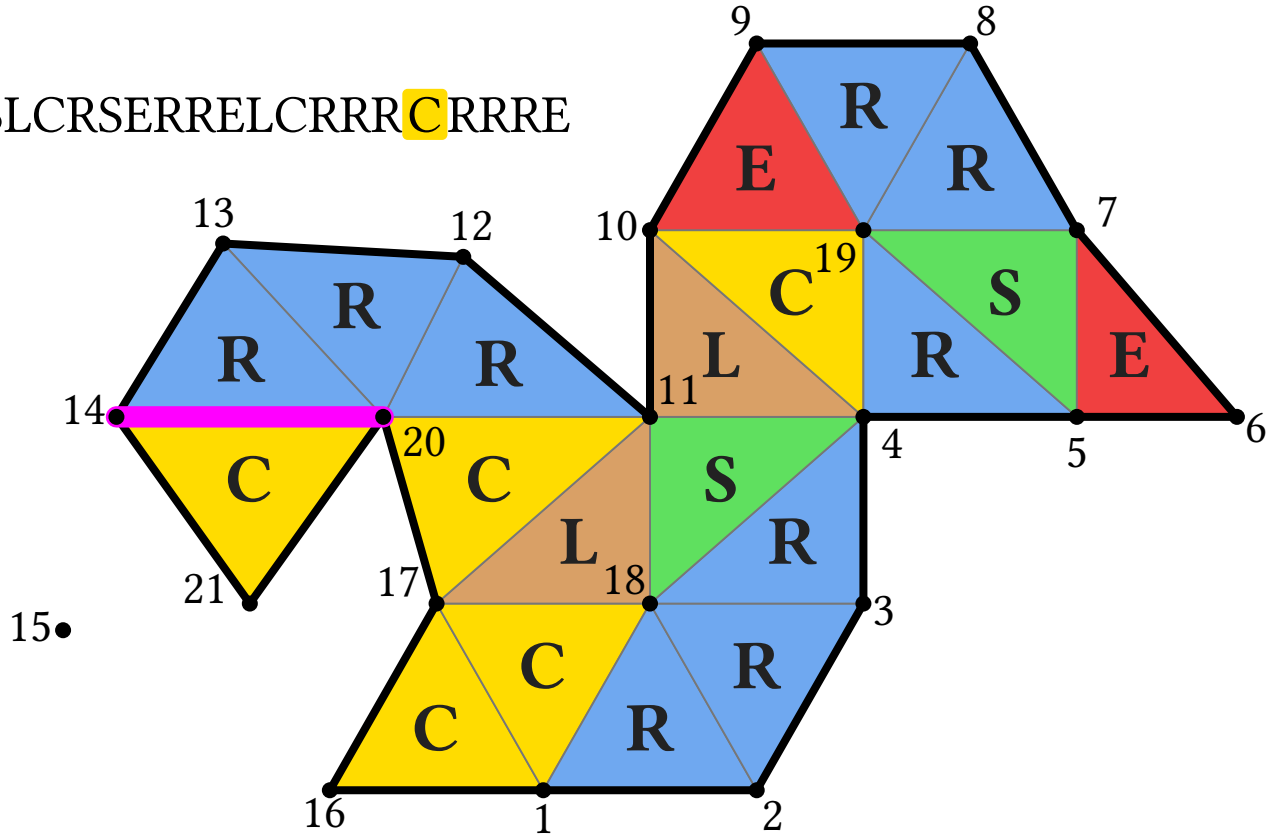
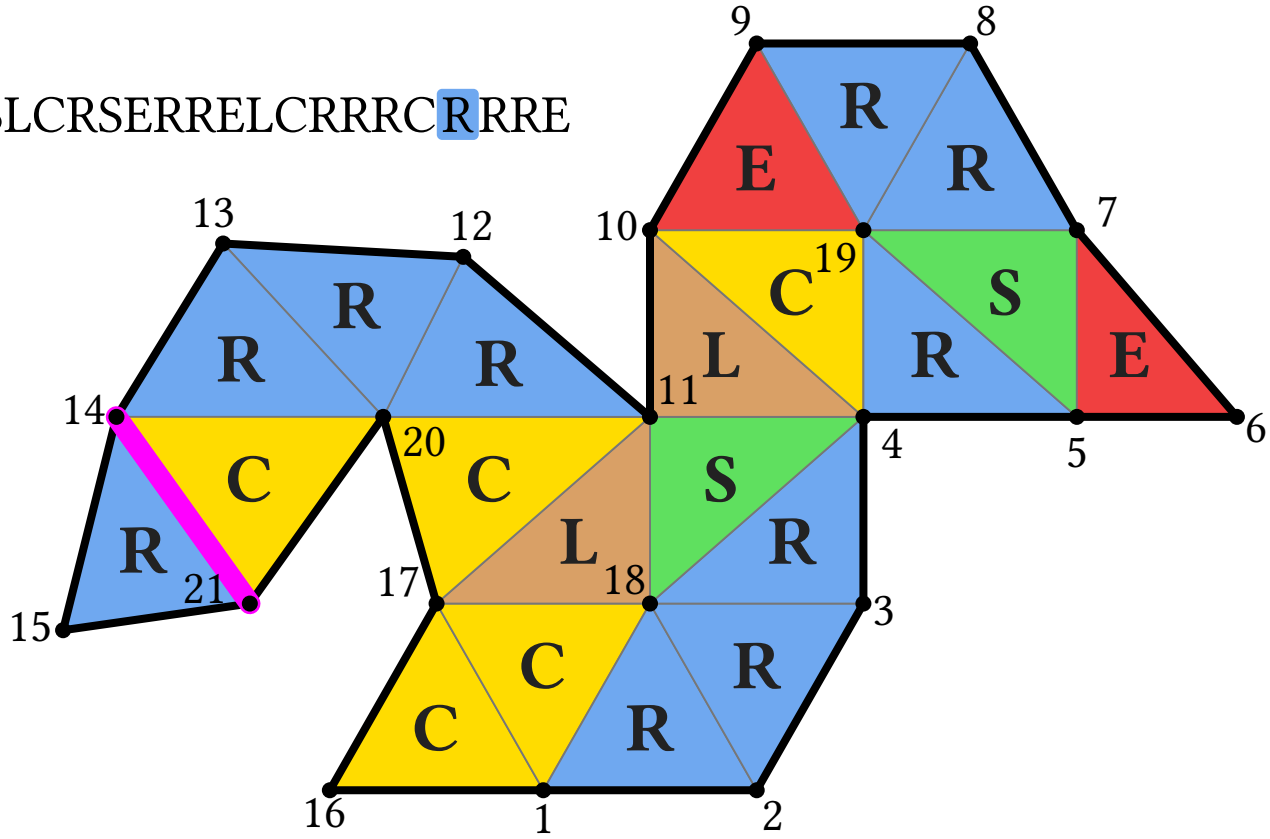


Figure 8: Dekomprese.

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Stack

Figure 8: Dekomprese.

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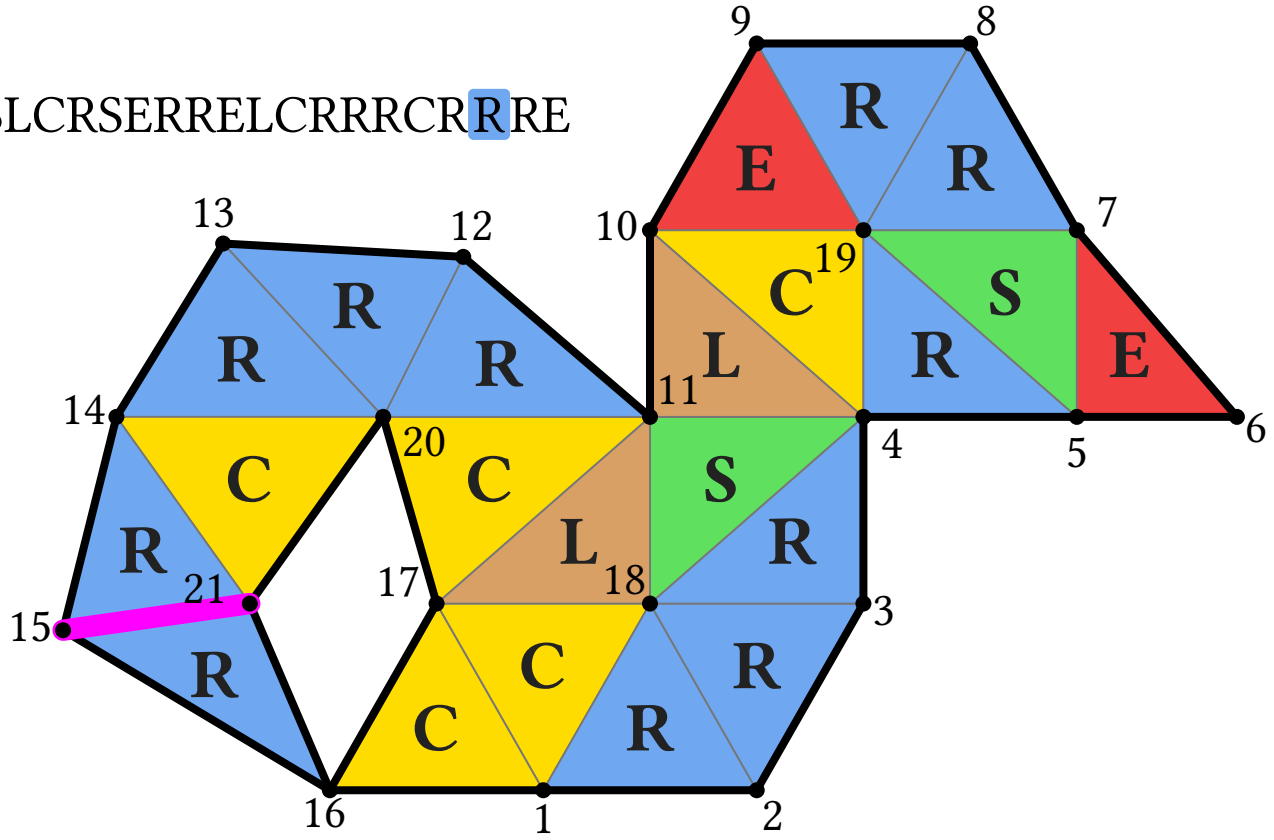


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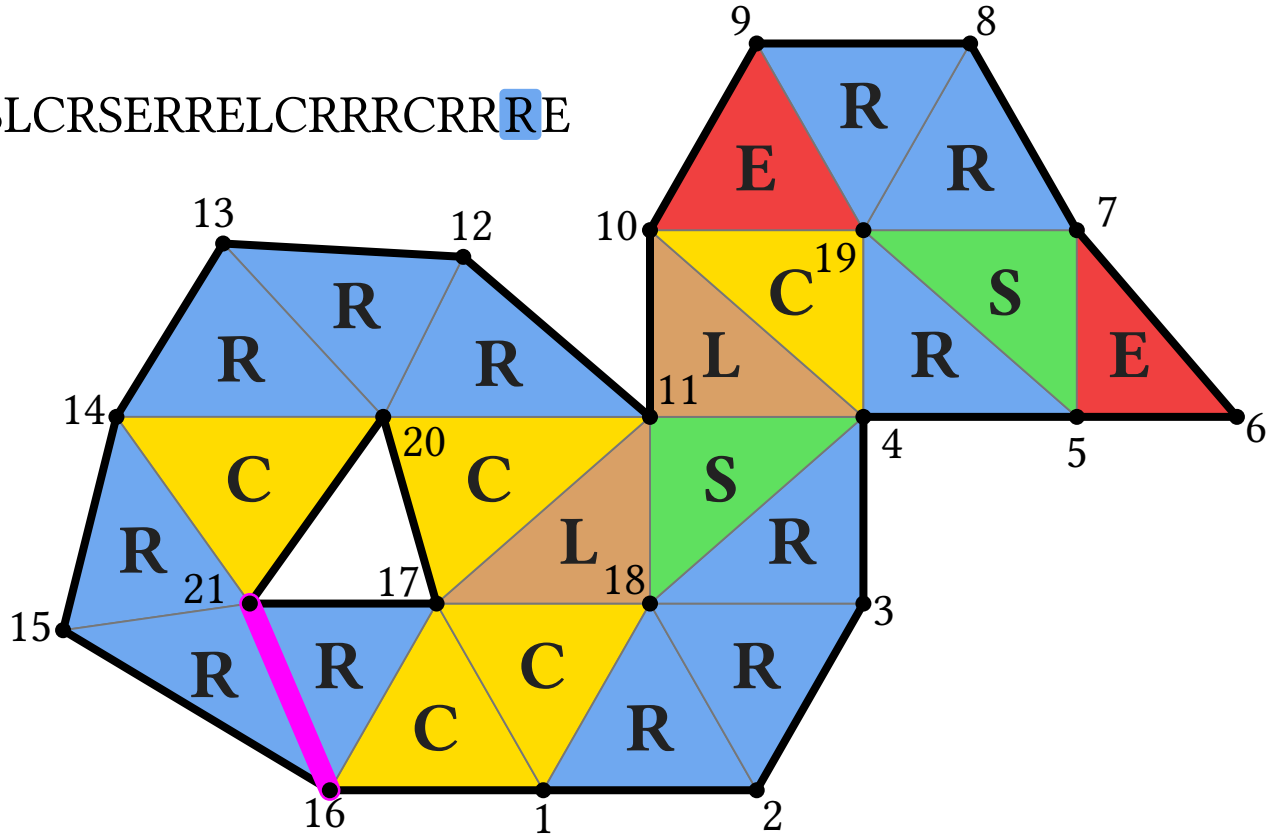


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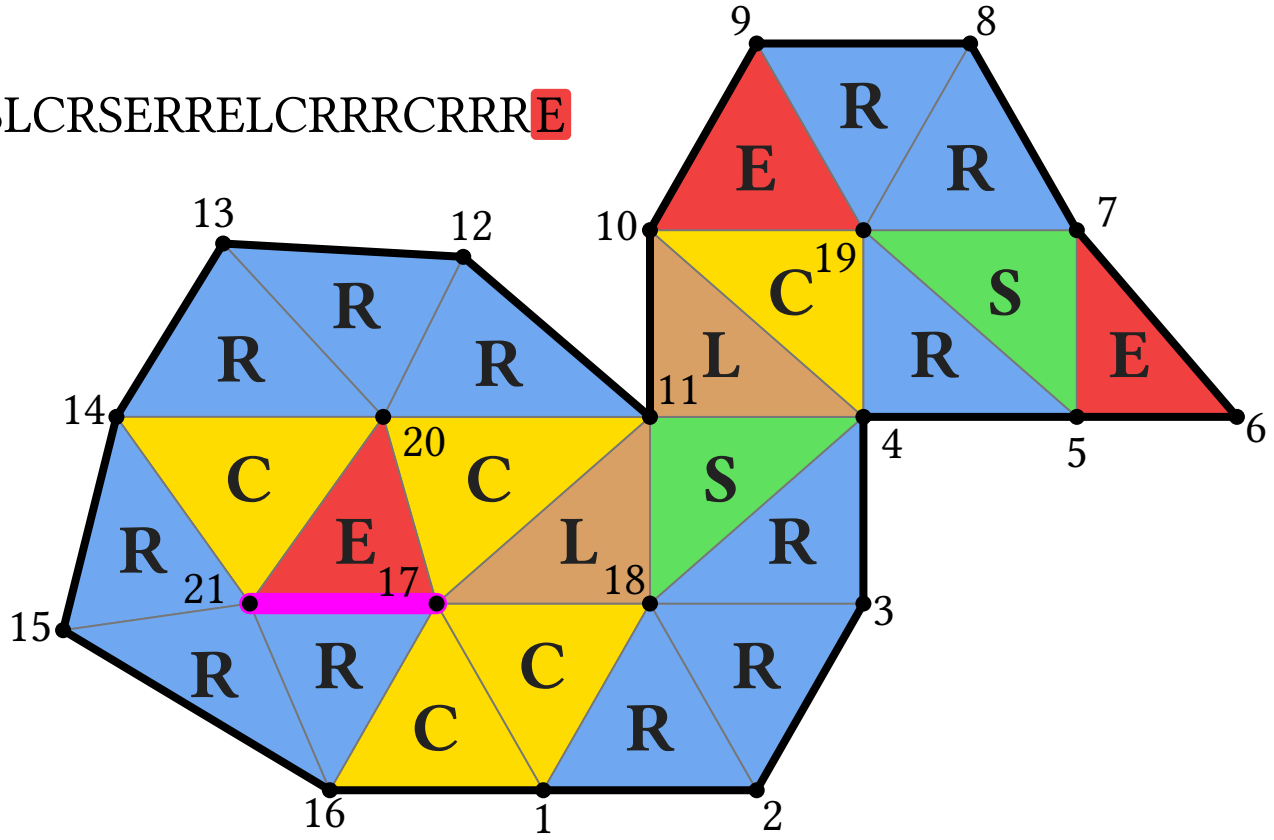


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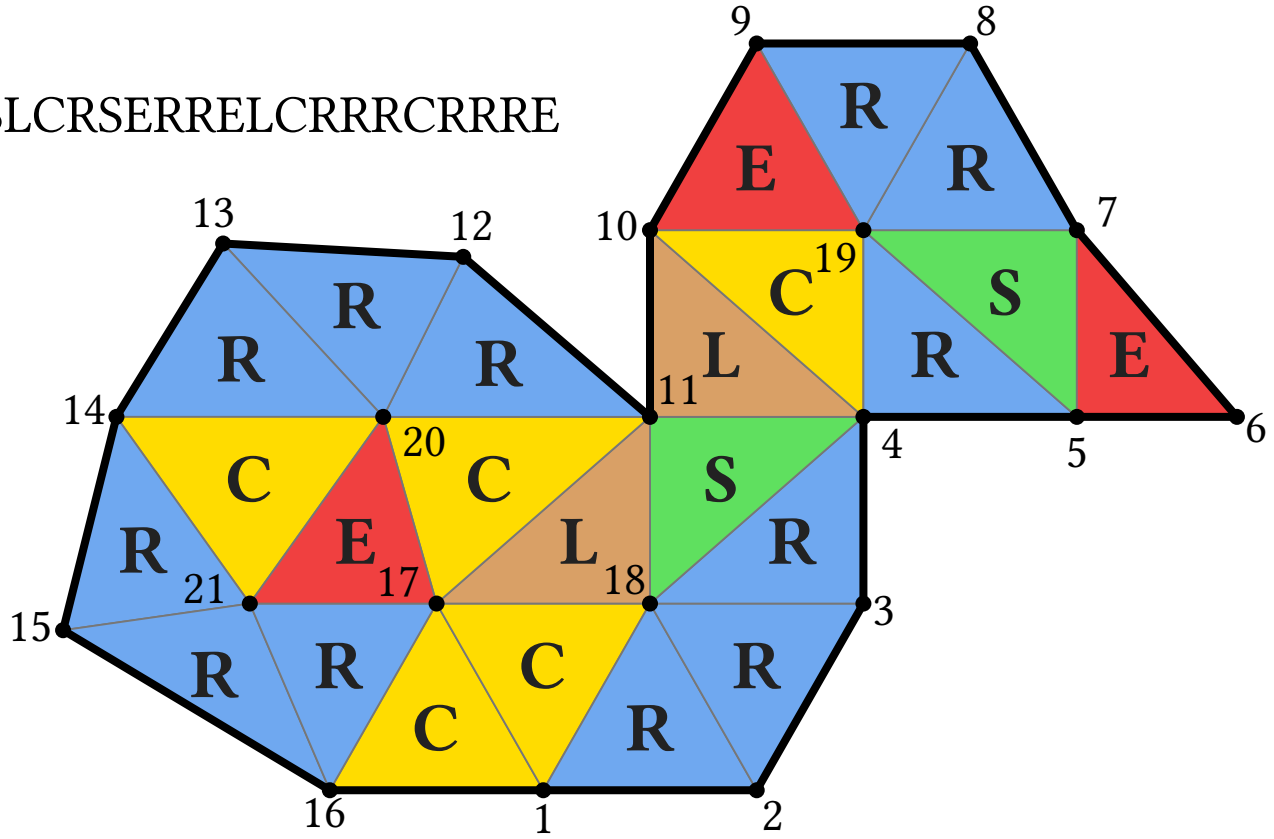


Figure 8: Dekomprese.

5. Vyšší genus

5.1 Řešení

- obecnější S
- navíc ukládání topologie

5.1 Řešení

- obecnější S
- navíc ukládání topologie

5.1.1 Díry

objevení díry => push topologie s délkou díry

5.1 Řešení

- obecnější S
- navíc ukládání topologie

5.1.1 Díry

objevení díry => push topologie s délkou díry

5.1.2 Handly

následuje S => push topologie se “zipem”

Historie =

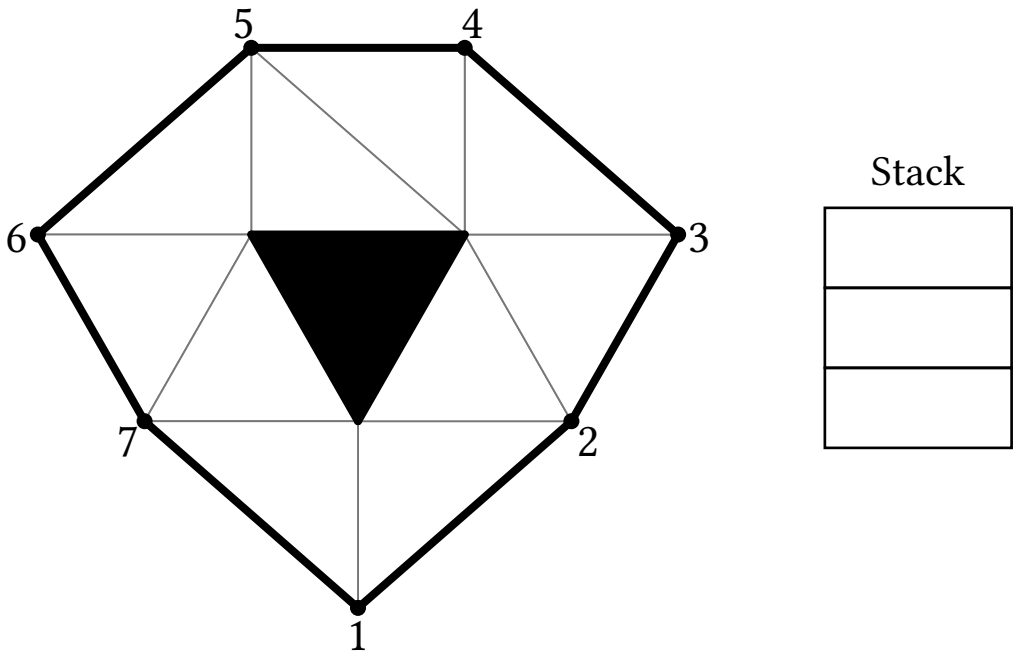


Figure 34: Složitější komprese.

Historie = S

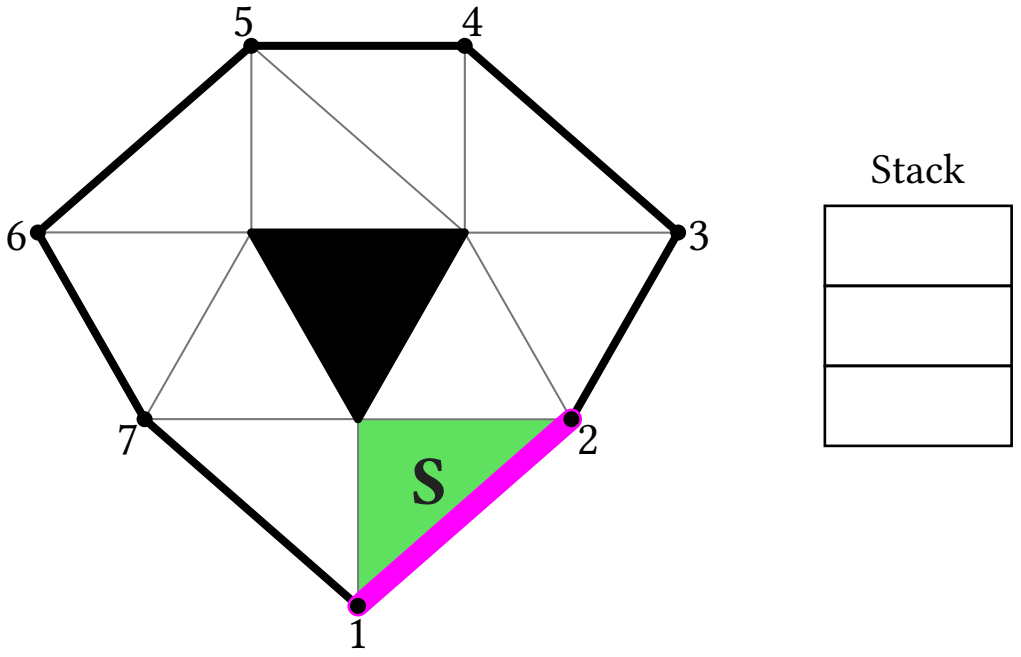


Figure 34: Složitější komprese.

Historie = S

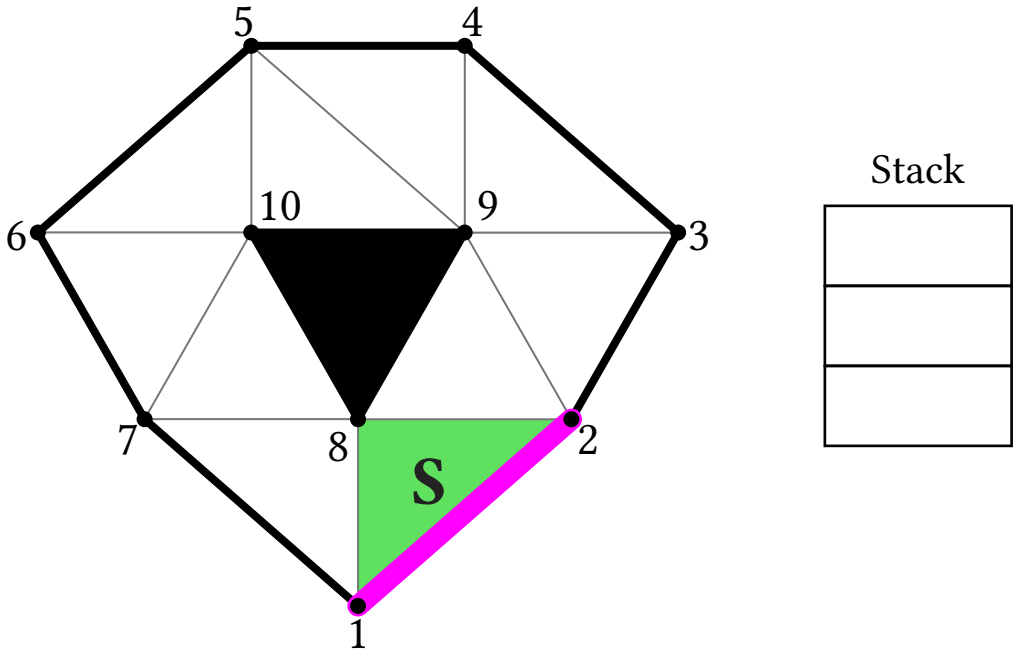


Figure 34: Složitější komprese.

Historie = SL

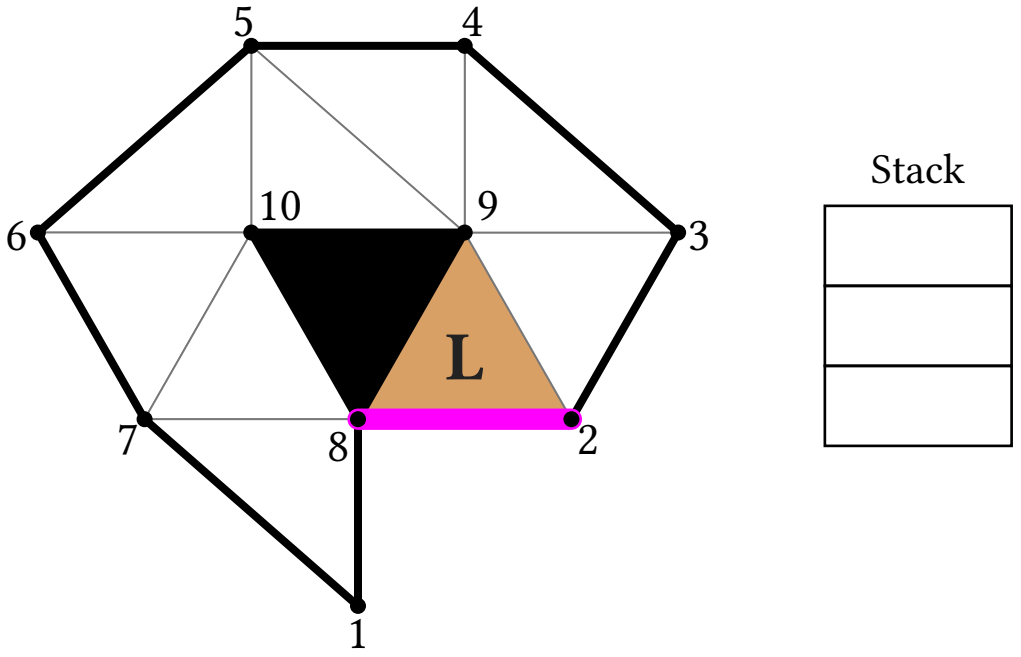


Figure 34: Složitější komprese.

Historie = SLR

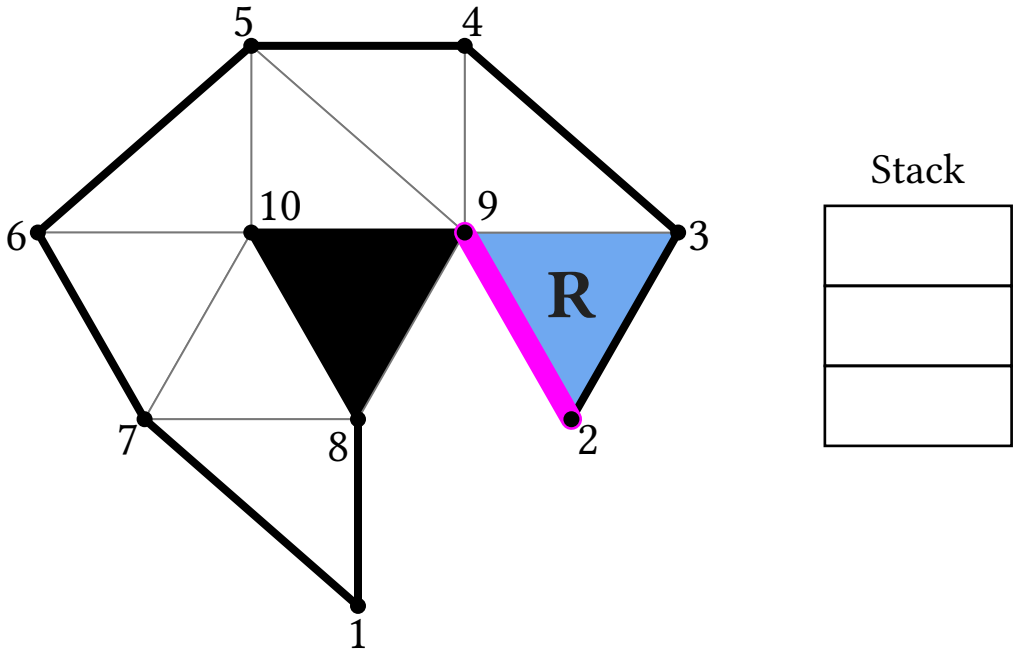


Figure 34: Složitější komprese.

Historie = SLRR

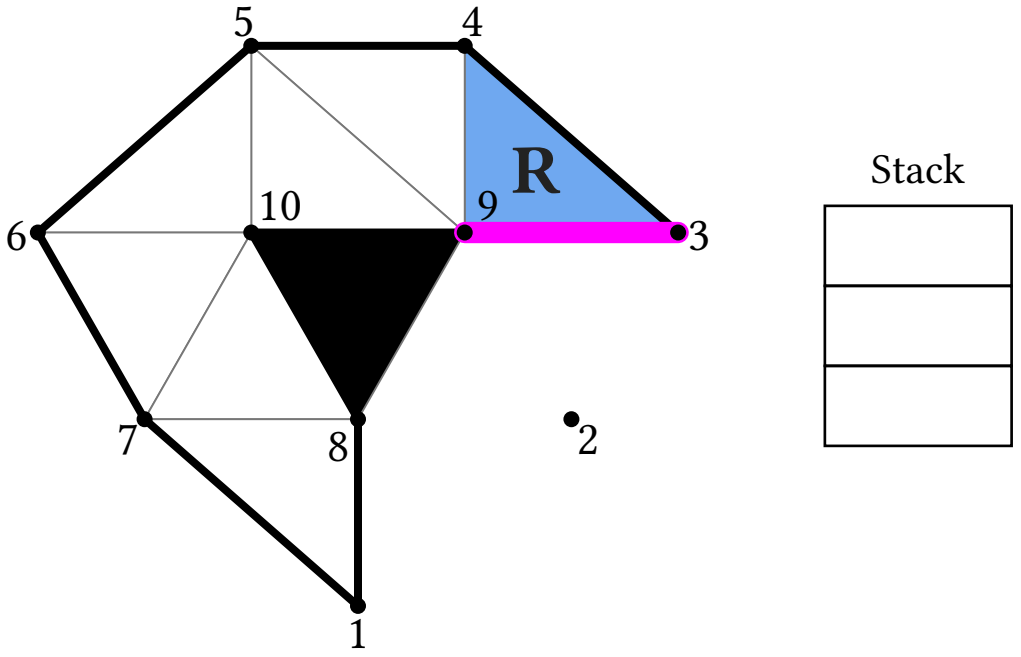


Figure 34: Složitější komprese.

Historie = SLRRR

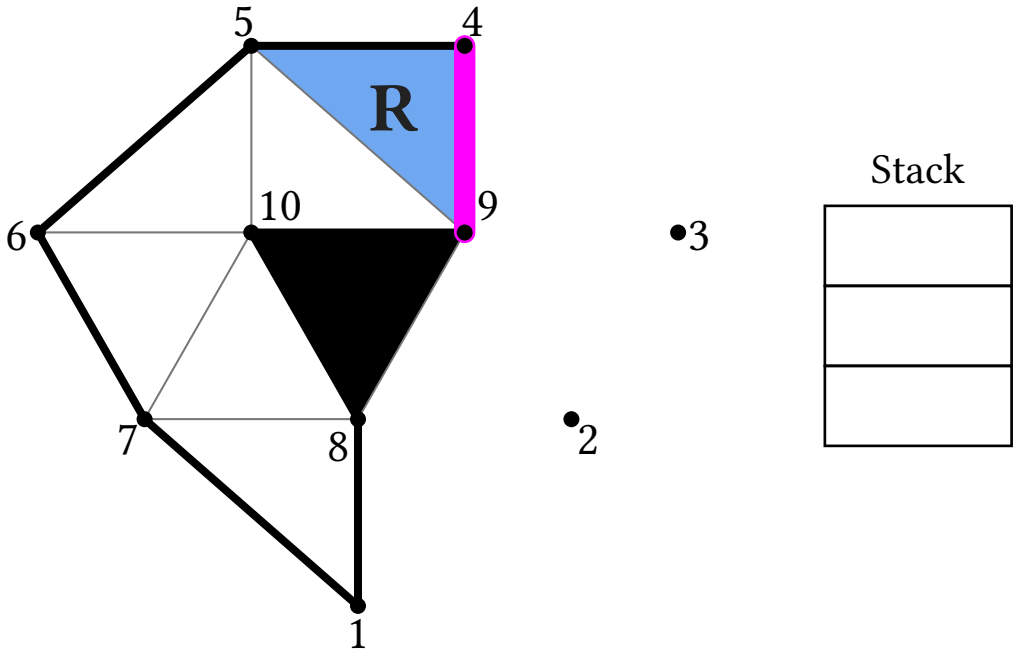


Figure 34: Složitější komprese.

Historie = SLRRRL

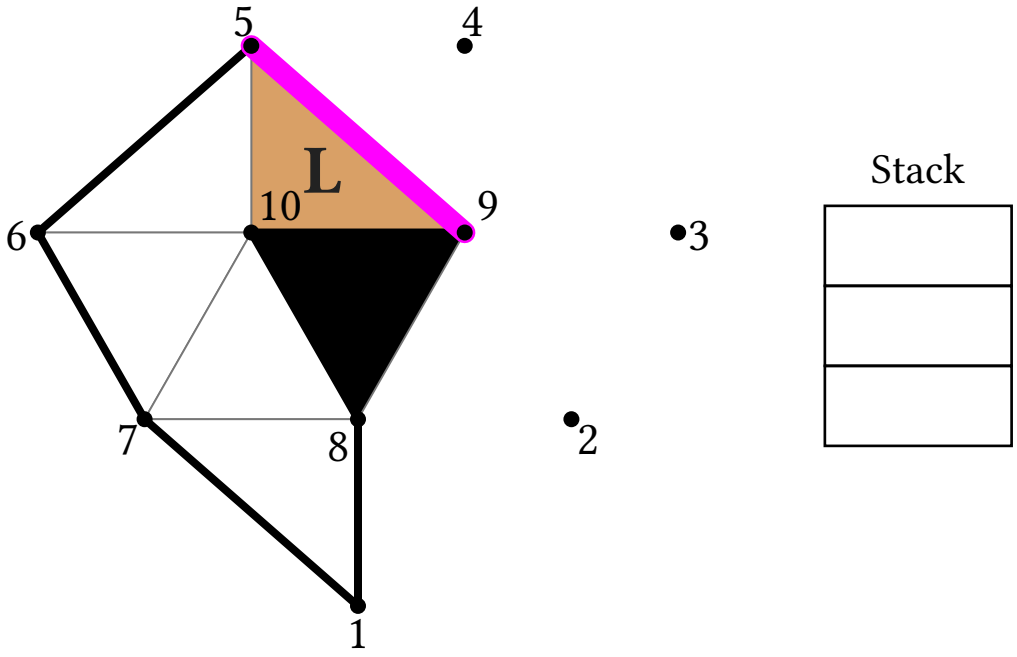


Figure 34: Složitější komprese.

Historie = SLRRRLR

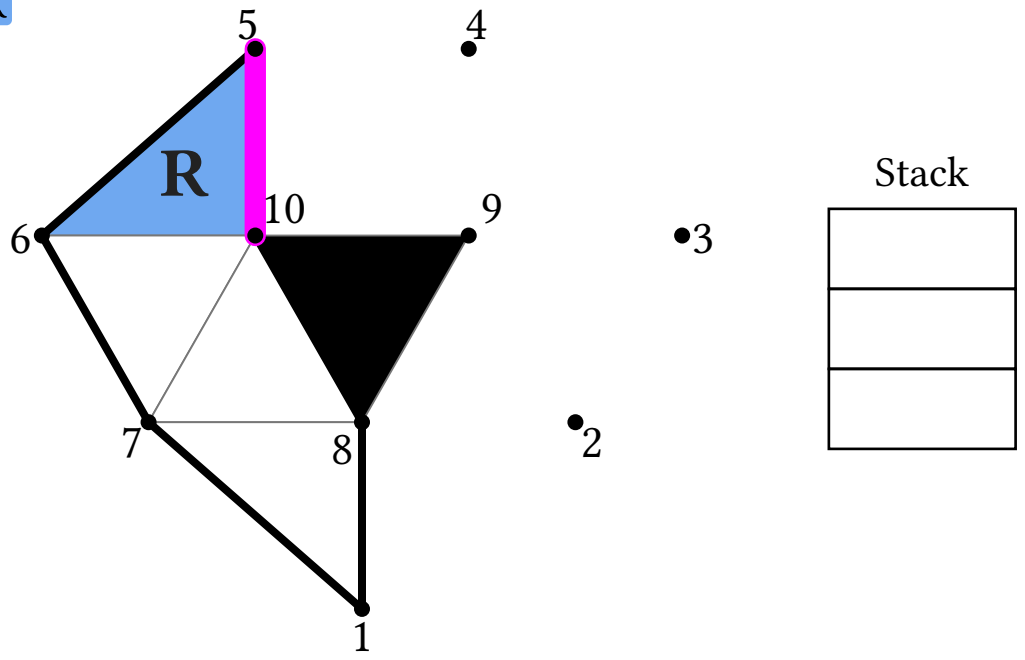


Figure 34: Složitější komprese.

Historie = SLRRRLRR

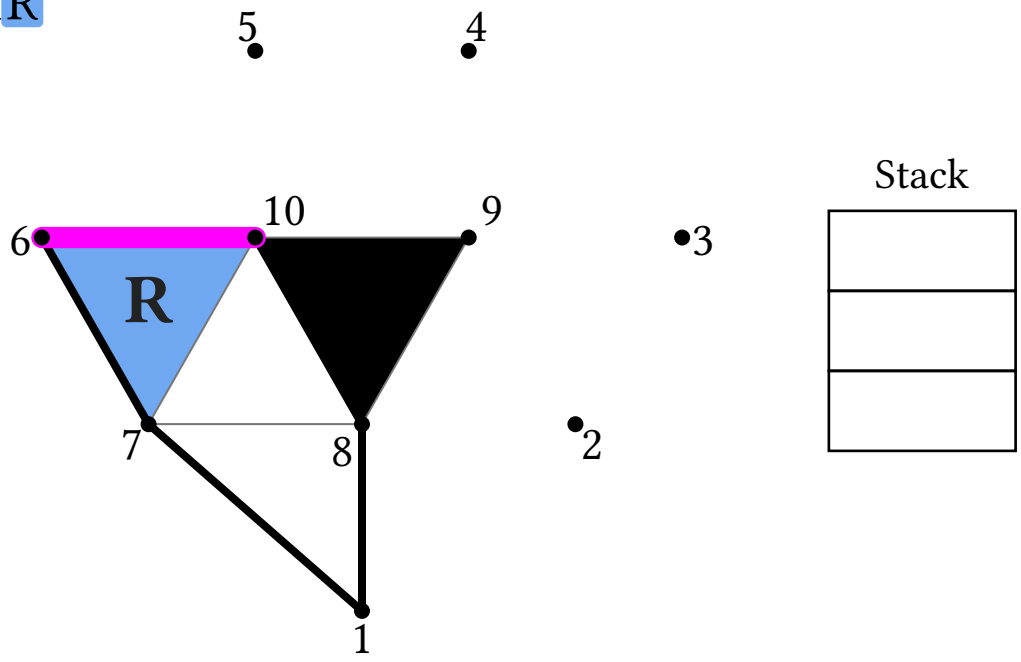


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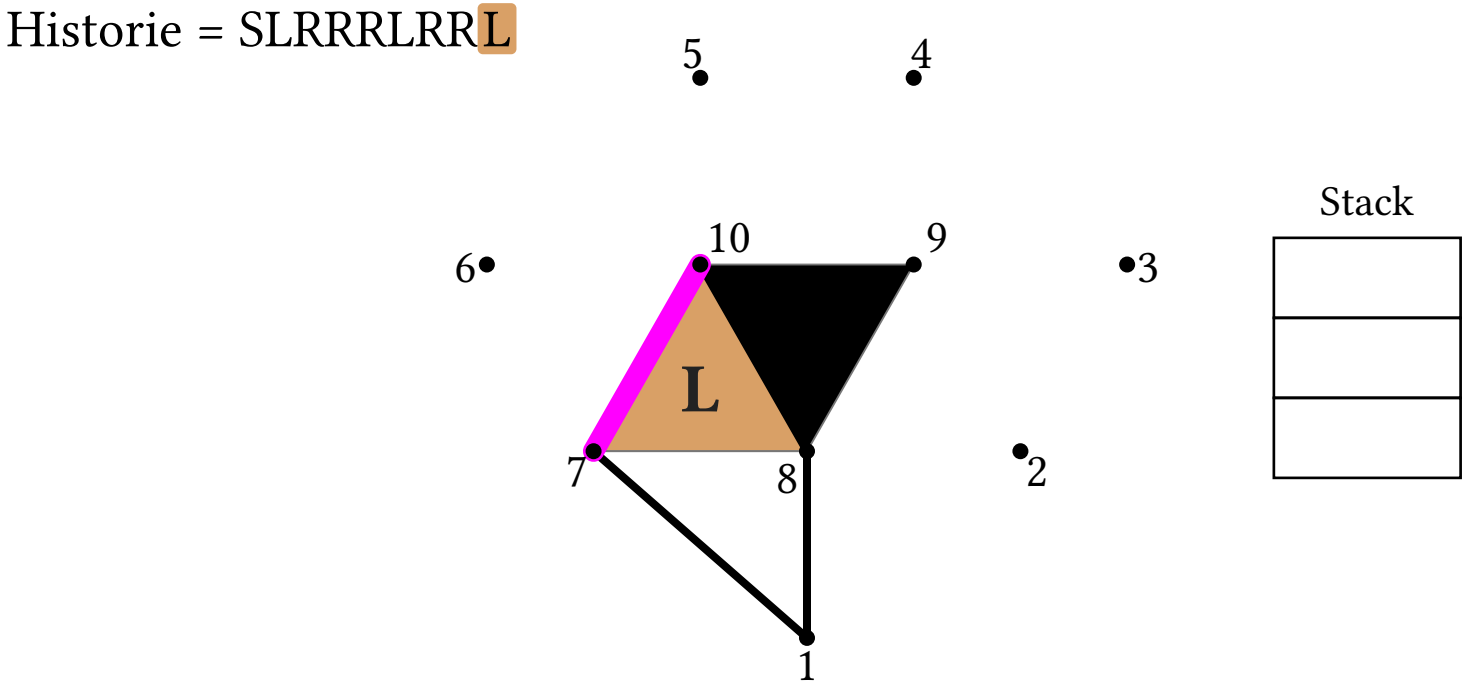


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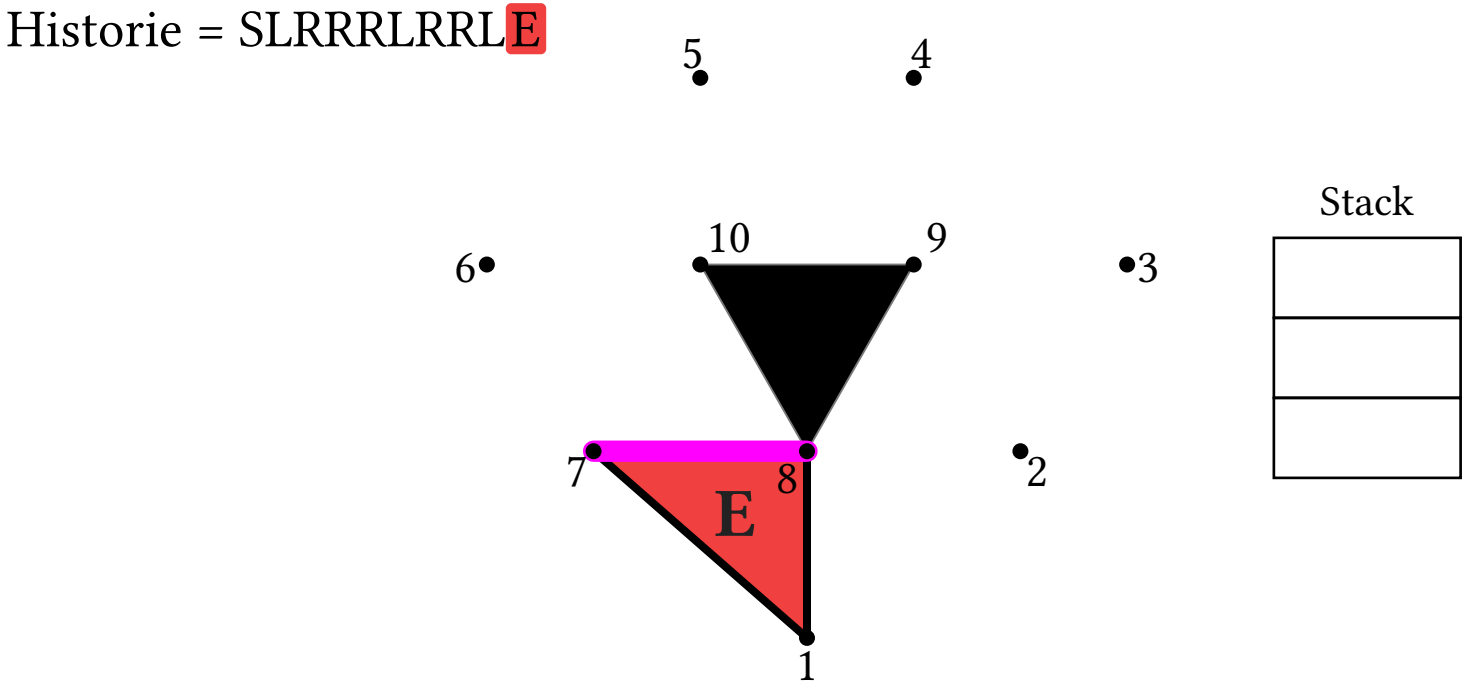


Figure 34: Složitější komprese.

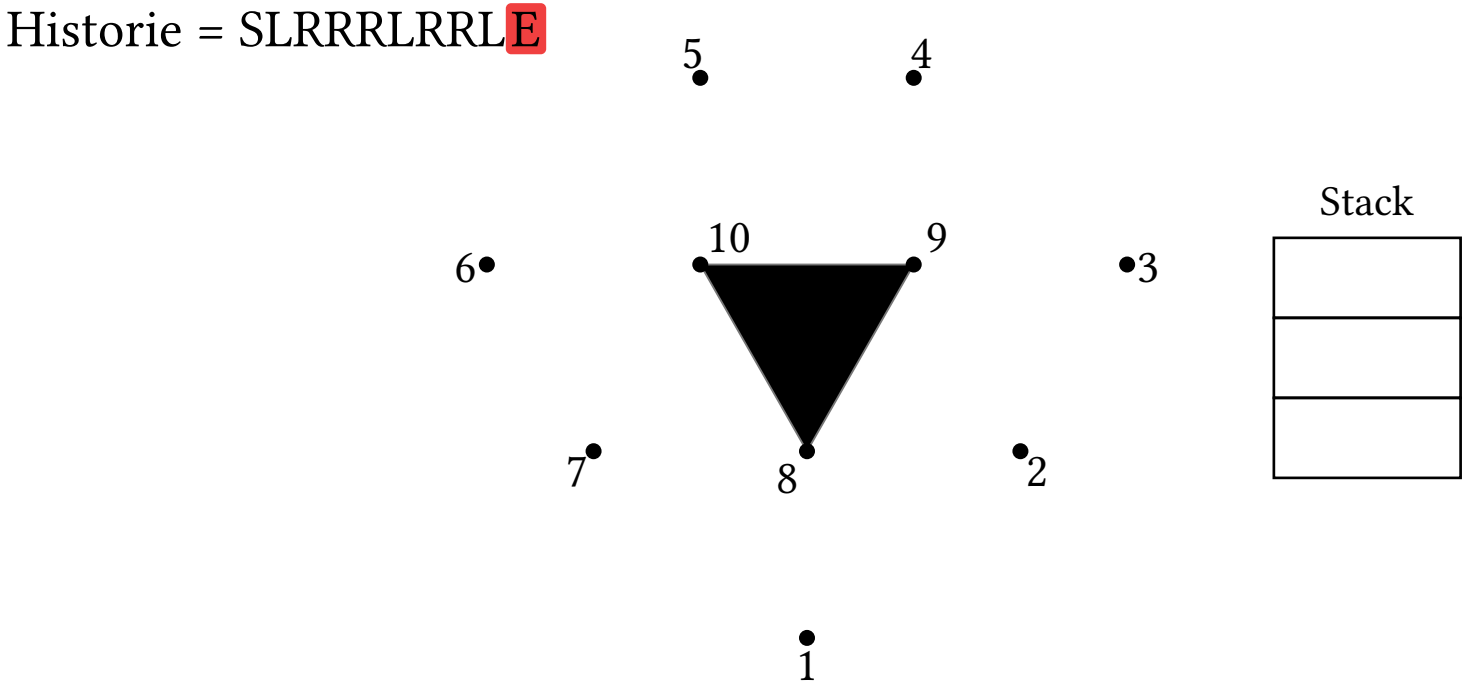


Figure 34: Složitější komprese.

6. Zdroje

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6.2 Obrázky (dobový kontext)

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6.3 Reference

Originální paper [1] Corner table [2] Významné vylepšení [3] Wrap & Zip [4] Inspirace animací [5]

[1] J. Rossignac, “Edgebreaker: Connectivity Compression for Triangle Meshes,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 5, no. 1, pp. 47–61, Jan. 1999, doi: 10.1109/2945.764870.

[2] J. Rossignac, A. Safonova, and A. Szymczak, “Edgebreaker on a Corner Table: A Simple Technique for Representing and Compressing Triangulated Surfaces,” in *Hierarchical and Geometrical Methods in Scientific Visualization*, G. Farin, B. Hamann, and H. Hagen, Eds., Berlin, Heidelberg: Springer, 2003, pp. 41–50. doi: 10.1007/978-3-662-07443-5_3.

[3] T. Lewiner, H. Lopes, J. Rossignac, and A. W. Vieira, “Efficient Edgebreaker for Surfaces of Arbitrary Topology,” in *Proceedings of SIBGRAPI 2004 (XVII Brazilian Symposium on Computer Graphics and Image Processing)*, Curitiba, Brazil, 2004, pp. 218–225. doi: 10.1109/SIBGRA.2004.1352965.

[4] J. Rossignac and A. Szymczak, “Wrap&Zip: Linear Decoding of Planar Triangle Graphs,” technical report GIT-GVU-99-08, Feb. 1999.

[5] “Edgebreaker: Connectivity Compression for Triangle Meshes.” Accessed: Nov. 03, 2026. [Online]. Available: https://graphics.stanford.edu/courses/cs468-08-winter/slides/sid2.pdf